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Ultragenyx Pharmaceutical Inc.
and Baylor Research Institute*

**UNITED STATES DISTRICT COURT
DISTRICT OF NEW JERSEY**

ULTRAGENYX PHARMACEUTICAL INC.
and BAYLOR RESEARCH INSTITUTE d/b/a
BAYLOR SCOTT & WHITE RESEARCH
INSTITUTE,

Plaintiffs,

v.

SOMERSET THERAPEUTICS LLC,
SOMERSET PHARMA, LLC, SOMERSET
THERAPEUTICS PRIVATE LIMITED f/k/a
SOMERSET THERAPEUTICS LIMITED, and
ODIN PHARMACEUTICALS LLC,

Defendants.

Civil Action No.

(Electronically Filed)

COMPLAINT FOR PATENT INFRINGEMENT

Plaintiffs Ultragenyx Pharmaceutical Inc. (“Ultragenyx”) and Baylor Research Institute d/b/a Baylor Scott & White Research Institute (“BRI”) (Ultragenyx and BRI, together, “Plaintiffs”), by their undersigned attorneys, bring this Complaint against Defendants Somerset Therapeutics LLC (“Somerset Therapeutics”), Somerset Pharma, LLC (“Somerset Pharma”),

Somerset Therapeutics Private Limited f/k/a Somerset Therapeutics Limited (“Somerset Private”), and Odin Pharmaceuticals LLC (“Odin”) (collectively, “Somerset” or “Defendants”), and hereby allege as follows:

NATURE OF THE ACTION

1. This is an action for patent infringement under the patent laws of the United States, 35 U.S.C. § 100, et seq., arising from Somerset’s filing of its Abbreviated New Drug Application (“ANDA”), No. 221271 (“Somerset’s ANDA”), with the U.S. Food and Drug Administration (“FDA”) seeking approval to commercially manufacture, use, offer for sale, sell, and/or import purported generic versions of triheptanoin oral liquid, 100% w/w (Dojolvi® (triheptanoin) Oral Liquid) (“Somerset’s ANDA Product”) prior to the expiration of U.S. Patent No. 8,697,748 (“the ’748 patent”) and U.S. Patent No. 12,551,461 (“the ’461 patent”) (together, “the Patents-in-Suit).

THE PARTIES

2. Plaintiff Ultragenyx is a pharmaceutical company focused on treating rare diseases. Ultragenyx is involved in the research and development of novel products for the treatment of rare and ultra-rare genetic diseases for which there are typically no approved treatments and there is a high unmet medical need.

3. Ultragenyx is a corporation organized and existing under the laws of the State of Delaware with its principal place of business at 60 Leveroni Court, Novato, California 94949.

4. Plaintiff BRI is a non-profit corporation organized and existing under the laws of the State of Texas with its principal place of business at 3310 Live Oak Street, Suite 501, Dallas, Texas 75204.

5. On information and belief, Defendant Somerset Therapeutics is a limited liability company organized and existing under the laws of the State of Delaware, having a principal place of business at 300 Franklin Square Drive, Somerset, New Jersey 08873. On information and belief,

Somerset Therapeutics is in the business of, among other things, manufacturing, promoting, marketing, selling, offering for sale, using, distributing, and importing into the United States, generic versions of branded pharmaceutical drugs for the U.S. market.

6. On information and belief, Defendant Somerset Pharma is a limited liability company organized and existing under the laws of the State of Delaware, having a principal place of business at 300 Franklin Square Drive, Somerset, New Jersey 08873. On information and belief, Somerset Pharma is in the business of, among other things, manufacturing, promoting, marketing, selling, offering for sale, using, distributing, and importing into the United States, generic versions of branded pharmaceutical drugs for the U.S. market.

7. On information and belief, Defendant Somerset Private is a corporation organized and existing under the laws of India, having a corporate headquarters at 54/1 Budihal, Nelamangala, Bengaluru-562123, Karnataka, India. On information and belief, Somerset Private is in the business of, among other things, manufacturing, promoting, marketing, selling, offering for sale, using, distributing, and importing into the United States, generic versions of branded pharmaceutical drugs for the U.S. market.

8. On information and belief, Defendant Odin is a limited liability company organized and existing under the laws of the State of Delaware, having a principal place of business at 300 Franklin Square Drive, Somerset, New Jersey 08873. On information and belief, Odin is in the business of, among other things, manufacturing, promoting, marketing, selling, offering for sale, using, distributing, and importing into the United States, generic versions of branded pharmaceutical drugs for the U.S. market.

THE PATENTS-IN-SUIT

9. BRI is the owner and assignee of the '748 patent, entitled "Glycogen or polysaccharide storage disease treatment method" (attached hereto as Exhibit A).

10. On April 15, 2014, the '748 patent was duly and legally issued to BRI as assignee of inventor Charles R. Roe.

11. BRI is the owner of all right, title, and interest in the '748 patent. Ultragenyx is the exclusive licensee of the '748 patent.

12. Ultragenyx is the owner and assignee of the '461 patent, entitled "Compositions of triglycerides and uses thereof" (attached hereto as Exhibit B).

13. On February 17, 2026, the '461 patent was duly and legally issued to Ultragenyx as assignee of inventors John Klopp, Gabrielle Morris, Emil Kakkis, and Steven Jungles.

14. Ultragenyx is the owner of all right, title, and interest in the '461 patent.

THE DOJOLVI® DRUG PRODUCT

15. Ultragenyx holds an approved New Drug Application ("NDA") under Section 505(a) of the Federal Food, Drug, and Cosmetic Act ("FDCA"), 21 U.S.C. § 355(a), for triheptanoin oral liquid (NDA No. 213687), which it sells under the trade name DOJOLVI®. DOJOLVI® is an FDA-approved medication for the treatment of adult and pediatric patients with molecularly confirmed long-chain fatty acid oxidation disorders (LC-FAOD).

16. Pursuant to 21 U.S.C. § 355(b)(1) and attendant FDA regulations, the Patents-in-Suit are listed in the FDA publication, "Approved Drug Products with Therapeutic Equivalence Evaluations" (the "Orange Book") with respect to DOJOLVI®.

17. The claims of the Patents-in-Suit cover, inter alia, pharmaceutical compositions of triheptanoin.

18. The FDA's Orange Book for DOJOLVI® currently lists the expiration of the '748 patent as April 28, 2029, and the expiration of the '461 patent as November 14, 2034.

JURISDICTION AND VENUE

19. Plaintiffs incorporate each of the preceding paragraphs 1–18 as if fully set forth herein.

20. This is a civil action for patent infringement arising under the patent laws of the United States, 35 U.S.C. § 100, et seq., including 35 U.S.C. § 271, for infringement of the Patents-in-Suit. This Court has jurisdiction over the subject matter of this action pursuant to 28 U.S.C. §§ 1331, 1338(a), 2201, and 2202.

21. On information and belief, Somerset Therapeutics has a principal place of business at 300 Franklin Square Drive, Somerset, New Jersey 08873. Somerset Therapeutics is registered to do business with the State of New Jersey’s Division of Revenue and Enterprise Services as a business operating in New Jersey under Business ID No. 0451084958 and is registered with New Jersey as a business that distributes prescription or non-prescription drugs or medical devices under Registration Number 5004837. Somerset Therapeutics is thus subject to personal jurisdiction in New Jersey.

22. On information and belief, Somerset Pharma has a principal place of business at 300 Franklin Square Drive, Somerset, New Jersey 08873. Somerset Pharma is registered to do business with the State of New Jersey’s Division of Revenue and Enterprise Services as a business operating in New Jersey under Business ID No. 0450400310 and is registered with New Jersey as a business that distributes prescription or non-prescription drugs or medical devices under Registration Number 5005665. Somerset Pharma is thus subject to personal jurisdiction in New Jersey.

23. On information and belief, Odin has a principal place of business at 300 Franklin Square Drive, Somerset, New Jersey 08873. Odin is registered to do business with the State of

New Jersey's Division of Revenue and Enterprise Services as a business operating in New Jersey under Business ID No. 0450315269. Odin is thus subject to personal jurisdiction in New Jersey.

24. On information and belief, Somerset Private is a corporation organized and existing under the laws of India, having a corporate headquarters 54/1, Budihal, Nelamangala, Bengaluru 562123, Karnataka, India. Somerset Private is subject to personal jurisdiction in this Court because, among other things, it (1) has purposefully availed itself of the privilege of doing business in New Jersey, including directly or indirectly through its related entities (e.g., Somerset Therapeutics, Somerset Pharma, and Odin), and (2) maintains pervasive, continuous, and systematic contacts with the State of New Jersey through the marketing, distribution, and sale of generic versions of branded pharmaceutical products in the State of New Jersey, directly and through its affiliates, and by deriving substantial revenue from the importation and sale of its products in the State of New Jersey.

25. Alternatively, Somerset Private is subject to personal jurisdiction in this Court pursuant to Federal Rule of Civil Procedure 4(k)(2) as (1) Plaintiffs' claims arise under federal law; (2) Somerset Private is a foreign defendant not subject to general personal jurisdiction in the courts of any state; and (3) Somerset Private has sufficient contacts with the United States as a whole, including, but not limited to, preparing and submitting ANDAs to the FDA and/or manufacturing, importing, offering to sell, and/or selling pharmaceutical products that are distributed throughout the United States, such that this Court's exercise of jurisdiction over Somerset Private satisfies due process.

26. Defendants are also subject to personal jurisdiction in New Jersey because, *inter alia*, Defendants have purposely availed themselves of the benefits and protections of New Jersey's laws such that they should reasonably anticipate being sued in this Judicial District. On

information and belief, Defendants develop, manufacture, import, market, distribute, use, offer to sell, and/or sell generic drugs throughout the United States, including New Jersey, and therefore transact business within New Jersey related to Plaintiffs' claims, and/or have engaged in systematic and continuous business contacts within New Jersey.

27. On information and belief, Defendants have systematic and continuous contacts with New Jersey; have established distribution channels for drug products in New Jersey; regularly and continuously conduct business in New Jersey, including by selling drug products in New Jersey, either directly or indirectly through their subsidiaries, agents, or affiliates; have purposefully availed themselves of the privilege of doing business in New Jersey; and derive substantial revenue from the sale of drug products in New Jersey.

28. On information and belief, Defendants derive substantial revenue from generic pharmaceutical products that are used and/or consumed within New Jersey, and that are manufactured by Defendants and/or for which Defendants are the named applicant on approved ANDAs. On information and belief, various products for which Defendants are the named applicant(s) on approved ANDAs are available at hospital and/or retail pharmacies in New Jersey.

29. On information and belief, Defendants intend a future course of conduct that includes acts of patent infringement in New Jersey. These acts have led and will continue to lead to foreseeable harm and injury to Plaintiffs in New Jersey and this Judicial District.

30. Defendants have committed an act of infringement in this Judicial District by filing ANDA No. 221271 with the intent to make, use, sell, offer for sale, and/or import Somerset's ANDA Product in or into this Judicial District, prior to the expiration of the Patents-in-Suit.

31. Defendants have taken the costly, significant step of applying to the FDA for approval to engage in future activities, including the marketing of Somerset's ANDA Product, which will be purposefully directed at New Jersey and elsewhere in the United States.

32. On information and belief, each of the Defendants participated, directly or indirectly, in the preparation and/or submission of Somerset's ANDA. On information and belief, Defendants will work in privity and/or concert either directly or indirectly with each other and/or through other related entities towards the regulatory approval, manufacturing, use, importation, marketing, offer for sale, sale, and distribution of generic pharmaceutical products, including Somerset's ANDA Product, throughout the United States, including in New Jersey and in this Judicial District, prior to the expiration of the Patents-in-Suit.

33. On information and belief, if Somerset's ANDA is approved, Defendants will manufacture, market, promote, sell, offer for sale, import, use, and/or distribute Somerset's ANDA Product within the United States, including in New Jersey, consistent with Somerset's practices for the marketing and distribution of other generic pharmaceutical products. On information and belief, Defendants regularly do business in New Jersey, and Defendants' practices with other generic pharmaceutical products have involved placing those products into the stream of commerce for distribution throughout the United States, including in New Jersey. On information and belief, Somerset's generic pharmaceutical products are used and/or consumed within and throughout the United States, including in New Jersey. On information and belief, Somerset's ANDA Product will be prescribed by physicians practicing in New Jersey, dispensed by pharmacies located within New Jersey, and used by patients in New Jersey. Each of these activities would have a substantial effect within New Jersey and would constitute infringement of the

Patents-in-Suit in the event that Somerset's ANDA Product is approved before the Patents-in-Suit expire.

34. On information and belief, Somerset Private intends to benefit directly if Somerset's ANDA is approved by participating in the manufacture, importation, distribution, and/or sale of the generic drug products that are the subject of Somerset's ANDA.

35. On information and belief, Somerset has not contested personal jurisdiction in this Judicial District and/or regularly invokes the jurisdiction of the courts of this Judicial District by pleading claims and counterclaims in pharmaceutical patent infringement actions in this Judicial District. *See, e.g., Xeris Pharmaceuticals, Inc., et al. v. Somerset Therapeutics, LLC, et al.*, C.A. No. 26-cv-2044, D.I. 19 (D.N.J. Mar. 27, 2026); *In re Selenious Acid Litigation*, C.A. No. 24-7791, D.I. 73 (D.N.J. Jan. 9, 2025); *American Regent, Inc. v. Somerset Therapeutics, LLC, et al.*, C.A. No. 24-1022, D.I. 12 (D.N.J. Apr. 1, 2024).

36. Venue is proper in this Judicial District pursuant to 28 U.S.C. §§ 1391 and/or 1400(b).

37. Venue is proper at least because, on information and belief, Somerset submitted its ANDA from its Somerset, New Jersey place of business and therefore Somerset has committed acts of infringement and has a regular and established place of business in New Jersey for the purposes of venue.

38. Venue is proper in this Judicial District with respect to Somerset Therapeutics because, *inter alia*, it maintains a regular and established place of business in this Judicial District and has committed an act of infringement in this Judicial District.

39. Venue is proper in this Judicial District with respect to Somerset Pharma because, *inter alia*, it maintains a regular and established place of business in this Judicial District and has committed an act of infringement in this Judicial District.

40. Venue is proper in this Judicial District with respect to Odin because, *inter alia*, it maintains a regular and established place of business in this Judicial District and has committed an act of infringement in this Judicial District.

41. Venue is proper in this Judicial District with respect to Somerset Private because, on information and belief, Somerset Private is a foreign corporation that may be sued in any judicial district where it is subject to the personal jurisdiction of the Court, including this Judicial District.

42. Moreover, on information and belief, Somerset has not contested venue in pharmaceutical patent infringement actions in this Judicial District. *See, e.g., Xeris Pharmaceuticals, Inc. et al v. Somerset Therapeutics, LLC et al*, C.A. No. 26-cv-2044, D.I. 19 (D.N.J. Mar. 27, 2026); *In re Selenious Acid Litigation*, C.A. No. 24-7791, D.I. 73 (D.N.J. Jan. 9, 2025); *American Regent, Inc. v. Somerset Therapeutics, LLC et. al.*, C.A. No. 24-1022, D.I. 12 (D.N.J. Apr. 1, 2024).

**ACTS GIVING RISE TO THIS SUIT: SOMERSET'S ANDA AND NOTICE OF
PARAGRAPH IV CERTIFICATION**

43. On information and belief, Somerset has submitted or caused the submission of Somerset's ANDA to the FDA under 21 U.S.C. § 355(j), to obtain approval to engage in the commercial manufacture, use, offer to sell, or sale within the United States or importation into the United States of Somerset's ANDA Product, as purported generic versions of DOJOLVI[®], prior to the expiration of the patents listed in the Orange Book, i.e., the Patents-in-Suit.

44. On information and belief, the FDA has not yet approved Somerset's ANDA.

45. By a letter dated February 18, 2026 (“Somerset’s Feb. 18, 2026 Notice Letter”), Somerset notified Plaintiffs of the submission of Somerset’s ANDA to the FDA. The purpose of this submission was to obtain approval under the FDCA to engage in the commercial manufacture, use, offer for sale, sale, and/or importation of Somerset’s ANDA Product prior to the expiration of the patents listed in the Orange Book, including the ’748 patent and the ’461 patent.

46. In Somerset’s Feb. 18, 2026 Notice Letter, Somerset acknowledged that the Reference Listed Drug for Somerset’s ANDA is DOJOLVI®. DOJOLVI® is indicated for the treatment of adult and pediatric patients with molecularly confirmed long-chain fatty acid oxidation disorders (LC-FAOD).

47. In Somerset’s Feb. 18, 2026 Notice Letter, Somerset stated that Somerset’s ANDA Product is a triheptanoin oral liquid.

48. By filing its ANDA, Somerset has represented to the FDA that Somerset’s ANDA Product is bioequivalent to Plaintiffs’ DOJOLVI®.

49. In Somerset’s Feb. 18, 2026 Notice Letter, Somerset also notified Plaintiffs that, as part of its ANDA, Somerset had filed a Paragraph IV Certification pursuant to 21 U.S.C. § 355(j)(2)(A)(vii)(IV) with respect to the ’748 patent.

50. Somerset’s Feb. 18, 2026 Notice Letter stated that Somerset had certified to the FDA that in Somerset’s opinion and to the best of its knowledge, the ’748 patent is invalid, unenforceable, and/or will not be infringed by the manufacture, use, offer for sale, sale, and/or importation of Somerset’s ANDA Product.

51. On February 17, 2026, Plaintiffs’ ’461 patent issued, and Plaintiffs submitted that patent for listing in the Orange Book on February 19, 2026.

52. On information and belief, Somerset amended its ANDA to the FDA to contain a certification pursuant to 21 U.S.C. § 355(j)(2)(A)(vii)(IV) asserting that the '461 patent is invalid, unenforceable, and/or will not be infringed by the manufacture, use, offer for sale, sale, and/or importation of Somerset's ANDA Product.

53. On information and belief, Somerset, through its own actions and through the actions of its agents, affiliates, and subsidiaries, prepared and submitted Somerset's ANDA, and intends to further prosecute Somerset's ANDA. On information and belief, if the FDA approves Somerset's ANDA, Somerset will manufacture, import, distribute, promote, market, offer for sale, or sell Somerset's ANDA Product within the United States, or will import Somerset's ANDA Product into the United States. On information and belief, if the FDA approves Somerset's ANDA, Somerset will actively induce or contribute to the manufacture, use, offer for sale, sale, or importation of Somerset's ANDA Product in or into the United States.

54. Plaintiffs bring this action within forty-five days of receipt of Somerset's Feb. 18, 2026 Notice Letter. Accordingly, Plaintiffs are entitled to a stay of FDA approval of Somerset's ANDA pursuant to 21 U.S.C. § 355(j)(5)(B)(iii).

COUNT I – INFRINGEMENT BY SOMERSET OF THE '748 PATENT

55. Plaintiffs incorporate each of the preceding paragraphs 1–54 as if fully set forth herein.

56. Somerset's submission of Somerset's ANDA with a Paragraph IV Certification for the purpose of obtaining approval to engage in the commercial manufacture, use, offer for sale, sale, and/or importation of Somerset's ANDA Product in or into the United States before the expiration of the '748 patent is an act of infringement of the '748 patent under 35 U.S.C. § 271(e)(2)(A).

57. There is a justiciable controversy between Plaintiffs and Somerset as to the infringement of the '748 patent.

58. Unless enjoined by this Court, upon FDA approval of Somerset's ANDA, Somerset will infringe one or more claims of the '748 patent under 35 U.S.C. § 271(a), including at least claim 1, by making, using, offering to sell, selling, and/or importing Somerset's ANDA Product in the United States.

59. Unless enjoined by this Court, upon FDA approval of Somerset's ANDA, Somerset will induce infringement of one or more claims of the '748 patent under 35 U.S.C. § 271(b), including at least claim 1, by making, using, offering to sell, selling, and/or importing Somerset's ANDA Product in the United States. On information and belief, upon FDA approval of Somerset's ANDA, Somerset will intentionally encourage acts of direct infringement with knowledge of the '748 patent and knowledge that its acts are encouraging infringement.

60. Unless enjoined by this Court, upon FDA approval of Somerset's ANDA, Somerset will contributorily infringe one or more claims of the '748 patent under 35 U.S.C. § 271(c), including at least claim 1, by making, using, offering to sell, selling, and/or importing Somerset's ANDA Product in the United States. On information and belief, Somerset has had and continues to have knowledge that Somerset's ANDA Product is especially adapted for a use that infringes one or more claims of the '748 patent, including at least claim 1, and that there is no substantial non-infringing use for Somerset's ANDA Product.

61. Plaintiffs will be substantially and irreparably damaged and harmed if Somerset's infringement of the '748 patent is not enjoined.

62. Plaintiffs do not have an adequate remedy at law.

63. This case is an exceptional one, and Plaintiffs are entitled to an award of their reasonable attorneys' fees under 35 U.S.C. § 285.

COUNT II – INFRINGEMENT BY SOMERSET OF THE '461 PATENT

64. Plaintiffs incorporate each of the preceding paragraphs 1–63 as if fully set forth herein.

65. Somerset's submission of Somerset's ANDA with a Paragraph IV Certification for the purpose of obtaining approval to engage in the commercial manufacture, use, offer for sale, sale, and/or importation of Somerset's ANDA Product in or into the United States before the expiration of the '461 patent is an act of infringement of the '461 patent under 35 U.S.C. § 271(e)(2)(A).

66. There is a justiciable controversy between Plaintiffs and Somerset as to the infringement of the '461 patent.

67. Unless enjoined by this Court, upon FDA approval of Somerset's ANDA, Somerset will infringe one or more claims of the '461 patent under 35 U.S.C. § 271(a), including at least claim 1, by making, using, offering to sell, selling, and/or importing Somerset's ANDA Product in the United States.

68. Unless enjoined by this Court, upon FDA approval of Somerset's ANDA, Somerset will induce infringement of one or more claims of the '461 patent under 35 U.S.C. § 271(b), including at least claim 1, by making, using, offering to sell, selling, and/or importing Somerset's ANDA Product in the United States. On information and belief, upon FDA approval of Somerset's ANDA, Somerset will intentionally encourage acts of direct infringement with knowledge of the '461 patent and knowledge that its acts are encouraging infringement.

69. Unless enjoined by this Court, upon FDA approval of Somerset's ANDA, Somerset will contributorily infringe one or more claims of the '461 patent under 35 U.S.C. § 271(c),

including at least claim 1, by making, using, offering to sell, selling, and/or importing Somerset's ANDA Product in the United States. On information and belief, Somerset has had and continues to have knowledge that Somerset's ANDA Product is especially adapted for a use that infringes one or more claims of the '461 patent, including at least claim 1, and that there is no substantial non-infringing use for Somerset's ANDA Product.

70. Plaintiffs will be substantially and irreparably damaged and harmed if Somerset's infringement of the '461 patent is not enjoined.

71. Plaintiffs do not have an adequate remedy at law.

72. This case is an exceptional one, and Plaintiffs are entitled to an award of its reasonable attorneys' fees under 35 U.S.C. § 285.

PRAYER FOR RELIEF

WHEREFORE, Plaintiffs request the following relief:

(a) A judgment that the Patents-in-Suit have been infringed under 35 U.S.C. § 271(e)(2) by Somerset's submission to the FDA of Somerset's ANDA;

(b) A judgment, pursuant to 35 U.S.C. § 271(e)(4)(A), ordering that the effective date of any FDA approval of ANDA No. 221271 for the commercial manufacture, use, or sale of Somerset's ANDA Product, or any other drug product that infringes or the use of which infringes the Patents-in-Suit, be not earlier than the expiration date of the Patents-in-Suit, inclusive of any extension(s) and additional period(s) of exclusivity;

(c) A preliminary and permanent injunction, pursuant to 35 U.S.C. §§ 271(e)(4)(B) and 283 and Rule 65, Fed. R. Civ. P., enjoining Somerset, and all persons acting in concert with Somerset, from the commercial manufacture, use, offer for sale, or sale in the United States, or importation into the United States of Somerset's ANDA Product, or any other

drug product covered by or whose use is covered by the Patents-in-Suit, prior to the expiration of the Patents-in-Suit, inclusive of any extension(s) and additional period(s) of exclusivity;

(d) A judgment declaring that the commercial manufacture, use, offer for sale, or sale in the United States, or importation into the United States of Somerset's ANDA Product, or any other drug product that is covered by or whose use is covered by the Patents-in-Suit, prior to the expiration of the Patents-in-Suit, inclusive of any extension(s) and additional period(s) of exclusivity, will infringe, induce the infringement of, and contribute to the infringement by others of the Patents-in-Suit;

(e) A declaration that Somerset's commercial manufacture, use, offer for sale, or sale in the United States, or importation into the United States, of Somerset's ANDA Product, or inducing or contributing to such conduct, would constitute infringement of one or more claims of the Patents-in-Suit by Somerset, including at least claim 1 of each, under one or more of 35 U.S.C. § 271(a), (b), and (c);

(f) An award of damages or other relief, pursuant to 35 U.S.C. § 271(e)(4)(C), if Somerset engages in the commercial manufacture, use, offer for sale, or sale in the United States, or importation into the United States of Somerset's ANDA Product, or any product that infringes the Patents-in-Suit, or induces or contributes to such conduct, prior to the expiration of the Patents-in-Suit, inclusive of any extension(s) and additional period(s) of exclusivity;

(g) A judgment that Somerset willfully and deliberately infringed the Patents-in-Suit;

(h) A declaration that this is an exceptional case and an award of attorneys' fees pursuant to 35 U.S.C. § 285;

(i) Costs and expenses in this action; and

(j) Such further and other relief as this Court may deem just and proper.

Dated: April 1, 2026

By: s/ Charles M. Lizza

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LOCAL CIVIL RULES 11.2 AND 40.1 CERTIFICATION

Pursuant to Local Civil Rules 11.2 and 40.1, I hereby certify that the matter in controversy involves the same plaintiffs, an overlapping patent, and defendants seeking approval to sell a generic version of the same drug product that is at issue in the matter captioned *Ultragenyx Pharmaceutical Inc. et al. v. Navinta LLC et al.*, No. 2:24-cv-09483 (ES) (MAH) (D.N.J.), currently pending in this Judicial District.

I further certify that, to the best of my knowledge, the matter in controversy is not the subject of any other action pending in any court or of any pending arbitration or administrative proceeding.

Dated: April 1, 2026

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EXHIBIT A

(12) **United States Patent**
Roe

(10) **Patent No.:** **US 8,697,748 B2**
(45) **Date of Patent:** **Apr. 15, 2014**

(54) **GLYCOGEN OR POLYSACCHARIDE
STORAGE DISEASE TREATMENT METHOD**

(75) Inventor: **Charles R. Roe**, Rockwall, TX (US)

(73) Assignee: **Baylor Research Institute**, Dallas, TX
(US)

(*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 94 days.

(21) Appl. No.: **13/343,578**

(22) Filed: **Jan. 4, 2012**

(65) **Prior Publication Data**

US 2013/0005818 A1 Jan. 3, 2013

Related U.S. Application Data

(62) Division of application No. 11/172,693, filed on Jul. 1,
2005, now Pat. No. 8,106,093.

(60) Provisional application No. 60/585,502, filed on Jul. 2,
2004.

(51) **Int. Cl.**
A61K 31/23 (2006.01)
A61K 31/20 (2006.01)

(52) **U.S. Cl.**
USPC **514/552**; 514/558; 514/560

(58) **Field of Classification Search**
USPC 514/552, 558, 560
See application file for complete search history.

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(57) **ABSTRACT**

A method for treating glycogen storage disease by administering an effective amount of a composition that includes ketogenic odd carbon fatty acids that ameliorate the symptoms of these diseases.

25 Claims, 9 Drawing Sheets

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METABOLIC FATE OF HEPTANOATE IN HUMANS

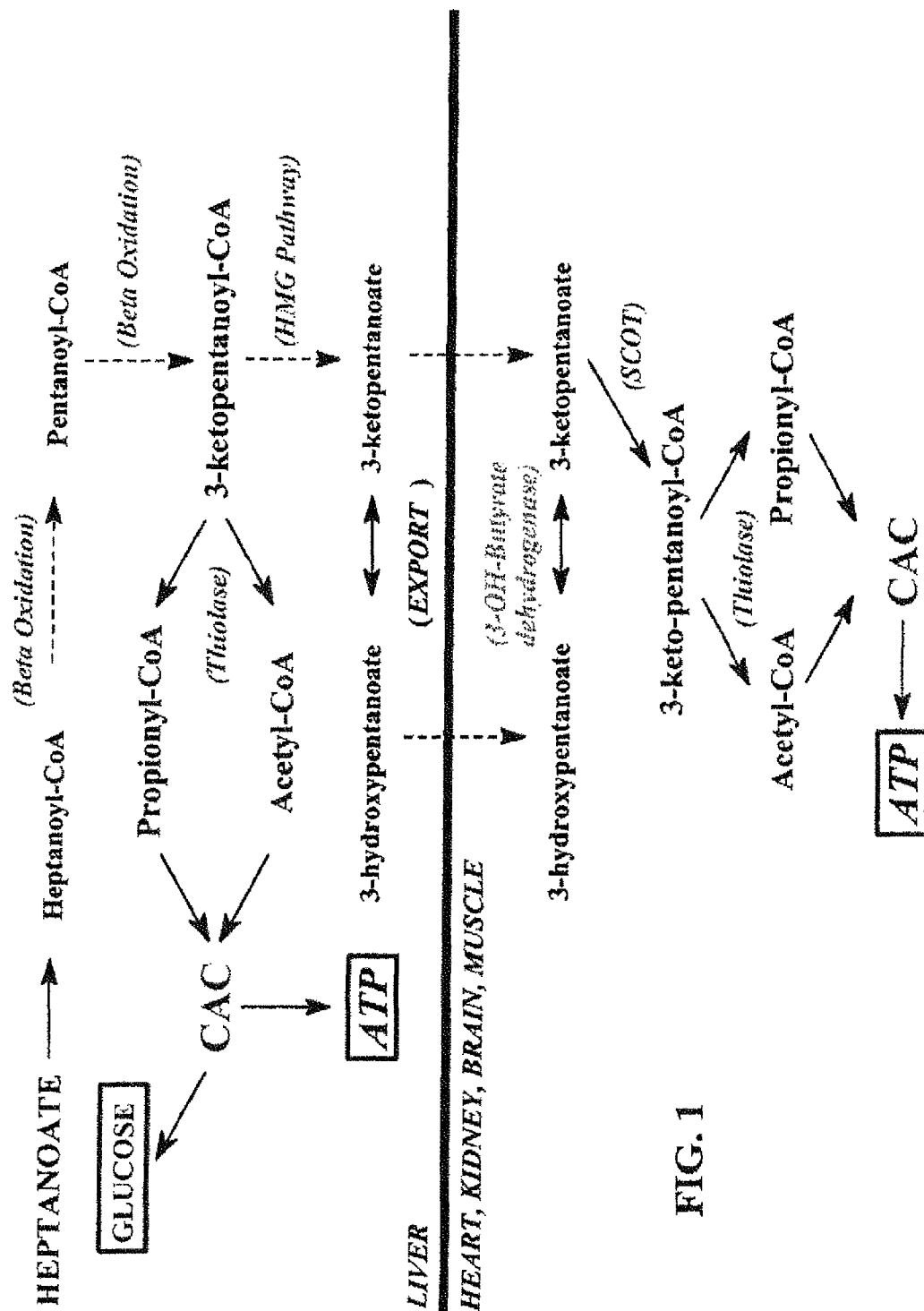


FIG. 1

ORGANELLAR INTEGRATION OF GLYCOGEN METABOLISM

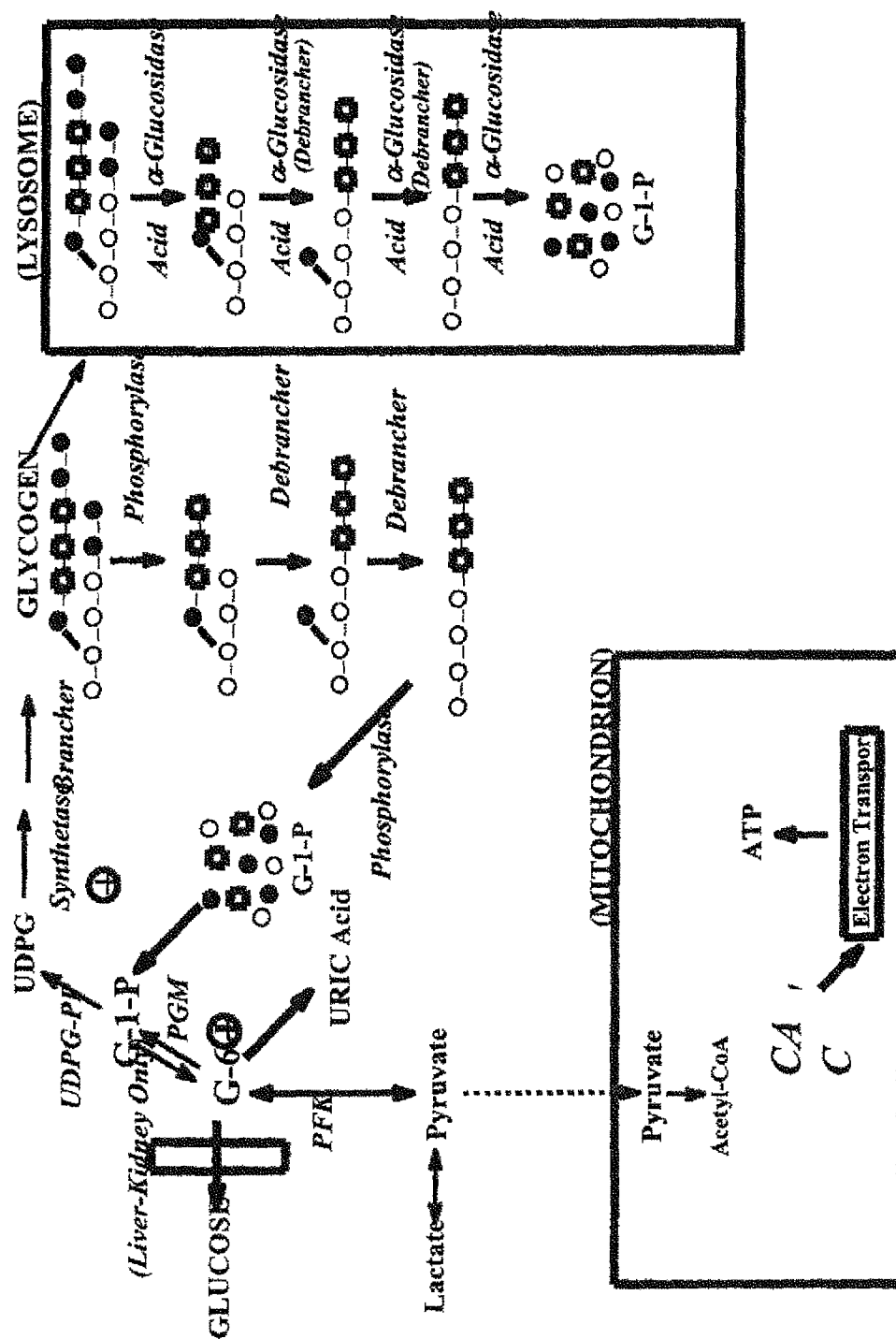


FIG. 2

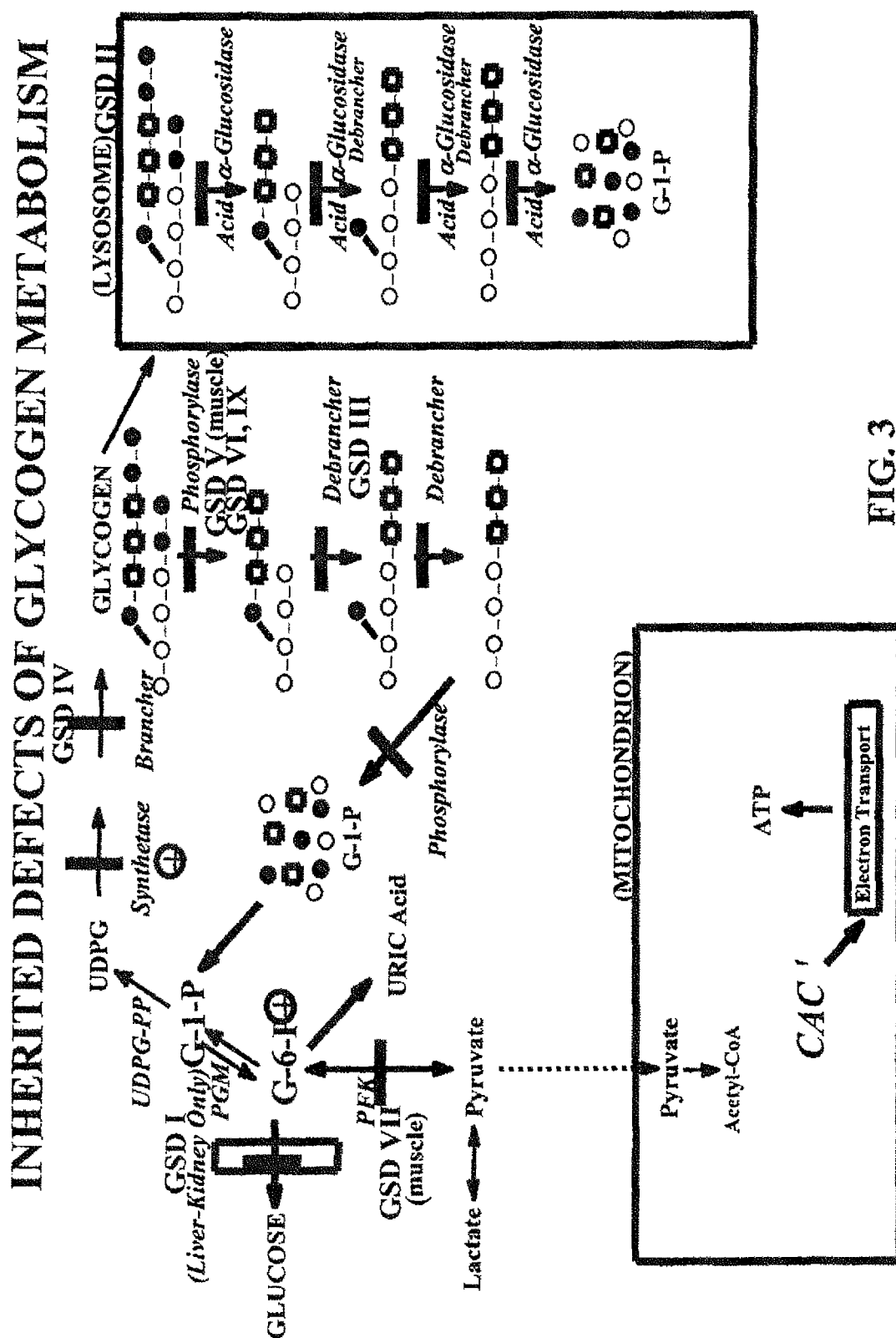


FIG. 3

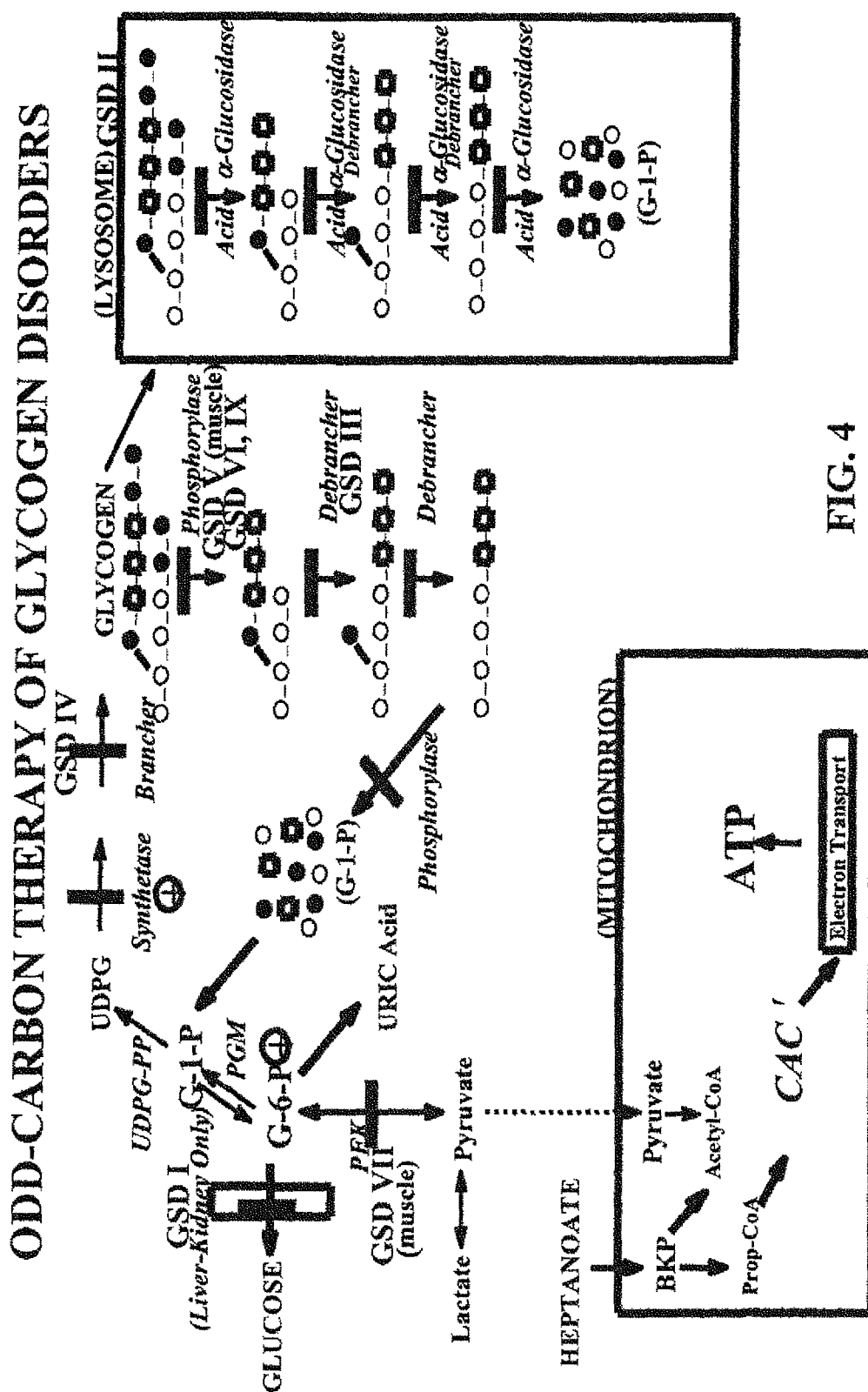


FIG. 4

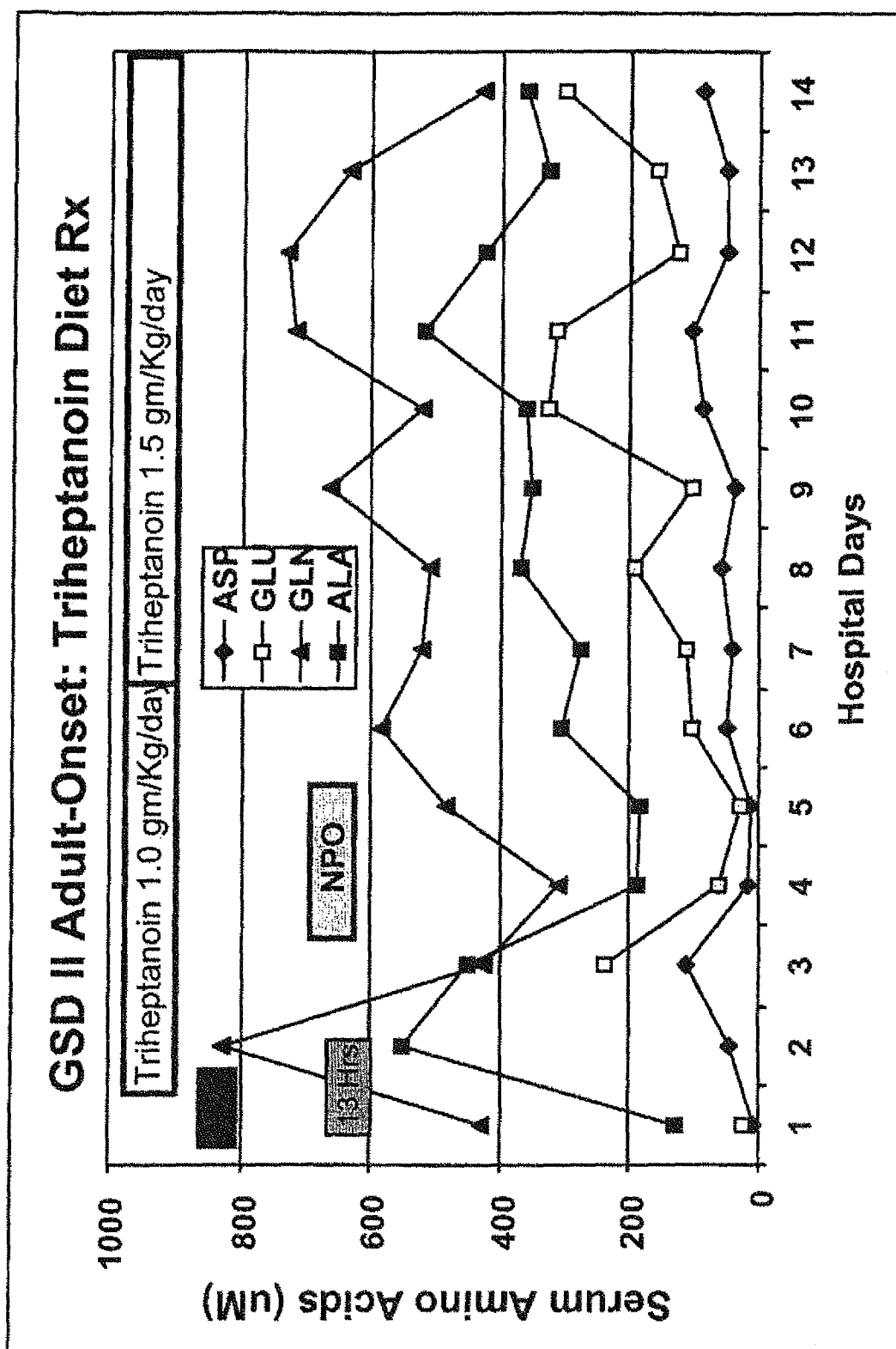


FIG. 5

OXIDATIVE SEQUENCE FOR HEPTANOATE

(Liver)

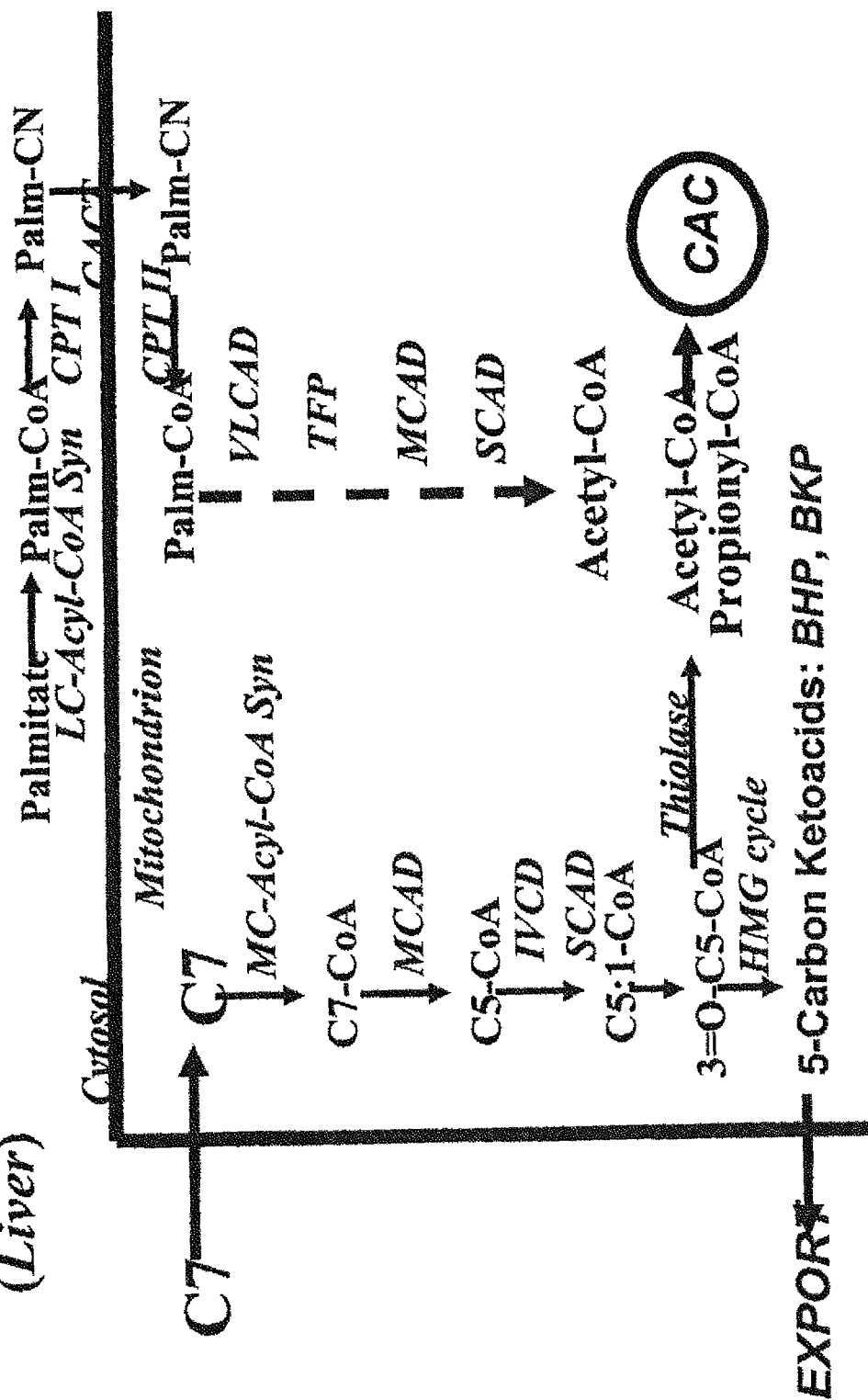
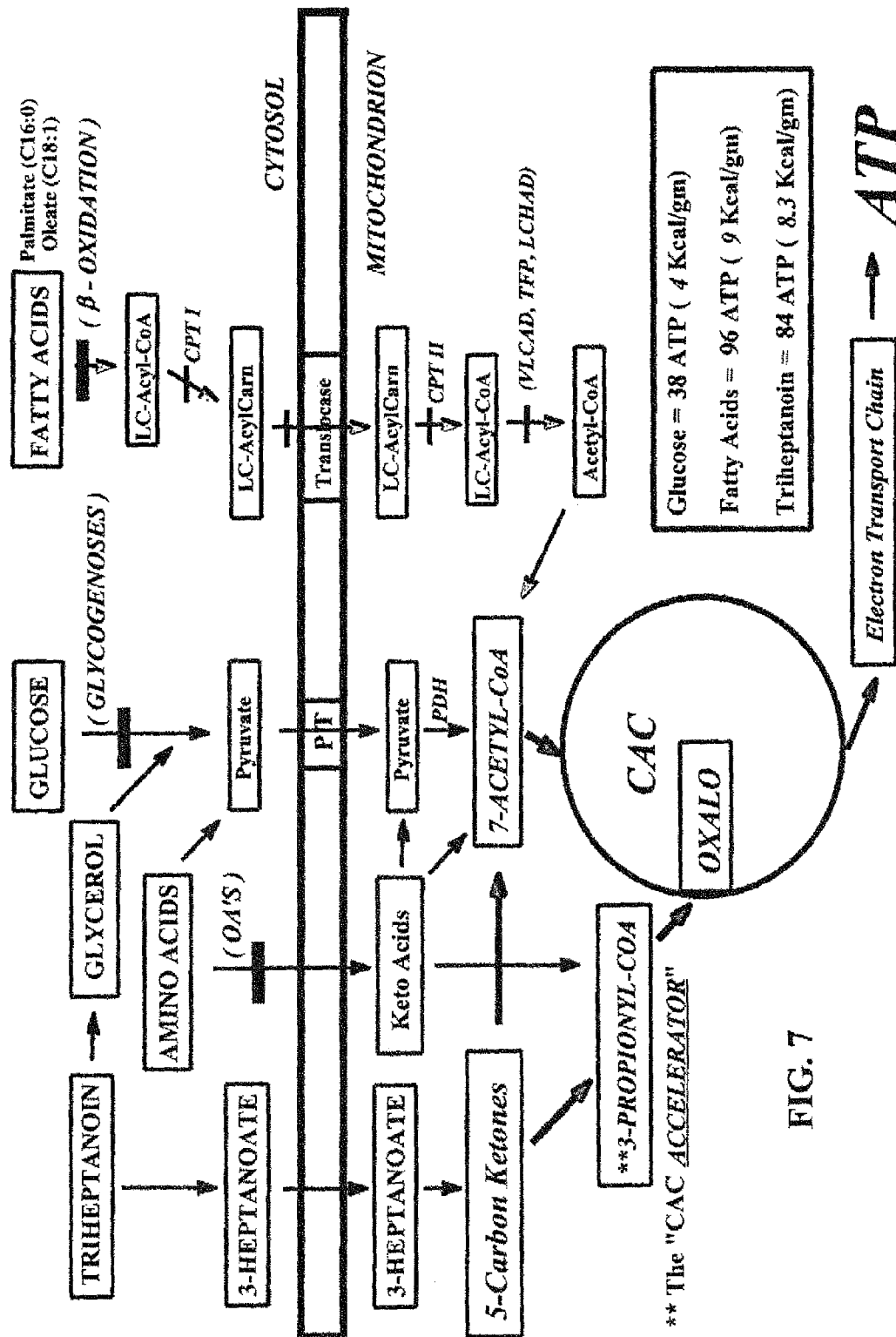


FIG. 6

Odd-Carbon Therapy & Inherited Metabolic Disease



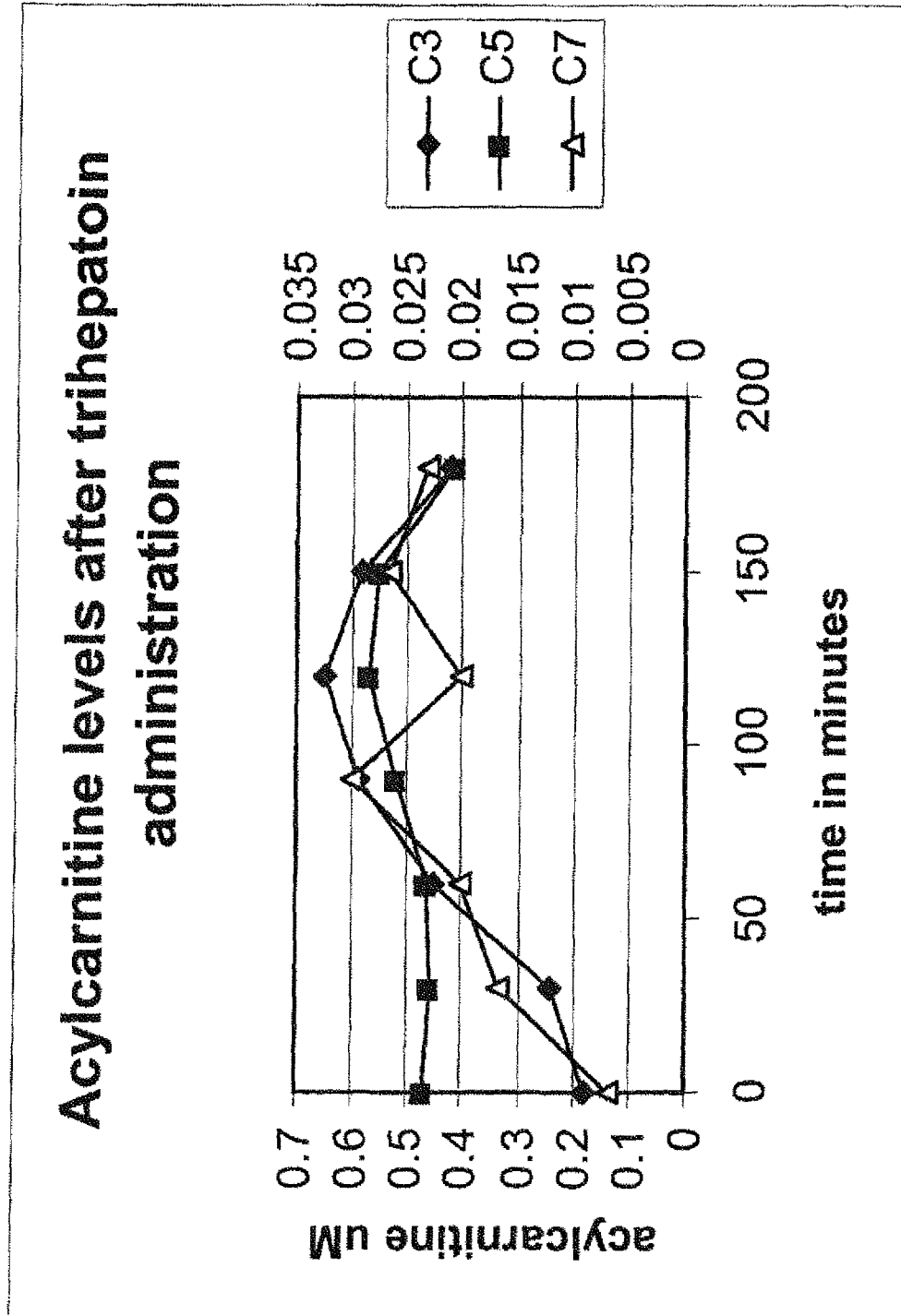


FIG. 8

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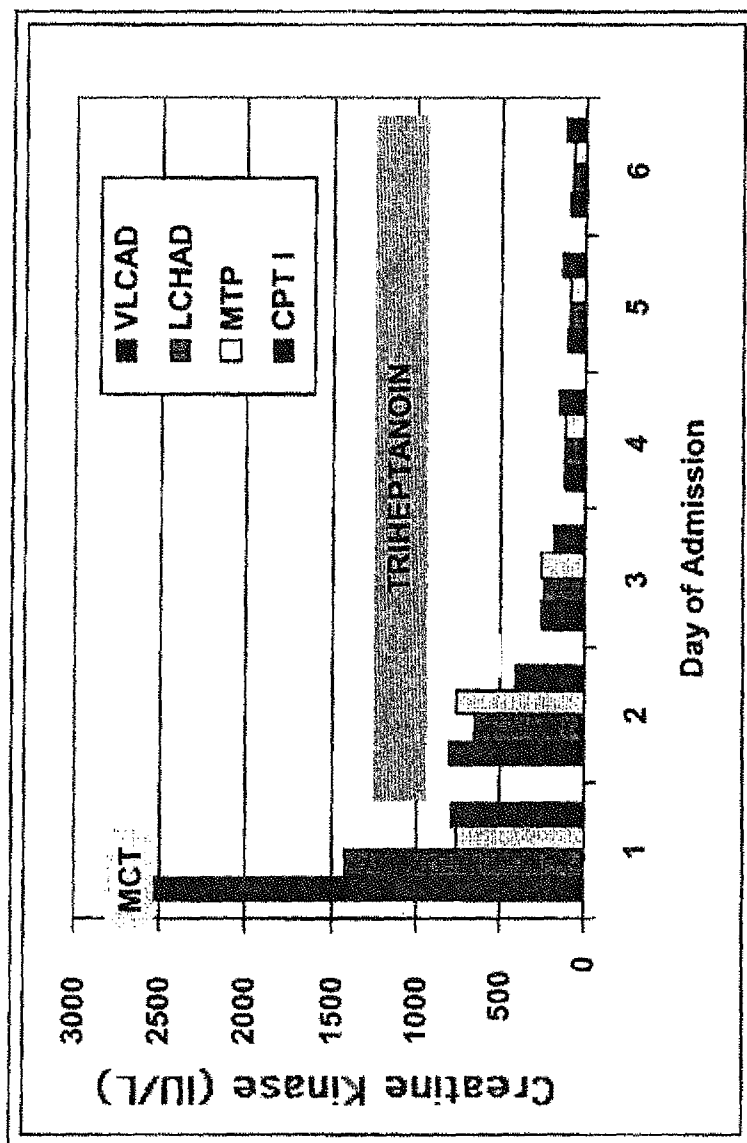


FIG. 9

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GLYCOGEN OR POLYSACCHARIDE STORAGE DISEASE TREATMENT METHOD

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a Divisional of U.S. patent application Ser. No. 11/172,693 filed Jul. 1, 2005 (now U.S. Pat. No. 8,106,093), which claims the benefit of U.S. Provisional Patent Application Ser. No. 60/585,502 filed Jul. 2, 2004.

TECHNICAL FIELD OF INVENTION

This invention relates to the treatment of glycogen or polysaccharide storage diseases affecting humans and other animals, and more particularly, to formulations that ameliorate the symptoms of these diseases.

BACKGROUND OF THE INVENTION

Without limiting the scope of the invention, its background is described in connection with polysaccharide storage diseases.

Glycogen-storage disease type II ("GSD II" or "Pompe's disease"), is a genetic disorder in humans resulting from the deficiency of acid alpha-glucosidase ("acid maltase"), a lysosomal hydrolase enzyme. The disease is characterized by the abnormal accumulation of glycogen in the lysosomes. (Hirschhorn, R and Reuser, A J J: Glycogen Storage Disease Type II: Acid α -Glucosidase Deficiency. In: The Metabolic and Molecular Bases of Inherited Diseases, 8th edition, Chapter 135, pp 3389-3420, McGraw-Hill, 2001). Polysaccharide storage diseases have also been reported in other animals, particularly in horses, cattle, and sheep.

With respect to humans, three major forms of GSD II have been described: infantile, juvenile, and adult-onset. See J. Ibrahim, et al., Glycogen Storage Disease Type II, <http://www.emedicine.com/PED/topic1866.htm> (2003). The infantile form of GSD II is described as presenting by 6 months of age and characterized by involvement of cardiac, skeletal and respiratory muscles with rapid progression to death by respiratory and cardiac failure. These inherited conditions are due to an enzyme-deficiency that occurs in the human population at about 1 in 14,000 to 1 in 60,000 live births. The infantile or neonatal form of the disease usually results in death by about twelve to eighteen months of age.

The juvenile (intermediate) form includes infants and children older than 6 months who present with weakness but generally have no cardiac disease. Adult-onset GSD II is slowly progressive and involves progressive muscle weakness that affects the accessory muscles of respiration and finally leads to respiratory failure and death. This form of the disease usually does not involve the heart. The adult-onset form of the disease may appear in the second or third decade of life, and as late as the sixth decade. The disease results in loss of weight and muscle mass. With the loss of muscle mass comes difficulty in breathing because the muscles have difficulty powering the breathing mechanism. The deficient enzyme, acid α -glucosidase, is not required for the vast majority of cellular glycogen because the main pathway for glycogen degradation is not deficient in GSD II disease, energy production is not impaired, and hypoglycemia does not occur. The main issue of deficiency of acid alpha-glucosidase enzymatic activity is the accumulation of structurally normal glycogen in lysosomes and cytoplasm in affected individuals. Excessive glycogen storage within lysosomes

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interrupts the normal functioning of other organelles and leads to cellular injury and dysfunction of the entire organ involved.

Presently, there is neither a cure for the disease, nor alteration of the clinical course of expected fatality, although some relief has been realized temporarily with high protein diets and with treatment for cardiac and respiratory failure. Alternatively, enzyme replacement therapy with recombinant human acid alpha-glucosidase, (rhGAA), an investigational enzyme replacement therapy for Pompe disease is now in clinical trials (www.genzyme.com and www.pompe.com).

Glycogen and other polysaccharide storage diseases also affect animals other than humans. For example, muscle disease in draft horses has been known for over 100 years, and in 1992 was correlated with polysaccharide storage diseases. Such disorders are commonly referred to as "Monday Morning Disease", "tying" or "locking up" or simply severe muscle disease have been reported. Draft horses which worked hard six days a week but were given a high grain diet and a day of rest on Sunday were found to be prone to massive muscle injury on Monday. Recently, the polysaccharide storage diseases have been found to be related to problems with giving a draft horse general anesthesia. In addition, glycogen branching enzyme deficiency has been reported as causing muscle weakness in Quarter Horses and related breeds. It is reportedly a separate disorder from polysaccharide storage myopathy and is a deficiency in the enzyme necessary for the formation of normal glycogen. S. Valberg, "Glycogen Storage Disorders of Quarter Horse Foals," J. Vet. Intern. Med. 15(6): 572-80 (2001).

A dietary solution has been suggested for the polysaccharide storage diseases using a diet with reduced starches and sugars and added fat as an alternative energy source. Providing fats as opposed to carbohydrates as an alternative source of energy to relieve tying up in horses has also been proposed. A subset of horses with chronic exertional rhabdomyolysis have an abnormal accumulation of glycogen (polysaccharide) stores in their muscles and one of the preventative measures suggested is feeding diets without grains and adding a fat supplement to maintain low blood glucose and insulin concentrations. L. Gray, "Polysaccharide Storage Myopathy (Glycogen Storage Disease)," Horse Previews, July 2002. These diets provide only partial relief, if at all, in these affected animals.

SUMMARY OF THE INVENTION

It has now been found that administering odd carbon fatty acids and/or ketones to humans and other animals can treat and ameliorate the effects of glycogen or polysaccharide storage diseases. More particularly, the present invention includes compositions, methods, feeds, diets, additives and the like for the treatment of polysaccharide storage diseases.

For example, the present invention includes a pharmaceutical composition for treating a glycogen storage disease that includes a pharmaceutically effective amount of an odd carbon fatty acid that is at least partially water-soluble for the treatment of the glycogen storage disease. The composition may also include a carrier, a diluent, e.g., a lipophilic diluent and/or an emulsifier. Non-limiting examples of useful emulsifiers include Imwitor 370, Imwitor 375, Imwitor 377, Imwitor 380 and Imwitor 829. In one embodiment, the odd carbon fatty acid is unneutralized and may be a C15, C7, C5 and mixtures or combinations thereof. For example, the odd carbon fatty acid may be 3-hydroxypentanoate, 3-ketopentanoate, tri-heptanoid, n-heptanoic acid and mixtures and combinations thereof. The compositions of the present inven-

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tion may be provided to a patient suspected of having a Glycogen Storage Disease, such as Pompe's disease. The patient may be suspected of having an anapleurotic disease, that is, a disease with symptoms that may be ameliorated with anapleurotic precursors that serve have therapeutic effects.

For example, the composition of the present invention may be an odd carbon fatty acid that is adapted for a dosage of between 0.1, 1, 1.5, 2, 3 or even 4 gr/kg/day. One example of an odd carbon fatty acid for use with patients may have an acid value of 0.1 or less mg KOH/gr, a hydroxyl value of 2.8 or less mg KOH/gr. The odd carbon fatty acid may have a purity of at least about 98 percent, e.g., triheptanoate that is at least about 97% pure.

The present invention also includes a method of treating a patient suffering from Pompe's disease by administering to a patient in need thereof a composition that with a human recombinant α -glucosidase in a pharmaceutically acceptable carrier, in an amount insufficient to treat the disease and concurrently providing the patient with a pharmaceutically effective amount of an odd carbon fatty acid to reduce the amount of the recombinant α -glucosidase necessary to treat the patient. The amount of human recombinant α -glucosidase may be from about 0.01 to about 100 milligrams per 100 kilograms of patient per month. The α -glucosidase and pharmaceutically effective amount of an odd carbon fatty acid may be provided in a single-dose, e.g., intravenously. The odd carbon fatty acid may dosed at between 1 to 2 gr/kg/day and/or be 30 to 40 percent of total daily Kcalories. Generally, a patient's total caloric intake is between about 1,200 to 3,000 calories per day.

Another embodiment of the present invention is a method of treating a disease in a mammal resulting from deficiencies of fatty acid oxidase disease by administering to the mammal a therapeutically effective amount of a pharmaceutical composition that includes a recombinant glycosylated enzymatically active α -glucosidase A or an enzymatically active fragment thereof and a pharmaceutically effective amount of an odd carbon fatty acid. By providing the patient with a supplemental source of anapleurotic carbon sources, the effective amount recombinant glycosylated enzymatically active α -glucosidase A or an enzymatically active fragment needed to treat a mammal is reduced by between about 20 and 80 percent. The mammal may be selected from humans, dogs, cats, horses, hamsters, rats, mice, sheep, goats and pigs. Depending on the course of treatment, specific disease, extent of the disease and other biochemical, metabolic and physiologic factors, the skilled artisan may be able to reduce the amount and total number of daily doses of pharmaceutical composition by half or less. The pharmaceutical composition may be administered orally, intravenously, intramuscularly, intranasally, intradermally, intraperitoneally, subcutaneously and/or as a suppository or by any other method known to those of skill in the pharmaceutical arts. The α -glucosidase A may further include a pharmacologically acceptable carrier, e.g., polyethylene glycol (PEG) and the amount of odd carbon fatty acid is sufficient to be detectable in serum as free heptanoate and/or metabolites thereof. By providing additional energetic support, the recombinant glycosylated enzymatically active α -glucosidase A or an enzymatically active fragment thereof is provided in a sub-optimal amount. The effective amount of the odd carbon fatty acid includes a nutritionally effective amount, that is, one that provides a measurable amount of change in a patient, e.g., decreased fatigue, increased energy, increased protein sparing and the like.

The present invention also includes a method of treating a glycogen storage disease by administering to a patient in need

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thereof a therapeutically effective amount of a pharmaceutical composition that includes an odd-chain fatty acid or ketone. The pharmaceutical composition is administered, e.g., at a daily oral administration dosage of between about 0.5 to about 2.0 grams, per day, per kilogram of patient weight. Useful dosage forms include those adapted for oral administration, e.g., powder, tablet, gelatin, gelcap, capsule, soft-gel, chewable or liquid form. For human patient's, therapeutically effective amount are between about 20 and 40% of the patient's daily caloric intake. The compositions may also be used for the treatment of muscle weakness by administering to a patient a therapeutically effective amount of a pharmaceutical composition with an odd-chain fatty acid or ketone, wherein the patient's muscle weakness is reduced and pulmonary function is increased. Muscle weakness may be the result of an underlying disease or may be caused by acute muscle usage, e.g., athletes, mountain climbers, spelunking, military operations, maneuvers, exercises and the like. The present invention will be particularly useful for chronic muscle weakness. The present invention may also be used in a treatment for urgency in urination and defecation by administering to a patient a therapeutically effective amount of a pharmaceutical composition with an effective amount of an odd-chain fatty acid or ketone that reduces the urgency to urinate and/or defecate.

Yet another method of the present invention includes a method for protein sparing in which a therapeutically effective amount of a pharmaceutical composition that includes an odd-chain fatty acid or ketone is administered to a patient in need of protein sparing. The composition may further include one or more vitamins, minerals, amino acids, lipids, nucleic acids, co-factors, pro-vitamins, and combinations of mixtures thereof, e.g., a nutritionally effective amount of branched amino acids.

The present invention may be formulated into immediate release product with an at least partially water-soluble ketogenic odd-chain fatty acid in an immediate release form that becomes immediately bioavailable in a subject. Bioavailability in a subject may be determined by serum levels of odd-chain fatty acids, organic salts, acylcarnitines and carnitine levels from, e.g., dried blood spots (dbs), saliva, urine, tears, sweat, plasma, amniotic or cerebro-spinal fluid and the like. The product may be adapted for parenteral, intravenous, oral, intramuscular, intraaortal, intrahepatic, intragastric, intranasal, intrapulmonary, intraperitoneal, subcutaneous, rectal, vaginal, intraosseal or dermal delivery. The product may be provided together or separately with one or more vitamins, minerals, amino acids, lipids, nucleic acids, co-factors, pro-vitamins, and combinations of mixtures thereof. Examples of odd chain fatty acids may be produced by solution precipitation methods, anti-solvent precipitation, spray drying, spray freezing, evaporative precipitation into an aqueous solution, wet milling, mechanical milling, vacuum-drying, vacuum-heating or lyophilization. The odd chain fatty acids may be particles that are amorphous, crystalline or semi-crystalline. An example is produced by evaporative precipitation into an aqueous solution and the formulation has a drug-to-excipient ratio greater than about 3:1. The odd chain fatty acids may be a provided as a ketogenic odd-chain fatty acid-to-excipient ratio that is greater than about 1:10, 1:1, 5:1, 7:1 or 10:1. For example, the ketogenic odd-chain fatty acid-to-emulsifier ratio is about 1:10, 1:1, 4:1, 7:1, 10:1, 13:1, 15:1, 25:1 or 40:1, e.g., C7:Imwitor.

The present invention may also be used in a method of providing nutritional support for surgery by providing a patient in need of supplemental metabolic support a nutritionally effective amount of a ketogenic odd-chain fatty acid of

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between about 30-35% of total Kcal/day. The nutritional supplements may also be used in a method for ameliorating the effects of physical exertion, the method includes administration to a person in need of such amelioration a pharmaceutically or nutritionally effective amount of one or more of the odd chain fatty acids described herein. The odd chain fatty acids will also find particular uses as a post-operative nutritional support that includes a nutritionally effective amount of a compound that provides both 2-carbon and 3-carbon citric acid cycle metabolites. An example of a 2-carbon metabolite is acetyl-CoA and the 3-carbon metabolite is propionyl-CoA. The composition may also include a nutritionally effective amount of one or more essential fatty acids and/or one or more essential saccharides.

The odd chain fatty acids may even be provided in a modified release product having two portions, wherein a first portion includes a first quantity of a ketogenic odd-chain fatty acid in an immediate release form which becomes fully bioavailable in the subject's stomach and a second portion with a second compound in a sustained release form wherein the ratio of the first quantity to the second quantity provides a serum Cmax in a human subject equivalent to the serum Cmax obtained when the first of one or more doses of a standard immediate release formulation having one third the amount of the ketogenic odd-chain fatty acid is dosed every four hours over a 12 hour period and wherein the product also provides therapeutically effective bioavailability for at least twelve hours after a single dose in a human subject according to serum analysis. The second compound may even be the ketogenic odd-chain fatty acid and/or one or more vitamins, minerals, amino acids, lipids, nucleic acids, co-factors, provitamins, and combinations of mixtures thereof.

A diet is provided by the present invention that includes a ketogenic odd-chain fatty acid with between about 5 to 40% of total Kcal/day, essential fats of between about 5-20% total Kcal/day, carbohydrates restricted to about 20 to 40% total Kcal/day of the diet and a protein content of between about 15 to 30% total Kcal/day. The total Kcal/day of each of the fatty acid, fats, carbohydrates and/or protein can be varied depending on the particular need and application. For example, the essential fats provides between about 10-15% of the total Kcal/day, the daily carbohydrate intake provides about 30% of the total Kcal/day, the daily protein intake provides about 20% of the total Kcal/day or the ketogenic odd-chain fatty acid provides between about 1-4 gm/Kg/day. The skilled artisan will recognize the different variations and combinations available.

The ketogenic odd-chain fatty acid includes one or more odd-chain fatty acids selected from C5, C7, C15, and mixtures or combinations thereof. In addition, the ketogenic odd-chain fatty acid includes triheptanoin, n-heptanoic acid, a triglyceride, or a salt or derivative thereof, or combinations thereof.

The present invention provides a beverage including water, carbohydrates, electrolytes and a ketogenic odd-chain fatty acid in a concentration of between about 0.5% to about 5.0%. The beverage includes ketogenic odd-chain fatty acid that have one or more odd-chain fatty acids selected from C5, C7, C15, and mixtures or combinations thereof. The ketogenic odd-chain fatty acid has a concentration of about 1.0% and in some instances, the fatty acid is emulsified. The beverage includes ketogenic odd-chain fatty acid that provides between about 20-30% of the daily caloric intake. The ingredients of the beverage include the following:

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Ingredient	Approximate Concentration
Potassium	2 meq/l
Sodium	26 meq/l
Glucose	4%
Pyruvate	1%
ketogenic odd-chain fatty acid	1 to 20%
Emulsifier	0.1 to 2.0%
water	balance.

A beverage is also provided that provides a source of immediate energy that has a composition by weight:

Total Carbohydrate	0.4-3.5%
Ketogenic odd-chain fatty acid	0.1 to 20%
Emulsifier	0.1 to 2.0%
Sodium Chloride	0.16-0.33%
Potassium Chloride	0.03-0.13%
Free Citric Acid	0.026-0.26%
Water	balance

In addition to beverages, the present invention includes a dry beverage concentrate for enhanced, immediate energy requirements in a mammalian body that includes a composition by weight in water: an emulsified, ketogenic odd-chain fatty acid 0.1-25%, total carbohydrate of 0.4-3.5%, sodium chloride 0.16-0.22%, potassium chloride 0.03-0.13%, and free citric acid 0.026-0.26%, the concentrate suitable for mixture with water to yield a beverage. The skilled artisan will recognize that the weight of the component may be varied depending on the particular application.

A food is also provided that includes a mixture of ingredients selected to make one or more snacks, soups, salads, cakes, cookies, crackers, breads, ice creams, yogurts, puddings, custards, baby foods, medicinal foods, sports bars, breakfast cereals and beverages and a ketogenic odd-chain fatty acid in a concentration of between about 0.5% to about 5.0% of the composition. The food composition includes grains, fruits, nuts and supplemental dietary fiber in the form of compressed flakes, supplemental dietary fiber and combinations thereof. The present invention provides a food composition that includes compressed flakes of supplemental dietary fiber in the form of apple fiber, corn bran, soy fiber, pectin, guar gum, gum ghatti, and gum arabic, as well as mixtures thereof. Some embodiments of the present invention include one or more binder materials that include rice flour, wheat flour, oat flour, corn flour, rye flour and potato flour, as well as mixtures thereof.

The present invention also includes a pre-cooked edible and chewable product selected from the group consisting of breakfast cereals, snacks, soups, salads, cakes, cookies, crackers, puddings, ice creams, yogurts, puddings, custards, baby foods, medicinal foods, sports bars, and beverages. Additionally, the composition can be extrusion cooked. The present invention includes an ingredient for fortification of a food product that has an emulsified ketogenic odd-chain fatty acid in a concentration of between about 0.5% to about 5.0% of the food supplement.

An orally administratable partially dry unit dosage is provided that includes at least about 10 to 1,000 mg of a ketogenic odd-chain fatty acid. When in the form of a nutritional supplement includes a nutritionally effective amount of a ketogenic odd-chain fatty acid that are metabolized into both 2-carbon and 3-carbon citric acid cycle metabolites, and the

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ketogenic odd-chain fatty acid is selected from the group consisting of a C5, C7, C15, and mixtures or combinations thereof.

The present invention also includes a method for increasing the immediately available energy supply to a cell by exposing the cell to an effective dosage of a ketogenic odd-chain fatty acid that represents 20-35% of the cellular requirement of kCals per day.

A method for increasing the immediately available energy supply to a mammal by providing the mammal with an effective dosage of a ketogenic odd-chain fatty acid that comprises 30-35% of the mammal's kCal per day is also included. The ketogenic odd-chain fatty acid includes one or more odd-chain fatty acids selected from C5, C7, C15, and mixtures or combinations thereof.

The present invention also includes an animal feed that has an odd chain fatty acid present in an amount effective to provide nutritive fat to an animal and a solid nutritive source. The animal feed is adapted for use by one or more of poultry, livestock, farm-raised fish, crabs, shrimp and fresh-water turtles. The solid nutritive source includes a source selected from the group consisting of soy, oats, sorghum, whole wheat, a nutritive wheat fraction, whole rice, a nutritive rice fraction, whole corn a nutritive corn fraction and whole barley, a nutritive barley fraction, vitamins and nutritive minerals. When used as an animal feed the odd chain fatty acid is present in an amount of at least about 5% to 40% by weight of the feed.

BRIEF DESCRIPTION OF THE DRAWINGS

For a more complete understanding of the features and advantages of the present invention, reference is now made to the detailed description of the invention along with the accompanying figures and in which:

FIG. 1 is a schematic drawing depicting the metabolic fate of the odd-carbon fatty acid, heptanoate (C7), derived from the triglyceride, Triheptanoin;

FIG. 2 is a schematic drawing depicting the organellar integration of glycogen metabolism in cells;

FIG. 3 is a schematic drawing depicting the possible enzyme deficiencies in inherited defects involving glycogen metabolism in humans. (GSD I-GSD VII);

FIG. 4 is a schematic drawing depicting the effect of odd-carbon therapy on glycogen metabolic disorders, including GSD II, in humans;

FIG. 5 depicts the effect of a triheptanoin-containing diet on the levels of selected amino acids in the blood of a patient with the adult onset form of glycogen storage disease II (acid maltase deficiency) with a diet containing Triheptanoin at a dose of 1.0 to 1.5 gm/kg/day);

FIG. 6 is a diagram that shows the oxidative sequence for heptanoate (liver);

FIG. 7 is a diagram that shows the entry points in metabolism, e.g., in the CAC of the odd chain fatty acids of the present invention; and

FIG. 8 is a graph that compared the acylcarnitine levels after triheptanoin treatment in horses; and

FIG. 9 is a graph that shows the effects of a resolution of rhabdomyolysis with dietary triheptanoin in various types of FODs.

DETAILED DESCRIPTION OF THE INVENTION

While the making and using of various embodiments of the present invention are discussed in detail below, it should be appreciated that the present invention provides many applicable inventive concepts that can be embodied in a wide

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variety of specific contexts. The specific embodiments discussed herein are merely illustrative of specific ways to make and use the invention and do not delimit the scope of the invention.

To facilitate the understanding of this invention, a number of terms are defined below. Terms defined herein have meanings as commonly understood by a person of ordinary skill in the areas relevant to the present invention. Terms such as "a", "an" and "the" are not intended to refer to only a singular entity, but include the general class of which a specific example may be used for illustration. As such, unless otherwise indicated, the terms "a" and "an" are taken to mean "one", "at least one" or "one or more". The terminology herein is used to describe specific embodiments of the invention, but their usage does not delimit the invention, except as outlined in the claims.

As used herein, the terms "subject" or "patient" are intended to include living organisms that may have one or more conditions generally referred to as polysaccharide storage diseases. Examples of subjects include humans, monkeys, horses, cows, sheep, goats, dogs, cats, mice, rats, and transgenic species thereof. Other examples of subjects include experimental animals such as mice, rats, dogs, cats, goats, sheep, pigs, and cows. A subject can be a human suffering from, or suspected of having, a polysaccharide storage disease, e.g., a glycogen storage disease such as Pompe's disease (GSD II).

As used herein, the phrases "therapeutically effective dosage" or "therapeutically effective amount" is an amount of a compound or mixtures of compounds, such as the odd-chain fatty acids and precursors or derivatives thereof, that reduce the amount of one or more symptoms of the condition in the infected subject by at least about 20%, at least about 40%, even more at least about 60%, 80% or even 100% relative to untreated subjects. Active compounds are administered at a therapeutically effective dosage sufficient to treat a condition associated with a condition in a subject. For example, the efficacy of a compound can be evaluated in patients or animal model systems that may be predictive of efficacy in treating the disease in humans or animals.

As used herein the term, "essential fatty acids" is used to describe fats and oils in foods are made up of basic units called fatty acids. In the body, these typically travel in three's as fatty acid chains attached to glycerol, forming a triglyceride. Based on their chemical structure, fatty acids are classified into 3 major categories: monounsaturated, polyunsaturated, or saturated fats. The oils and fats that people and animals eat are nearly always mixtures of these 3 types of fatty acids, with one type predominating. Two specific types of polyunsaturated fatty acids, linoleic and alpha-linolenic, are called essential fatty acids. They must be present in the diet in adequate amounts because they are considered necessary for proper nutrition and health. Linoleic acid (LA) is an omega-6 fatty acid and is found in many oils, e.g., corn, safflower, soybean and sunflower, whole grains and walnuts. Alpha-linolenic acid (ALA) is a plant precursor of docosahexanoic acid (DHA). Sources of ALA include seaweeds and green leaves of plants (in very small amounts), soybeans, walnuts, butternuts, some seeds (flax, chia, hemp, canola) and the oils extracted from these foods.

As used herein, the term "nutritionally effective amount" is used to mean the amount of odd chain fatty acids that will provide a beneficial nutritional effect or response in a mammal. For example, as with a nutritional response to vitamin- and mineral-containing dietary supplements varies from mammal to mammal, it should be understood that nutritionally effective amounts of the odd chain fatty acids will vary.

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Thus, while one mammal may require a particular profile of vitamins and minerals present in defined amounts, another mammal may require the same particular profile of vitamins and minerals present in different defined amounts. Such is the case with the nutritionally effective amounts of the odd chain fatty acids of the present invention, in which the supplementation may be used to add C3 and C2 carbon chains into the liver and/or the heart, muscle, brain and kidney.

When provided as a dietary supplement or additive, the odd chain fatty acids of the invention has been prepared and administered to mammals in powdered, reconstitutable powder, liquid-solid suspension, liquid, capsule, tablet, caplet, lotion and cream dosage forms. The skilled artisan in the science of formulations can use the odd chain fatty acids disclosed herein as a dietary supplement that may be formulated appropriately for, e.g., irrigation, ophthalmic, otic, rectal, sublingual, transdermal, buccal, vaginal, or dermal administration. Thus, other dosage forms such as chewable candy bar, concentrate, drops, elixir, emulsion, film, gel, granule, chewing gum, jelly, oil, paste, pastille, pellet, shampoo, rinse, soap, sponge, suppository, swab, syrup, chewable gelatin form, chewable tablet and the like, can be used.

Due to varying diets among people, the dietary odd chain fatty acids of the invention may be administered in a wide range of dosages and formulated in a wide range of dosage unit strengths. It should be noted that the dosage of the dietary supplement can also vary according to a particular ailment or disorder that a mammal is suffering from when taking the supplement. For example, a person suffering from chronic fatigue syndrome or fibromyalgia will generally require a dose different than an athlete that is wanting to attain a nutritional benefit. An appropriate dose of the dietary supplement can be readily determined by monitoring patient response, i.e., general health, to particular doses of the supplement. The appropriate doses of the supplement and each of the agents can be readily determined in a like fashion by monitoring patient response, i.e., general health to particular doses of each.

The odd chain fatty acids may be administered simultaneously or sequentially in one or a combination of dosage forms. While it is possible and even likely that the present dietary supplement will provide an immediate overall health benefit, such benefit may take days, weeks or months to materialize. Nonetheless, the present dietary odd chain fatty acid supplement will provide a beneficial nutritional response in a mammal consuming it.

The odd-chain fatty acids of the present invention may be administered, e.g., orally or by subcutaneous, intravenous, intraperitoneal, etc., administration (e.g. by injection). Depending on the route of administration, the active compound may be neutralized, made miscible, at least partially or fully water-soluble or even coated in a material to protect the odd-chain fatty acids from the action of bases, acids, enzymes or other natural conditions that may interfere with their effectiveness, uptake or metabolic use.

To administer the therapeutic compound by other than parenteral administration, it may be necessary to coat the compound with, or co-administer the compound with, a material to prevent its inactivation. For example, the therapeutic compound may be administered to a subject in an appropriate carrier, for example, emulsifiers, liposomes, or a diluent. Pharmaceutically acceptable diluents include saline and aqueous buffer solutions. The therapeutic odd-chain fatty acids may be dispersed in glycerol, liquid polyethylene glycols, and mixtures thereof and in oils. Under ordinary conditions of storage and use, these preparations may contain a preservative to prevent the growth of microorganisms.

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Pharmaceutical compositions that include the odd-chain fatty acids of the present invention suitable for injectable use may include sterile aqueous solutions, dispersions and sterile powders for the extemporaneous preparation of sterile injectable solutions or dispersion. In all cases, the composition must be sterile and must be fluid to the extent that easy syringability exists. It must be stable under the conditions of manufacture and storage and must be preserved against the contaminating action of microorganisms such as bacteria and fungi.

The odd-chain fatty acids may be provided with a carrier in a solvent or dispersion medium containing, for example, water, ethanol, polyol (for example, glycerol, propylene glycol, and liquid polyethylene glycol, and the like), suitable mixtures thereof, and vegetable oils. The proper fluidity can be maintained, for example, by the use of a coating such as lecithin, by the maintenance of the required particle size in the case of dispersion and by the use of surfactants. Prevention of the action of microorganisms can be achieved by various antibacterial and antifungal agents, for example, parabens, chlorobutanol, phenol, ascorbic acid, thimerosal, and the like. In many cases, it will be preferable to include isotonic agents, for example, sugars, sodium chloride, or polyalcohols such as mannitol and sorbitol, in the composition. Prolonged absorption of the injectable compositions can be brought about by including in the composition an agent which delays absorption, for example, aluminum monostearate or gelatin.

The odd-chain fatty acids may be provided in one or more controlled sizes and characteristics with one or more water-soluble polymers depending on the size and structural requirements of the patient, e.g., the particles may be small enough to traverse blood vessels when provided intravenously. Either synthetic or naturally occurring polymers may be used, and while not limited to this group, some types of polymers that might be used are polysaccharides (e.g. dextran, ficoll), proteins (e.g. poly-lysine), poly(ethylene glycol), or poly(methacrylates). Different polymers, because of their different size and shape, will produce different diffusion characteristics for the odd-chain fatty acids in the target tissue or organ.

Sterile injectable solutions can be prepared by incorporating the therapeutic compound in the required amount in an appropriate solvent with one or a combination of ingredients enumerated above, as required, followed by filtered sterilization. Generally, dispersions are prepared by incorporating the therapeutic compound into a sterile carrier which contains a basic dispersion medium and the required other ingredients from those enumerated above. In the case of sterile powders for the preparation of sterile injectable solutions, methods of preparation include: vacuum drying, spray freezing, freeze-drying and the like, which yield a powder of the active ingredient (i.e., the therapeutic compound) plus any additional desired ingredient from a previously sterile-filtered solution thereof.

The odd-chain fatty acids can be orally administered, for example, with an inert diluent or an assimilable edible carrier. The therapeutic compound and other ingredients may also be enclosed in a hard or soft shell gelatin capsule, compressed into tablets, or incorporated directly into the subject's diet. The odd-chain fatty acids may be incorporated with excipients and used in the form of ingestible tablets, buccal tablets, troches, capsules, elixirs, suspensions, syrups, wafers, and the like. The amount of odd-chain fatty acids in the compositions and preparations may, of course, be varied depending on, e.g., the age, weight, gender, condition, disease and course of treatment of the individual patient. Pediatric doses are likely to differ from adult doses as will be known to

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the skilled artisan. The amount of the therapeutic compound in such therapeutically useful compositions is such that a suitable dosage will be obtained.

A dosage unit for use with the odd chain fatty acids disclosed herein may be a single compound or mixtures thereof with other compounds, e.g., amino acids, nucleic acids, vitamins, minerals, pro-vitamins and the like. The compounds may be mixed together, form ionic or even covalent bonds. For pharmaceutical purposes the odd chain fatty acids (e.g., C5, C7 and C15) of the present invention may be administered in oral, intravenous (bolus or infusion), intraperitoneal, subcutaneous, or intramuscular form, all using dosage forms well known to those of ordinary skill in the pharmaceutical arts. Depending on the particular location or method of delivery, different dosage forms, e.g., tablets, capsules, pills, powders, granules, elixirs, tinctures, suspensions, syrups, and emulsions may be used to provide the odd chain fatty acids of the present invention to a patient in need of therapy that includes a number of conditions, e.g., polysaccharide storage diseases, fatigue, low energy, wasting and the like. The odd chain fatty acids may also be administered as any one of known salt forms.

The total daily amount of odd chain fatty acids will vary depending on the condition and needs of a patient. For example, the odd chain fatty acids may be provided as a supplemental source of immediate, short-term, mid-term or long-term energy and may be provided in formulations that are immediately available, slow release or extended release. The dosage amount may be measured in grams per day, as a percentage of kCalories consumed in a day, as a percentage of the total daily caloric intake, as part of a fixed, a modified or a diet that changes over time. For example, a patient may need immediate intervention that "spikes" the amount of odd chain fatty acids to approach or reach ketosis. These "ketogenic" odd chain fatty acids will then be varied to not have other side effects, e.g., start with 40% of total caloric intake per day and then reduced over time as the patient's condition, symptoms, clinical course and/or metabolic conditions improves. The range of percentage caloric intake may vary from between about 0.01, 0.1, 1, 2, 5, 10, 15, 20, 22, 25, 30, 35, 40 or even higher percent, which may include one or more of the odd chain fatty acids (e.g., C5, C7 or C15 (available from, e.g., Sassol, Germany)). One way to measure the effect and/or dosing of the odd chain fatty acids is to measure the amount that is detectable in body solids or fluids, e.g., biopsies and blood, respectively. A wide variety of odd chain fatty acids metabolites may be detected from multiple sources, e.g., urine, tears, feces, blood, sweat, breath and the like.

For example, when using C7 as the source of odd chain fatty acids these can be provided in the form of a triglyceride, e.g., tri-heptanoin. The triglyceride triheptanoin is provided in a concentration sufficient to provide a beneficial effect is most useful in this aspect of the present invention. The seven-carbon fatty acid may be provided, e.g.:

Infants	1-4 g/kg	35% kcalories
Children	3-4 g/kg	33-37% kcalories
Adolescent	1-2 g/kg	35% kcalories
Adults	0.1-2 g/kg	35% kcalories

Goals have been set using 4 g/kg (within ideal body weight (IBW) range) for infants, children, and some adolescents. Goals have been set using 2 g/kg (within IBW range) for adolescents. Goals have been set using 2 g/kg (within IBW

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range) for adults; but toleration is 1-1.2 g per kg (which is 35% kcal of estimated needs).

The odd chain fatty acids are typically administered in admixture with suitable pharmaceutical salts, buffers, diluents, extenders, excipients and/or carriers (collectively referred to herein as a pharmaceutically acceptable carrier or carrier materials) selected based on the intended form of administration and as consistent with conventional pharmaceutical practices. Depending on the best location for administration, the odd chain fatty acids may be formulated to provide, e.g., maximum and/or consistent dosing for the particular form for oral, rectal, topical, intravenous injection or parenteral administration. While the odd chain fatty acids may be administered alone or pure, they may also be provided as stable salt form mixed with a pharmaceutically acceptable carrier. The carrier may be solid or liquid, depending on the type and/or location of administration selected.

Techniques and compositions for making useful dosage forms using the present invention are described in one or more of the following references: Ansel, *Introduction to Pharmaceutical Dosage Forms* 2nd Edition (1976); Remington's *Pharmaceutical Sciences*, 17th ed. (Mack Publishing Company, Easton, Pa., 1985); *Advances in Pharmaceutical Sciences* (David Ganderton, Trevor Jones, Eds., 1992); *Advances in Pharmaceutical Sciences Vol 7*. (David Ganderton, Trevor Jones, James McGinity, Eds., 1995); *Aqueous Polymeric Coatings for Pharmaceutical Dosage Forms* (Drugs and the Pharmaceutical Sciences, Series 36 (James McGinity, Ed., 1989); *Pharmaceutical Particulate Carriers: Therapeutic Applications: Drugs and the Pharmaceutical Sciences*, Vol 61 (Alain Rolland, Ed., 1993); *Drug Delivery to the Gastrointestinal Tract* (Ellis Horwood Books in the Biological Sciences. Series in Pharmaceutical Technology; J. G. Hardy, S. S. Davis, Clive G. Wilson, Eds.); *Modern Pharmaceutics Drugs and the Pharmaceutical Sciences*, Vol 40 (Gilbert S. Banker, Christopher T. Rhodes, Eds.), and the like, relevant portions of each incorporated herein by reference.

Odd chain fatty acids may be administered in the form of an emulsion and/or liposome, e.g., small unilamellar vesicles, large unilamellar vesicles and multilamellar vesicles, whether charged or uncharged. Liposomes may include one or more: phospholipids (e.g., cholesterol), stearylamine and/or phosphatidylcholines, mixtures thereof, and the like. Examples of emulsifiers for use with the present invention include: Imwitor 370, Imwitor 375, Imwitor 377, Imwitor 380 and Imwitor 829.

The odd chain fatty acid vesicles may also be coupled to one or more soluble, biodegradable, bioacceptable polymers as drug carriers or as a prodrug. Such polymers may include: polyvinylpyrrolidone, pyran copolymer, polyhydroxypropylmethacrylamide-phenol, polyhydroxyethylaspartamidophenol, or polyethyleneoxide-polylysine substituted with palmitoyl residues, mixtures thereof, and the like. Furthermore, the vesicles may be coupled one or more biodegradable polymers to achieve controlled release of the odd chain fatty acids. Biodegradable polymers for use with the present invention include, e.g., polylactic acid, polyglycolic acid, copolymers of polylactic and polyglycolic acid, polyepsilon caprolactone, polyhydroxy butyric acid, polyorthoesters, polyacetals, polydihydropyrans, polycyanoacylates, and crosslinked or amphipathic block copolymers of hydrogels, mixtures thereof, and the like.

In one embodiment, gelatin capsules (gelcaps) may include the odd chain fatty acid in its native state. For oral administration in a liquid dosage form, the oral drug components may be combined with any oral, non-toxic, pharmaceutically acceptable inert carrier such as an emulsifier, a diluent

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or solvent (e.g., ethanol), glycerol, water, and the like. Examples of suitable liquid dosage forms include oily solutions or suspensions in water, pharmaceutically acceptable fats and oils, alcohols or other organic solvents, including esters, emulsions, syrups or elixirs, suspensions, solutions and/or suspensions reconstituted from non-effervescent granules and even effervescent preparations reconstituted from effervescent granules. Such liquid dosage forms may contain, for example, suitable solvents, preservatives, emulsifying agents, suspending agents, diluents, sweeteners, thickeners, and melting agents, mixtures thereof, and the like.

Liquid dosage forms for oral administration may also include coloring and flavoring agents that increase patient acceptance and therefore compliance with a dosing regimen. In general, water, a suitable oil, saline, aqueous dextrose (e.g., glucose, lactose and related sugar solutions) and glycols (e.g., propylene glycol or polyethylene glycols) may be used as suitable carriers for parenteral solutions. Solutions for parenteral administration include generally, a water soluble salt of the active ingredient, suitable stabilizing agents, and if necessary, buffering salts. Antioxidizing agents such as sodium bisulfite, sodium sulfite and/or ascorbic acid, either alone or in combination, are suitable stabilizing agents. Citric acid and its salts and sodium EDTA may also be included to increase stability. In addition, parenteral solutions may include pharmaceutically acceptable preservatives, e.g., benzalkonium chloride, methyl- or propyl-paraben, and/or chlorobutanol. Suitable pharmaceutical carriers are described in multiple editions of Remington's Pharmaceutical Sciences, Mack Publishing Company, a standard reference text in this field, relevant portions incorporated herein by reference.

For direct delivery to the nasal passages, sinuses, mouth, throat, esophagus, trachea, lungs and alveoli, the odd chain fatty acids may also be delivered as an intranasal form via use of a suitable intranasal vehicle. For dermal and transdermal delivery, the odd chain fatty acids may be delivered using lotions, creams, oils, elixirs, serums, transdermal skin patches and the like, as are well known to those of ordinary skill in that art. Parenteral and intravenous forms may also include pharmaceutically acceptable salts and/or minerals and other materials to make them compatible with the type of injection or delivery system chosen, e.g., a buffered, isotonic solution.

To the extent that the odd chain fatty acids may be made into a dry powder or form, they may be included in a tablet. Tablets will generally include, e.g., suitable binders, lubricants, disintegrating agents, coloring agents, flavoring agents, flow-inducing agents and/or melting agents. For example, oral administration may be in a dosage unit form of a tablet, gelcap, caplet or capsule, the active drug component being combined with a non-toxic, pharmaceutically acceptable, inert carrier such as lactose, gelatin, agar, starch, sucrose, glucose, methyl cellulose, magnesium stearate, dicalcium phosphate, calcium sulfate, mannitol, sorbitol, mixtures thereof, and the like. Suitable binders for use with the present invention include: starch, gelatin, natural sugars (e.g., glucose or beta-lactose), corn sweeteners, natural and synthetic gums (e.g., acacia, tragacanth or sodium alginate), carboxymethylcellulose, polyethylene glycol, waxes, and the like. Lubricants for use with the invention may include: sodium oleate, sodium stearate, magnesium stearate, sodium benzoate, sodium acetate, sodium chloride, mixtures thereof, and the like. Disintegrators may include: starch, methyl cellulose, agar, bentonite, xanthan gum, mixtures thereof, and the like.

Capsules. Capsules may be prepared by filling standard two-piece hard gelatin capsules each with 10 to 500 milli-

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grams of powdered active ingredient, 5 to 150 milligrams of lactose, 5 to 50 milligrams of cellulose and 6 milligrams magnesium stearate.

Soft Gelatin Capsules. The odd chain fatty acids may be dissolved in an oil, e.g., a digestible oil such as soybean oil, cottonseed oil or olive oil. Non-digestible oils may also be used to have better control over the total caloric intake provided by the oil. The active ingredient is prepared and injected by using a positive displacement pump into gelatin to form soft gelatin capsules containing, e.g., 100-500 milligrams of the active ingredient. The capsules are washed and dried.

Tablets. A large number of tablets are prepared by conventional procedures so that the dosage unit was 100-500 milligrams of active ingredient, 0.2 milligrams of colloidal silicon dioxide, 5 milligrams of magnesium stearate, 50-275 milligrams of microcrystalline cellulose, 11 milligrams of starch and 98.8 milligrams of lactose. Appropriate coatings may be applied to increase palatability or delay absorption.

To provide an effervescent tablet, appropriate amounts of, e.g., monosodium citrate and sodium bicarbonate, are blended together and then roller compacted, in the absence of water, to form flakes that are then crushed to give granulates. The granulates are then combined with the active ingredient, drug and/or salt thereof, conventional beading or filling agents and, optionally, sweeteners, flavors and lubricants.

Injectable solution. A parenteral composition suitable for administration by injection is prepared by stirring sufficient active ingredient in deionized water and mixed with, e.g., up to 10% by volume propylene glycol, salts and/or water to deliver a composition, whether in concentrated or ready-to-use form. Given the nature of the odd chain fatty acids (alone, partially or fully-soluble in water) the amount and final concentration of the odd chain fatty acids may be varied such that the liquid may be provided intravenously using syringes and/or standard intravenous liquids or fluids. The solution will generally be made isotonic with sodium chloride and sterilized using, e.g., ultrafiltration.

Suspension. An aqueous suspension is prepared for oral administration so that each 5 ml contain 100 mg of finely divided active ingredient, 200 mg of sodium carboxymethyl cellulose, 5 mg of sodium benzoate, 1.0 g of sorbitol solution, U.S.P., and 0.025 ml of vanillin.

Mini-tablets. For mini-tablets, the active ingredient is compressed into a hardness in the range 6 to 12 Kp. The hardness of the final tablets is influenced by the linear roller compaction strength used in preparing the granulates, which are influenced by the particle size of, e.g., the monosodium hydrogen carbonate and sodium hydrogen carbonate. For smaller particle sizes, a linear roller compaction strength of about 15 to 20 KN/cm may be used.

Kits. The present invention also includes pharmaceutical kits useful, for example, for providing an immediate source of alternative cellular energy, e.g., before, during or after surgery. The dosage will generally be prepared sterile and ready-to-use, e.g., one or more containers that may be broken (e.g., sealed glass ampoules), pierced with a syringe for immediate administration or even a pressurized container. Such kits may further include, if desired, one or more of various conventional pharmaceutical kit components, such as, for example, containers with one or more pharmaceutically acceptable diluents, carriers, additional containers, etc., as will be readily apparent to those skilled in the art. Printed instructions, either as inserts or as labels, indicating quantities of the components to be administered, guidelines for administration, and/or guidelines for mixing the components, may also be included in the kit. It should be understood that although the specified materials and conditions are important in practicing the

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invention, unspecified materials and conditions are not excluded so long as they do not prevent the benefits of the invention from being realized.

Pharmaceutical Dosage Forms. The odd chain fatty acids of the present invention may be provided in liquid form or may also be provided in a capsule, gelcap or other encapsulated form. Generally, one composition of the present invention is prepared by adding, e.g., half of the Kaolin clay or other carrier into the blended followed by addition of a first active salt form, e.g., the salt form that is less soluble in the final liquid suspension, e.g., as an emulsion in water. This process is particularly suitable for very large mixtures, e.g., 500, 1,000, 3,000 or even 5,000 liters.

One particular method of delivery of the odd chain fatty acids of the present invention is in a tablet, capsule or gelcap that is coated for enteric delivery. Enteric coating relates to a mixture of pharmaceutically acceptable excipients that is applied to, combined with, mixed with or otherwise added to a carrier to deliver the medicinal content, in this case one or more odd chain fatty acids (e.g., C5, C7, C15, mixtures and combinations thereof) through the stomach unaltered for delivery into the intestines. The coating may be applied to a compressed or molded or extruded tablet, a gelatin capsule, and/or pellets, beads, granules or particles of the carrier or composition. The coating may be applied through an aqueous dispersion or after dissolving in appropriate solvent. Additional additives and their levels, and selection of a primary coating material or materials will depend on the following properties: resistance to dissolution and disintegration in the stomach; impermeability to gastric fluids and drug/carrier/enzyme while in the stomach; ability to dissolve or disintegrate rapidly at the target intestine site; physical and chemical stability during storage; non-toxicity; easy application as a coating (substrate friendly); and economical practicality. Methods for enteric coating are well known in the art.

Remington's Pharmaceutical Sciences, discloses that enteric polymer carries generally include carboxyl groups and hydrophobic groups in the molecule and the enteric polymer is dissolved in a solvent having a specific pH value through the dissociation of the carboxyl groups. For instance, commercially available hydroxypropylmethyl cellulose acetate succinate is a derivative of hydroxypropylmethyl cellulose which is substituted with carboxyl groups (succinoyl groups) and hydrophobic groups (acetyl groups). Alginate acid, sodium alginate other natural materials may also be used to provide an enteric coating.

Other additives and excipients may then be added to the formulation of the partially water soluble carrier-active odd chain fatty acids mixture, e.g., adding Povidone (e.g., Povidone 30), Xanthan gum (or other gums) and Sorbitol to a mixture of Kaolin Clay to provide a specific example of one formulation of the present invention. As will be apparent to those of skill in the art, the actual amount of the partially-excipient soluble active salt (e.g., non or partially water soluble) may be varied in accordance with the dissolution characteristics of the active, which may be further varied by addition of agents that affect the solubility and/or dissolution of the active in, e.g., water. As regards a pediatric formulation, the amount of active may be reduced in accordance with the dosage form approved for pediatric use.

One example of a liquid odd chain fatty acid(s) pharmaceutical composition may be prepared with the following components:

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Components	Weight
Odd chain fatty acid(s)	1.0 Kg
emulsifier (e.g., Imwitor 375)	100 gr
Purified water (USP)	2.0 Kg

The formulation may further include, e.g.:

Glycerin (USP)	500.0 ml
Sorbitol Solution, 70% (USP)	500.0 ml
Saccharin Sodium (USP)	10.0 gr
Citric Acid (USP)	10.0 gr
Sodium Benzoate (NF)	6.0 gr
Kollidon 30	330.0 gr
Xanthan Gum 200 Mesh	20.0 gr
Bubble Gum Flavor	11.1 gr
Methylparaben	1.0 gr
Propylparaben	100 mg
Propylene Glycol (USP)	75 ml
Additional ddH ₂ O	QS to 5 liters.

With appropriate increases of the above for scale-up.

A batch of mixed release odd chain fatty acids in an enveloped preparation on a carrier, e.g., beads, may be prepared with the following components:

Components	Weight
Emulsified odd chain fatty acids	8.0 mg
Carrier	51.7 mg
Calcium Stearate	4.0 mg
Talc	4.0 mg
Pharmaceutical Glaze	5.5 mg

When combining odd chain fatty acids (C5, C7 and/or C15), these may be formulated as follows. A capsule for extended release of a first active and extended release of a second active in an enveloped formulation, in a single capsule:

First Bead	Weight	Second Bead	Weight
odd chain fatty acid C7	6.0 mg	odd chain fatty acid C15	2.0 mg
Bead	162.9 mg	Bead	108.5 mg
Lacquer	6 mg	Lacquer	3.3 mg
Talc	12.6 mg	Talc	5 mg
Calcium Stearate	12.6 mg	Calcium Stearate	5 mg
Capsule	1		

When combining the odd chain fatty acids, these may be formulated as follows. A capsule for extended release of a first active and extended release of a second active in an enveloped formulation, in a single capsule:

First Bead	Weight	Second Bead	Weight
odd chain fatty acid C5	6.0 mg	odd chain fatty acids C7	2.0 mg
Bead	162.9 mg	Bead	108.5 mg
Lacquer	6 mg	Lacquer	3.3 mg

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-continued

First Bead	Weight	Second Bead	Weight
Talc	12.6 mg	Talc	5 mg
Calcium Stearate	12.6 mg	Calcium Stearate	5 mg
Mini-capsule	1		

A formulation for extended release of odd chain fatty acids of a second active in an enveloped formulation, in a gelcap:

Component	Weight	Component	Weight
odd chain fatty acid	6.0 mg	odd chain fatty acid	2.0 mg
Bead	162.9 mg	Bead	108.5 mg
Lacquer	6 mg	Lacquer	3.3 mg
Talc	12.6 mg	Talc	5 mg
Calcium Stearate	12.6 mg	Calcium Stearate	5 mg
Gelcap	1		

A formulation for rectal release of odd chain fatty acids in a suppository:

Component	Weight
Odd chain fatty acids	100 mg
Carrier	10 mg
Talc	12.6 mg
Calcium Stearate	12.6 mg
beeswax/glycerol	1-2 gr

An enteric-coated soft gelatin capsule that includes the odd chain fatty acids (with or without an emulsifier) is made by coating the odd chain fatty acids with a lipophilic material to obtain granules, mixing the granules obtained in step with an oily matrix, antioxidants and preservatives to form a lipid suspension, mixing the lipid suspension within a soft gelatin film, and coating the soft gelatin film to obtain an enteric coated soft gelatin capsule.

The odd chain fatty acid(s), stearic acid and triethanolamine are heated and mixed to form an emulsified fluid. The resulting emulsified fluid is mixed well by a homogenizer to obtain an emulsified suspension and enterically coated. Examples of formulations include:

Component	Weight
Odd Chain Fatty Acids	360.0 g
Stearic acid	78.6 g
Ethanolamine	21.4 g

Component	Weight
Odd Chain Fatty Acids	360.0 g
Stearic acid	30.0 g
Triethanolamine	20.0 g

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Component	Weight
Odd Chain Fatty Acids	400.0 g
Stearic acid	77.0 g
Ethanolamine	23.0 g
Cetyl alcohol	50.0 g

Component	Weight
Odd Chain Fatty Acids	245.0 g
Stearic acid	38.5 g
Ethanolamine	11.5 g
Cetyl alcohol	50.0 g
Carboxymethyl cellulose	25.0 g

TABLE 1

Recommended Daily Nutrient Intake Ranges
RECOMMENDED DAILY NUTRIENT INTAKE
RANGES WITH FOD DEFECT

AGE	NUTRIENT			
	Protein	Energy	Fluid	C7
	% of energy	kcal/kg/day	mL/kg	% Kcal/d
INFANTS				
0-<3 mo	10-12%	120	150-125	35%
3-6 mo	10-12%	115	160-130	35%
6-9 mo	10-12%	110	145-125	35%
9-12 mo	10-12%	105	135-120	35%
	g/kg	kcal/kg/day	mL/day	% Kcal/d
Children				
1-3 years	2-2.8	102	900-1800	35%
4-6 years	2	90	1300-2300	35%
7-10 years	1.5	70	1650-3300	35%
WOMEN				
11-14 years	1	47	1500-3000	35%
15-18 years	0.8	40	2100-3000	35%
>18 years	0.8	20-25	1400-2500	35%
MEN				
11-14 years	1	55	2000-3700	35%
15-18 years	0.9	45	2100-3900	35%
>18 years	0.8	20-25	2000-3300	35%

If patient is >20% ideal body weight (IBW), use upper range IBW to calculate needs

It has now been found that administration of odd carbon fatty acids that are converted or metabolized to the 5 carbon ketone bodies 3-hydroxypentanoate (BHP) and 3-ketopentanoate (BKP) is an effective treatment for GSDII in humans and polysaccharide storage diseases in horses and other animals. In addition, administration of the 5 carbon ketone bodies BHP and/or BKP in the form of either free ketone bodies or in other forms capable of providing the free ketones in vivo after administration is also an effective treatment for these diseases. Such forms of BHP and/or BKP as a triglyceride form, a polymeric form or a salt form could be used. The method of the invention spares the patient's own muscle protein because the patient no longer requires amino acids from the muscle for an energy source. In addition, the patient no longer requires the glycogen stored in the lysosomes as an energy source. The glycogen stored in the lysosomes will gradually decrease, because there is an intact glycogen deg-

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radative pathway in the cytosol that can compensate for the defective lysosomal pathway due to the absence of acid maltase (alpha glucosidase) in GSD II. Lysosomal "acid maltase" also has debrancher enzyme activity as well as alpha glucosidase activity. In the cytosolic glycogenoses, glycogen can enter the lysosome and degradation can be carried out by acid maltase since it has both glucosidase and debrancher activities.

The effective therapeutic agent in vivo is a product of the metabolism of C15, C7 or C5, or any other odd-carbon fatty acid that breaks down metabolically to BHP and BKP in the subject animal. FIG. 1 depicts the metabolic fate of C7 in humans. In addition, the C5 ketone bodies BHP and BKP, as stated above, could also be used directly for therapy. The degradation of C15, C7, & C5 all lead to the production of BHP and BKP and their export to other organ systems from the liver, as shown in FIG. 1.

Referring now to FIG. 2, the glycogen molecule is a polymer produced from the union of glucose molecules in a straight chain of glucose molecules joined together in α 1,4 linkage as well as branching of glucose residues in an α 1,6 linkage between glucose residues. This structure is represented schematically by a chain having a branch in this figure involving glucose residues depicted in red, blue and green. The branch (α 1,6 linkage) is depicted in (red). In the cytosol, a phosphorylase enzyme is required to remove the blue residues. A debrancher enzyme is required to remove and relocate the green glucose residues to the end of the α 1,4 chain. The debrancher enzyme also clips the 1 to 6 branched residue as glucose-1-phosphate and the final product is a straight chain of glucose residues (blue) in α 1,4 linkage. This chain is completely degraded to glucose-1-phosphate residues. Table 2 shows the profound effect of C7 therapy on amino acid levels in the blood. After 13 and 41 hours the levels of Alanine, for example, had greatly increased. The patient was then off the C7 diet during a gastrostomy and the level decreased. Once the C7 diet was re-initiated, the Alanine levels rapidly increased. FIG. 5 shows the metabolism of C7 in the cytosol and the lumen of the mitochondria and demonstrates the locations and entry of the ketogenic odd chain fatty acids of the present invention.

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Now referring to FIG. 3, the (red) bars show the location of the various inherited deficiencies causing Glycogen Storage Disease in humans. If any of these defects are present, it affects the ability of the subject to receive adequate energy from stored glycogen or polysaccharide. See: Chen, YT: Glycogen Storage Diseases. In: The Metabolic and Molecular Bases of Inherited Diseases, 8th edition, Chapter 71, pp 1521-1551, Mc-Graw-Hill, 2001.

FIG. 4 shows how odd-carbon therapy serves to circumvent the problems of glycogen metabolism that may be present. Administration of odd-carbon fatty acids (for example as triglycerides or as acids such as heptanoate) provides an alternate energy source, in the form of intramitochondrial acetyl-CoA and Propionyl-CoA, that can be utilized directly by the Krebs or "citric acid" cycle regardless of the existence of any one of the defects of glycogen metabolism (shown as red bars). Therefore, the demand for energy from glycogen is eliminated.

FIG. 5 is a graph that shows the effect of a triheptanoin-containing diet on the levels of select amino acids in an adult onset glycogen storage disease—Type II patient (GSD II or Pompe's Disease) following treatment with the Triheptanoin diet. The characteristic deficits of Alanine and Glutamine were restored within 13 hours with only the Triheptanoin diet, which documents the rapid elimination of the need for degradation of body muscle protein by provision of energy from Triheptanoin. The rapid reduction of the serum levels of these amino acids is obvious when the source of Triheptanoin was interrupted (NPO) awaiting the placement of a gastrostomy tube. Following that event and the resumed delivery of Triheptanoin via the gastrostomy tube, these serum amino acids were rapidly restored to normal.

FIG. 6 shows the steps and location of, e.g., C7 metabolism. The abbreviations on the right hand side of the arrows shows the location of various FODs that are mitigated and/or avoided using the odd chain fatty acids of the present invention as a single source of fats or as a supplement. FIG. 7 summarized the "CAC acceleration" of the present invention. By providing additional C2-CoA and C3-CoA into the CAC, the odd chain fatty acids of the present invention circumvent and avoid the enzymes that are deficient or lacking in FOD

TABLE 2

Plasma Amino Acids (uM/L) During Initiation of Anaplerotic Therapy, Adult-Onset Glycogen Storage Disease II (Pompe's).								
		Event:						
		Admit	C7 = 1.0 gm/Kg		NPO*		C7 = 1.5 gm/Kg	
		Time:						
		Baseline	13 hrs	41 hrs	65 hrs	84 hrs	108 hrs	132 hrs
ALANINE	(162-572)	129	551	450	189	184	307	277
GLUTAMINE	(424-720)	430	827	424	313	483	584	519
LEUCINE	(60-204)	104	154	266	158	108	238	220
VALINE	(108-295)	188	333	384	200	183	304	269
ISOLEUCINE	(39-119)	58	81	138	87	61	144	128

*NPO for gastrostomy - IV glucose only;

**Essential amino acids

In the lysosome, another pathway exists for glycogen degradation. The lysosomal α -glucosidase enzyme (acid maltase) is not only a glucosidase, but also has the capability to act as a debrancher enzyme as well. These characteristics of the lysosomal α -glucosidase permit complete degradation of lysosomal glycogen. These features are illustrated schematically in FIG. 2.

Importantly, the odd chain fatty acids of the present invention may be provided as an adjunct to current enzyme replacement therapies, e.g., intravenous PEGylated- α -glycoside or other like enzymes. By providing additional support for patients, a larger pool of patient may be addressed, with reduced cost, increased compliance, reduced morbidity and mortality and increased patient satisfaction.

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The odd chain fatty acids of the present invention not only prevented lipolysis they were also found to eliminate chronic infections in FOD patients. To spare intra-mitochondrial Coenzyme A, simple carnitine supplementation is required. Importantly, it was found that the odd chain fatty acids optimized the supply of propionate and acetate to the CAC and augmented ATP Production. Furthermore, unlike the use of supplemental propionate, the present invention did not sterilize the gut. The present invention provides a readily available energy source via an alternate pathway and may even be used to evaluate and protect siblings from disease.

The basic principle of the alternative metabolic pathway of the present invention is to reduce the need for carbohydrate in the form of stored glycogen or polysaccharide as substrate for the Citric Acid (Kreb's) Cycle (CAC). Meeting the energy needs of the body with Triheptanoin or other odd carbon fatty acids (C15, C5 or BHP) can permit the gradual reduction of excessive stored glycogen or polysaccharide resulting from any of the various types of glycogen storage disease. This is accomplished by the intact alternate metabolic pathway for glycogen degradation that exists in either the cytosol and in the lysosome. Patients with GSD II must remain on a diet containing odd carbon fatty acids, daily, for life, in order to prevent further occurrence of the clinical complications of the disease.

The amount of odd carbon fatty acid or ketone to provide is 25-35% of total Kcal/day for humans and ~10-20% of total Kcal/day for horses and other animals. The amount given can be calculated based on the weight of the patient in kilograms (Kg). The effective dose for adult humans and horses when triheptanoin is utilized, for example, is 1-2 gm/Kg/day. The dose for children when triheptanoin is utilized is approximately 1 to 4, preferably 3-4 gm/Kg/day.

The odd carbon fatty acid or ketone may be administered enterally or parenterally. Enteral administration includes oral administration and administration via a nasogastric tube or via a gastrostomy tube. The administered substance may be in a food or beverage or a nutritional composition. It may be desirable to control or monitor the total caloric intake of the subject and this may be accomplished through preparation of an appropriate liquid diet. A parenterally administered preparation may be made by methods known in the art for adding fatty acids as triglyceride emulsions to parenteral nutritional preparations.

For horses as well as such animals as cattle and sheep, one may use from about 1-2 gm/kg/day of triheptanoin and sufficient amounts of any other odd carbon fatty acid or ketone to provide about 10 to 20% of the total Kcal/day. The dose must be maintained at the optimal level related to Kg body weight for the best therapeutic result. This is especially important for a young growing animal or human child. Symptoms return when the dose per kg/day decreases as a result of growth and weight gain. It can be given to animals in a variety of ways, as with humans. It can be mixed with liquids, given straight via tube, or absorbed into "feed pellets" like "alfalfa" or other animal feeds. Parenteral therapy for animals is not a frequent treatment approach at this time, but it could be used very effectively with horses and other large animals when illness or stress results in serious metabolic crisis.

Organic Acids may be measured as follows. Organic acids in urine specimens are determined with isolation by Liquid Partition Chromatography (LPC) and quantitative Gas Chromatography-Mass Spectrometry (GCMS) using an internal standard. The specimen is derivatized to form pentafluorobenzyl oximes (PFBO) of oxo-acids and the organic acids isolated by Liquid Partition chromatography on silicic acid hydrated with dilute sulfuric acid and eluted with mixtures of

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tert-Amyl alcohol and chloroform. After drying, the organic acids are derivatized to form volatile trimethylsilyl (TMS) derivatives for separation by capillary Gas Chromatography (GC) with temperature programming. Detection is by Mass Spectrometry (MS) with identification of the organic acids by their mass spectra. Organic acids are quantified by peak areas of reconstructed ion chromatograms with internal standards and calibration curves.

Acylcarnitine Profiles. Acylcarnitines and carnitine levels in dried blood spots (dbs), plasma and amniotic fluid by automated electrospray tandem mass spectrometry Acylcarnitines are extracted from blood spots, plasmas and amniotic fluid with methanol containing stable isotopically labeled internal standards. The extract is derivatized to form butyl esters. An automated sample introduction system delivers the samples to the electrospray tandem mass spectrometer (ESI-MS/MS) operating in the multiple channel analysis mode in which acylcarnitines are identified, profiled, and quantified with stable isotopically labeled internal standards. This test also provides quantification of free, sum of acylcarnitines, and total carnitine levels.

Example 1

Treatment of Human Subject Diagnosed with Acid Maltase Deficiency

Clinical Description. A 42 year old caucasian female had onset of progressive muscle weakness and difficulty breathing dating back nearly 20 years. She was seen at 40 years for muscle weakness and a muscle biopsy revealed increased glycogen deposition in lysosomes and cytosol (2.0%—normal 1.03+/-0.18%), autophagic vacuolization, and increased acid phosphatase staining. An assay for acid- α -glucosidase confirmed the deficiency as adult-onset GSD II (0.46—normal 8.13+/-2.1 nm MU-hydrolyzed/min/gm). No therapy was initiated.

During the subsequent 2 years, the patient experienced progressive muscle weakness, affecting strength and endurance, swallowing and producing urgency for urination and defecation. The patient's breathing became a major issue and weight had decreased from ~127 to 101 lbs (46 Kg) over this interval. Pulmonary evaluation a day prior to admission revealed MIP=-70 cm HOH; MEP=+60 cm HOH; FVC=40% of predicted (1.4 liters); blood gases: pH 7.38, pCO₂ 68, pO₂ 56 (room air). Home O₂ was started and upon respiratory arrest was intubated and unsuccessfully extubated leading to emergency admission lasting one month. Following informed consent, the patient agreed to enter an approved Institutional Review Board (IRB) Protocol for dietary therapy with triheptanoin oil.

Clinical and Biochemical Course on Therapy. The patient was admitted with left lower lobe atelectasis and infiltrate, placed on antibiotics, carnitine supplement (20 mg/Kg/day) and diet containing triheptanoin was given by nasogastric tube. Esophageal dysmotility was documented. A percutaneous endoscopic gastrostomy (PEG) was placed on the sixth hospital day.

The diet was based on vivonex and ~1500 Kcal was given by nasogastric tube, on admission. The composition (% Kcal) was: Protein 29%, carbohydrate (CHO) 30%, total lipid 41% (of which triheptanoin (C7) accounted for 26%). The C7 dose of 1 gm/Kg/day (8.3 Kcal/gm) was gradually raised from 26% to 35% and finally 40% of total Kcal corresponding to 1.0, 1.5, and 2.0 gm C7 per Kg per day. Carbohydrate was correspondingly reduced from 30 to 27 to 11% of total Kcal/

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day. Total caloric intake was raised from ~1500 to 1700 to 1900 per day with these dietary changes during her 33 day admission.

Within 13 hours of beginning the C7 diet, pCO₂ fell from 68 to 43 (normal 35-45), and a long history of urgency for urination and defecation had ceased and remained normal for the rest of the admission. The patient began ambulating by the 10th day that improved steadily, showering with assistance by the 28th day. Pneumonia on day 17 responded to therapy and was extubated on day 23. She was discharged on bilevel positive airway pressure (BIPAP), the C7 diet, and carnitine supplement.

Following discharge, the patient returned to work half-time at 5 weeks and was working full-time by 10 weeks. Swallowing was sufficiently improved that gastrostomy feeds were stopped at 7 months and the PEG was removed at 10 months. By 15 months post-discharge, patient weight had increased from 46 Kg on admission to 60 Kg and has remained stable. Patient maintains normal activities without any significant weakness but continues to use BIPAP at night. Patient pCO₂ levels remain in the normal range, however, her full vital capacity remains at 40-45% of predicted.

Metabolic Observations. Routine blood chemistries on admission were largely normal except for pCO₂ of 68.1 (normal 35-45 units). ALT and AST and LDL were mildly elevated while serum CPK was normal. Hemoglobin was 11 gm/dl and creatinine was reduced slightly at 0.4 (normal 0.5-1.0). Urinalysis revealed ketosis. Potassium was reduced, slightly (3.5, normal 3.6-5.0 units). Blood acylcarnitine analysis revealed normal total carnitine with increased acetylcarnitine (ketosis), and a surprising decrease in propionylcarnitine (1.25, normal <2.71 uM). There were no other acylcarnitine abnormalities.

Quantitative urinary organic acids revealed only ketosis with a urine pH=8.5. The excretion of 3-OH-butyrate (BOB) and acetoacetate (AcAc) was 1633 and 1211 mmol/mol creatinine, respectively, (normal: <17 & 7 respectively). The ratio of BOB:AcAc was decreased from 2.4 to 1.3 suggesting an alteration in the intra-mitochondrial redox state. During hospitalization and follow-up, this ratio was less than 1.17 (range 0.40-1.17) during 4 of 5 episodes of mild to moderate ketosis. However, more recently with addition of an amino acid supplement to the C7 diet, the levels and ratio became normal (3.0). Citric acid cycle intermediate excretion fell into the normal ranges for adults.

The most significant abnormalities were observed with plasma amino acid analysis. The levels of Alanine and Glutamine were low compared to the adult normal ranges. While the levels of the branched-chain amino acids (BCAA=Leucine, Valine, & Isoleucine) were within the adult normal ranges. The reduced levels of Alanine and Glutamine have been observed often in this disease and has led to the high protein—low carbohydrate diet and/or alanine supplementation. This patient did not receive any supplementary alanine. In only thirteen hours after initiating the C7 diet by nasogastric tube, the plasma alanine level had increased 4-fold. Glutamine increased nearly 2-fold as did the branched chain amino acids (BCAAs). The levels of these amino acids remained in the normal range while receiving the C7 diet. However, the diet was interrupted on day 4 & 5 while awaiting placement of the PEG for gastrostomy feeding. The levels of each of these amino acids promptly fell to admission levels (Table 2: 65 & 84 hour samples). With resumption of the diet via gastrostomy, all levels returned to (or above) the normal ranges in 24 hours. These amino acid levels remained normal on this odd-carbon triglyceride diet therapy. Further analysis of all of the measured amino acids revealed the same sudden

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increase with the C7 diet and the same sudden decrease when the diet was interrupted. As C7 was increased from 26 to 40% of total caloric intake (from 1.0 to 2.0 gm/Kg/day). All amino acids remained normal and propionylcarnitine levels had increased without any evidence for propionyl overload from the metabolism of heptanoate.

Furthermore, the present invention may be used to provide a viable source of energy to patients with compromised ketogenesis, gluconeogenesis and even the urea cycle, due to FODs. The odd chain fatty acids help support the sub-optimal function of the Citric Acid Cycle, gluconeogenesis, the urea cycle and ATP generation by the respiratory chain. The anaplerotic substrates support the Citric Acid Cycle, e.g., by providing pyruvate to malate and oxaloacetate; glutamate to ketoglutarate and heptanoate to propionyl-CoA to succinyl-CoA. These types of support may be provided alone or in combination. Furthermore, the present invention may be used with, e.g., amino acid supplementation to reduce or prevent amino acid scavenging as an alternative source of energy. As the patient improves, muscle and other tissues require additional amino acid support, which may be supplemented along with the present invention using amino acid supplements. One example of an amino acid supplement for use with the present invention is Amino-Vital, which contains branched chain amino acids (BCAA) via branched chain ketone acids (BCKA) that may also supplement the C2-CoA and C3-CoA. Additional co-factors and vitamins (e.g., Biotin and B-12) may be added to the patient's diet to improve the metabolism of Propionyl-CoA to Succinyl-CoA and/or entry into the CAC as Acetyl-CoA.

Example 2

Treatment of Human Subject Diagnosed with GSDII

Clinical History & Description. A 66 year old white male whose history of muscle weakness dated back to his youth when he noticed weakness that caused limitations in sports activities. In his 40's, he became much more aware of weakness that primarily affected his back and legs. Spirometry, at that time, revealed reduced lung capacity. At age 54, a muscle biopsy was obtained that showed glycogen deposition in lysosomes and cytoplasm but was not increased (0.76%). Acid maltase was 0.94 (normal: 8.13+/-2.1). Debrancher enzyme was also assayed in this biopsy and was 1.40 (range 1.88-4.24). Shortly after that, he experienced shortness of breath at high altitude and has used bilevel positive airway pressure (BIPAP) since that time also for sleep apnea. Patient muscle weakness was progressive and he has been using a motorized scooter for the past 10 years. Patient weakness was generalized, he had great difficulty standing, and walked with an irregular gait, using a cane. Patient had no history of impaired swallowing but did complain of urgency for urination and defecation.

Clinical Course. Following informed consent, the patient agreed to enter an approved IRB Protocol for dietary therapy with triheptanoin oil. Initial pulmonary studies revealed severe restriction of his vital capacity (2.41 L, 43% of predicted). Maximal inspiratory and expiratory pressures were 60 cm of water. Initial arterial blood gases revealed pH=7.39, pCO₂=51. And pO₂=62.

The oral diet consisted of ~2900 Kcal on admission. The composition (% Kcal) was: Protein 16%, carbohydrate (CHO) 40%, total lipid 44% (of which triheptanoin (C7) represented 22%). The C7 dose of 1 gm/Kg/day (8.3 Kcal/gm) was gradually raised from 26% to 30% of total Kcal corresponding to 1.0, and 1.5, gm C7 per Kg per day. Carbo-

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hydrate was correspondingly reduced from 43 to 22% of total Kcal/day. Total caloric intake was maintained at ~2800 per day during his 30 day admission.

Initially, the patient required assistance and supervision with grooming, upper and lower extremity dressing, bathing, and toileting. The patient also required maximal assistance with ambulation and transfers from bed to wheelchair. With intensive physical therapy and diet, at discharge, he had become independent with grooming, upper and lower extremity dressing, toileting, and locomotion with a manual wheelchair. He continued to require supervision, but without assistance, with transfers from bed to wheelchair and to toilet and shower. He had clearly made improvements in strength and endurance with exercise. Pulmonary function testing showed an increase of 700 ml in total vital capacity when compared to his studies on admission.

The patient was discharged on the triheptanoin diet with supplements of citrate (bicitra, 15 cc four times a day) and amino acid supplement ("Amino-Vital", Ajinomoto, Inc) 4.8 gms four times a day. At four months after discharge, he continues this regimen and physical therapy three times per week.

Metabolic Observations. Routine blood chemistries on admission were largely normal except for pCO₂ of 51 (normal 35-45 units). ALT and AST and lipid profile were normal while serum CPK was mildly increased (218, normal: 38-174 IU/L). His BUN was increased (30, range 9-20 units) and creatinine was reduced slightly at 0.6 (normal 0.7-1.2 units). All other chemistries were normal. Blood acylcarnitine analysis revealed normal total carnitine and a reduced level of propionylcarnitine (1.08, normal <2.71 uM). There were no other acylcarnitine abnormalities.

Patient diet was initially adjusted to 2800-3000 Kcal/day composed of protein 16%, CHO 40%, total lipid 44% of which C7 represented 22% of his daily Kcal intake. The C7 composition was increased to 30-31% during the admission. His initial response to C7 revealed a marked decrease in urinary pH (from 7.5 to 6.0) and acylcarnitine monitoring revealed a marked increase in propionylcarnitine (1.08 to 5.86 uM) and pimelate, heptanoate and methylmalonate (MMA) excretion increased significantly (321, 307, and 38 mol/mol creatinine, respectively). These changes indicated impaired oxidation of C7 and reduced utilization of propionyl-CoA for the citric acid cycle (CAC). Following supplementation with citrate (bicitra) the urine pH returned to 7.5, and pimelate, heptanoate, and MMA decreased immediately to 58, 50, and 24, respectively. The propionylcarnitine blood level was also reduced to 3.74 uM. These changes reflected more normal metabolism of dietary C7. Citrate supplementation was continued throughout the treatment.

Although there were 5 periods when the quantitative urinary organic acids revealed mild ketosis, the ratio of BOB: AcAc was always normal while on citrate supplementation. Unlike patient 1, there was no indication of an alteration in the intra-mitochondrial redox state. Although citric acid cycle intermediate excretion was in the normal adult ranges, there were some abnormalities: isocitrate was initially increased (134-nl=<82) while succinate and α -ketoglutarate levels were reduced (6 [nl<42], and 4 [nl<75], respectively). The sum of the excretion of the CAC intermediates was 391 (upper limit=1185). This profile persisted during the study.

The most significant abnormalities were observed with plasma amino acid analysis. The levels of Alanine and Glutamine were low compared to the adult normal ranges. While the levels of the branched-chain amino acids (BCAA=Leucine, Valine, & Isoleucine) were within the adult normal ranges. The reduced levels of Alanine and Glutamine

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have been observed often in this disease and has led to the high protein—low carbohydrate diet and/or alanine supplementation. This patient did not receive any supplementary alanine. In only thirteen hours after initiating the C7 diet by nasogastric tube, the plasma alanine level had increased 4-fold. Glutamine increased nearly 2-fold as did the BCAA's. The levels of these amino acids remained in the normal range while receiving the C7 diet. However, the diet was interrupted on day 4 & 5 while awaiting placement of the PEG for gastrostomy feeding. The levels of each of these amino acids promptly fell to admission levels. With resumption of the diet via gastrostomy, all levels returned to (or above) the normal ranges in 24 hours. These amino acid levels remained normal on this odd-carbon triglyceride diet therapy. Further analysis of all of the measured amino acids revealed the same sudden increase with the C7 diet and the same sudden decrease when the diet was interrupted. As C7 was increased from 26 to 40% of total caloric intake (from 1.0 to 2.0 gm/Kg/day). All amino acids remained normal and propionylcarnitine levels had increased without any evidence for propionyl overload from the metabolism of heptanoate.

Example 3

Treatment of Horses

Three horses affected with biopsy proven, polysaccharide storage myopathy have been successfully treated with the triheptanoin diet. The first was an 8 year old mare who was unable to walk or exercise without severe muscle pain, rhabdomyolysis, and "locking up." The owners were approaching "putting the animal down" (killing it). Within 30 days of triheptanoin diet incorporated into feed pellets, this animal has become asymptomatic and has returned to full training and riding without any symptoms.

A second horse was encountered at 4 months with proven polysaccharide storage myopathy (PSSM) that was not able to walk adequately or nurse from its mother. The owners were prepared to kill the animal but decided to try the triheptanoin diet first. Within 24 hours of initiating the triheptanoin diet, the foal was running and playing with other young horses without any evidence of muscle pain, etc. All blood enzyme levels related to muscle disease (including serum creatine phosphokinase) had normalized. As this foal gained weight, in a normal rate for a young horse, symptoms returned since the dose of C7 had not been adjusted for the weight gain. Adjustment of the dose back to 2 grams/Kg body weight eliminated all symptoms and there is no further evidence of disease.

A third horse, also an 8 year old mare, developed PSSM and "locked up." She was no longer available to ride and could no longer jump. She started triheptanoin at 1.0 gm/Kg/day added to her diet pellet feeds and is currently asymptomatic with normal blood chemistry levels compared to pre-treatment after only one week of diet treatment.

Example 4

Treatment of FOD in Horses and Diets for Treatment of Fatigue

Horses: Eight healthy, fit 4 year old thoroughbred geldings may be used in a study. These horses are all sound and trained to the treadmill. The ability of horses to absorb and digest C7 oil was first determined. Three horses were fasted for 12 hours prior to administration of 0.5 gm/kg of triheptanoin via nasogastric tube. Samples for blood acylcarnitine profile were

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drawn at 30, 60, 90, 120, 150, and 180 minutes after triheptanoin administration. C7 acyl carnitine peaked at 90 minutes after administration and peak serum concentration of C5 and C3 acylcarnitines occurred 120 minutes after administration (FIG. 9).

In humans C7 (triheptanoin) is hydrolyzed to heptanoate which is then oxidized in the liver to form β -hydroxypentionate and β -ketopentionate, both 5 carbon ketone bodies (C5). C5 is metabolized to C3 (propionyl CoA) and C2 or acetyl CoA. These fatty acids are transported in the bloodstream as acylcarnitines, which are easily taken up in muscle tissues and cross mitochondrial membranes to enter the CAC.

The five carbon ketones B-hydroxypentionate and B-ketopentionate produced from the breakdown of triheptanoin in the liver are represented in our preliminary results as plasma C5 acylcarnitine. Horses have a poorly developed ability to produce ketone bodies, however the presence of 5 carbon acylcarnitines in our horses, suggests that they were able to metabolize triheptanoin into the 5 carbon ketone. C3 acylcarnitines were also increased in our preliminary studies suggesting the break down of the C5 to propionyl CoA (C3).

Palatability of C7 has also been studied, in which the C7 was provided as triheptanoin or mixed with corn oil into the diet of horses. It was found that when horses were fed the oil mixed into hay cubes soaked in water prior to feeding, they consistently ate all of the feed.

Diets: A variety of diets may be formulated. For example, the diets may be isocaloric and formulated to meet the minimum daily requirements for all nutrients. In one example of horse feed is made that includes 8.5 kg of grass hay/grass hay cubes, 0.5 kg of ration balancer, free choice salt and 750 mls of oil per day with a total digestible energy of 23 MCal/500 kg horse per day. The oil will be either triheptanoin (provided by Dr. Roe) or corn oil (purchased) and will be divided into 3 feedings.

Exercise study: Horses will generally be rested for one day prior to and one day after the exercise study. On day 14 of each diet an IV catheter will be placed and horses will perform a 90 min submaximal exercise study on a flat treadmill 120 min after consuming the oil. A loosely fitting mask with a high flow rate may be placed over the horse's nose and samples of expired gas collected every 15 minutes during exercise using an open circuit calorimeter. Oxygen consumption (VO_2) and carbon dioxide production (VCO_2) will be measured and respiratory exchange ratio (RER) will be calculated.

The exercise study protocol may include a 5 min warm up at a walk, followed by:

Block 1: 15 min 35% VO_2 max (previously measured in these horses), 5 min walk

Block 2: 15 min 35% VO_2 max, 5 min walk

Block 3: 15 min 35% VO_2 max, 5 min walk

Block 4: 15 min 35% VO_2 max, 5 min walk

Block 5: 10 min 75% VO_2 max

Block 6: 5 min walk

Blood samples: Blood samples may be drawn prior to the initiation of the trial; prior to the exercise study; and at the end of the highest speed during each exercise block of the exercise study. Blood samples for glucose, lactate, ketones, FFA, amino acids and acylcarnitine concentrations will be obtained. Glucose and lactate will be assayed on an automated lactate analyzer and ketones and FFA will be assayed spectrophotometrically using the appropriate kits. Plasma amino acid and acylcarnitine levels may also be measured by mass spectrometry from four randomly selected horses; all other values will be analyzed in all study subjects.

Muscle Biopsy: Gluteal muscle biopsy specimens may be obtained by use of a percutaneous needle biopsy technique

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prior to initiation of the diet trial; prior to and immediately upon completion of the exercise study; and 24 hours after exercise study. Samples are obtained within a 2 inch square at a standardized site along the gluteal medius muscle and alternating sides will be used for each sample. Biopsy specimens will be immediately frozen in liquid nitrogen and stored at -80°C .

Muscle Analysis: Frozen muscle specimens will be dissected free of blood and connective tissue. Glycogen will be assayed fluorometric as glucose residues remaining after 1-2 mg portions of muscle tissue are boiled for 2 h in 1 M HCl. Lactate, pyruvate, G-6-P, and CAC intermediates will be assayed in a separate portion of the muscle sample (8). One 4 mg portion of muscle will be homogenized by crushing with a glass rod in 1.5 M perchloric acid. The supernatant obtained after centrifugation will be neutralized with KHCO_3 , centrifuged again and the remaining supernatant removed for analysis of metabolites using fluorometric techniques and purine nucleotides using high-performance liquid chromatography (HPLC). Concentrations of CP, ATP, ADP, AMP and IMP in muscle specimens as well as in external standards will be analyzed by use of a reverse phase column (10). Separation of nucleotides will be achieved with a flow rate of 1.0 mL/min, UV-detection at 254 nm, and an oven temperature of 40°C .

Urine Samples: A urine collection device is placed the evening prior to exercise on the 4 horses randomly selected for plasma acylcarnitines and amino acid analyses and will again be placed on the horses after exercise. An aliquot of voided urine will be collected for measurement of urine organic acids by mass spectrometry.

Example 4

Beverages and Food Supplements

In one embodiment, the odd chain fatty acids of the present invention may be provided as a nutritional supplement and/or as part of an overall patient diet. As a non-limiting example of a foodstuff that may include the present invention in a pharmaceutical or nutritional amount, the odd chain fatty acids may be formulated into a snack bar, such as that described in U.S. Pat. No. 4,777,045, issued to Vanderveer, relevant portions incorporated herein by reference. A high bran snack is taught that includes a typical formulation of graham (whole wheat) flour, 40 percent; rice flour, 10 percent; whole wheat bran flour, 50 percent; calcium carbonate, 1.25 percent; reduced iron, 0.013 percent; and riboflavin (as a tracer), 0.02 percent and between 0.1 and 10% odd chain fatty acids. Water is added per 80 ounces of dry materials to provide a semi-solid flowable product that may be fed into a ribbon blender and then into a cooker extruder, e.g., a twin-screw cooker extruder. Each of the ingredients in the formulation has a processing, nutritional or therapeutic purpose. The extruded pieces may be further flavored, e.g., coated with coconut oil and powdered flavorant.

A whole wheat and bran snack may be 50 weight percent of wheat bran, with approximately 50 percent of whole wheat flour, including 0.2 weight percent of riboflavin (20 mg per 100 gm of finished product), have a desired size, density and shape for the subsequent operation of adding oil, flavor and eventually packaging. Samples of finished products, with 4 different flavors, applied at a level of 3 weight percent to the extruded base product, with the addition of 3 weight percent of odd chain fatty acids. The extent and formulation of the odd chain fatty acids (e.g., with an emulsifier) may improved adhesion of the flavor particles, flavor acceptability and

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mouth feel organoleptically. The bar may further include vitamins, minerals and even protein and carbohydrates to provide a so-called "power-bar."

Another bar may be a granola bar with supplemental dietary fiber as taught by Linscott in U.S. Pat. No. 4,871,557, relevant portions incorporated herein by reference. A granola bar is taught that includes supplemental dietary fiber added to the granola bar in the form of compressed flakes, as well as the method of making such a granola bar, which may further include the odd chain fatty acids of the present invention. Briefly, a mixture of granola ingredients such as grains, fruits, nuts and compressed flakes are mixed with the granola ingredients and the odd chain fatty acids. The compressed flakes of supplemental dietary fiber are combined with water and a binder material, such as rice flour, and then extruded. The extrudate is dried and then ground to the desired particle size.

Although sources of supplemental dietary fiber can contribute both soluble and insoluble fiber, sources generally known to contribute insoluble fiber include but are not limited to soy fiber, apple fiber, corn bran, wheat bran, oat bran, barley bran, rye bran, triticale bran, cellulose, pea fiber, sugar beet fiber, and peanut fiber. Sources generally known to contribute soluble fiber include but are not limited to gum arabic, gum ghatti, guar gum, pectins, psyllium, carrageenans, xanthan, tragacanth, karaya, locust bean gum, agar, and alginates. These non-digestible fibers provide additional control over the exact amount of kilocalories are provided to a patient as the primary source of energy from fat is provided by the odd chain fatty acids.

The extrusion step may be performed by conventional techniques in conventional extrusion apparatus, e.g., a damp mixture is heated to a temperature between about 300 and about 330° F., e.g., 315° F. during the extrusion process. The damp mixture is then extruded at a pressure of between about 100 and about 900 p.s.i. A die through which the mix is extruded may include a square, rectangular, triangular, oval or round hole with a diameter of, e.g., 0.5 inches. Generally, extrudates are particle sized before being dried in an oven at 270° F. for about 20 minutes. Often, the dried particles will have a moisture content of about 7 percent.

The odd chain fatty acids may also be provided as beverage. The odd chain fatty acids may be added to a beverage such as that taught in U.S. Pat. No. 4,981,687, relevant portions incorporated herein by reference. The beverage taught is used to improve physiological responses to exercise, however, it uses standard sources of carbohydrates. The addition of the odd chain fatty acids taught herein provides a beverage that not only reduces or prevents adverse physiological effects of physical exercise or environmental exposure, but provides a new and/or additional sources of readily available energy. The fluid will generally include water, electrolytes, and odd chain fatty acids and is non-toxic to man or animals. Sugar may also be provided or sugar substitutes may be used without a concomitant loss of new energy production. As such, the odd chain fatty acids substitute the energy source and can be absorbed rapidly through the gastrointestinal tract, prevents decreases in blood volume.

The odd chain fatty acids will provide a beneficial physiological effect on cells during exercise by providing a substitute and/or additional readily available energy source that may increase blood volume and cardiac output, improved skin blood flow, prevention or delay of onset of hyperthermia, increased rate of movement of electrolytes across the gastrointestinal wall, reduction in the breakdown of proteins and associated metabolism of essential amino acids, and decreased time needed for repair of body tissue following strenuous exercise.

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When the fluid of the subject invention is administered, the body's physiological response to exercise or environmental exposure is greatly enhanced compared to the response when the body receives no fluids, receives only water, or receives fluid such as GATORADE®, which contains electrolytes and a sugar source in addition to water. The composition described here can be used to ameliorate the adverse effects of physical exertion or environmental exposure. The effect of the odd chain fatty acids on physical exertion may be measured by determining plasma volume, respiratory quotient, rectal temperature, pulse rate and/or cardiac output; combined with either enhanced endurance or performance, lower perceived difficulty of a physical task, or an enhanced ability to withstand heat exposure or chronic exposure to cold. The beverage may include: water, odd chain fatty acids (1-4%), potassium (2 meq/l), sodium (26 meq/l) and phosphate (4 meq/l).

Addition of a small amount of odd chain fatty acids, given at frequent intervals, may improve performance and endurance because it enhances entrance of both acetyl CoA and propionyl CoA into the Krebs cycle, thereby bypassing the need for additional oxaloacetate (OAA). The Krebs cycle is a well-known, but very complicated, biochemical pathway which provides a working muscle with its energy source. A detailed description of the Krebs cycle can be found in most biochemistry textbooks.

The beverage of the present invention may also be used to improve the physiologic response of any animal undergoing exercise or being subjected to high temperature conditions. For example, humans, horses, dogs, cats, mules, oxen, camels, emu, bison, nilguy, elephants, sheep, cows, chickens, turkey, goats, llamas, alpaca and pigs are a few of the animals that may benefit from the administration of the novel fluid composition described here. The fluid of the subject invention can also be used to alleviate or prevent dehydration which is known to result from chronic exposure to cold temperatures.

The odd chain fatty acids of the present invention may also be provided with dietetic beverages. An example of a dietetic beverage that may include the odd chain fatty acids includes, e.g., U.S. Pat. No. 4,042,684 that teaches a beverage for supplementing the dietetic requirements of sugar and essential salts in a mammalian body depleted through vigorous physical activity. The odd chain fatty acids may be added and substitute for simple carbohydrates in an aqueous that includes sodium chloride, potassium chloride, and free citric acid in physiologic ranges.

The combination of odd chain fatty acids and essential salts in a mammalian body will be particularly useful for athletes that have depleted short-term, immediately available energy through vigorous physical activity. For example, an athlete engaged in strenuous activity requires a ready source of energy for endurance, and replacement of both body fluids and essential salts lost through perspiration. Likewise, individuals working in a hot, humid atmosphere have similar requirements to maintain efficiency and productivity. Long term activity initiates bypass metabolism that the odd chain fatty acids of the present invention help to supplement to increase protein sparing.

Typical diets and sports beverages provide intake of sugars and sugar-precursor materials such as carbohydrates. When metabolized in the digestive tract, fuel values are obtained through enzyme and acid attack on these complex sugars. However, the process requires time to achieve the desired boost in metabolism, which is short lived and leads to protein and/or amino acid scavenging to supplement the citric acid cycle. Increased metabolic demands exceed the steady-state ability of this natural metabolic process.

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Rather than rely on well-known sugar metabolism, e.g., fructose and glucose to achieve an energy store capable of providing needed fuel values, odd chain fatty acids provide the fully utilizable energy at the cellular level for metabolism in either the liver or muscles, brain, kidney and heart. One problem with commonly available sports beverages is that, e.g., glucose is easily and rapidly transported out of the digestive system into the blood whereas fructose is more passively and slowly transported. Once into circulation, the fructose is somewhat more efficient insofar as initial transport requires less energy and subsequent utilization for energy proceeds more readily. Thus, both immediate and longer lasting benefits may be attainable. The odd chain fatty acids disclosed herein may be provided with or partial or complete substitution for those complex saccharides that have differential breakdown and uptake. The odd chain fatty acids are provided along with salt constituents, either as salts with the odd chain fatty acids themselves, or in the form of additional salts, e.g., sodium chloride and potassium chloride. The quantity and relative proportion of the salts may be selected to achieve an isotonic liquid without a strong acid taste. Thus, the salt components provide replacement for those essential ions lost in perspiration while, at the same time, yielding a highly palatable beverage.

The beverage of the present invention may be compounded with water and suitably bottled and stored. Alternatively, all the components, save for water, may be prepared in advance as a concentrate for ease of handling and transportation. Also, the beverage may be prepared with carbonated water should such be desirable.

During athletic competition, or other strenuous physical activity, the individual may replace lost body fluids and essential salts while sustaining a high level of energy through consumption of the beverage of the present invention. Moreover, contrary to many of the prior art formulations for such beverages, e.g., extended citrus juices, intake of the beverage of the present invention is not accompanied by a "full" feeling that would present a serious hindrance in athletic competition. Thus, the present dietetic beverage successfully replaces and/or maintains individual body requirements in accordance with the physiological needs thereof and further provides additional storage of energy.

Food Compositions and Additives. The odd chain fatty acids of the present invention may also be formulated into an edible matrix and/or food composition. One such chewable matrix is taught in U.S. Pat. No. 6,723,358 that teaches the encapsulation of edible products, relevant portions incorporated herein by reference. An encapsulated product is obtained by mixing a plasticizer, a ground, free-flowing particulate mixture with the odd chain fatty acids, at least one starch, mixed and heated. A chewable texture is obtained when the starch is maintained substantially ungelatinized. A flavorful product may also be obtained and made part of an overall patient diet without destroying a heat sensitive encapsulant because the oil and starch are heated to develop flavor at high temperatures prior to mixing with a heat sensitive encapsulant. The encapsulated component may be at least one biologically active component, pharmaceutical component, nutraceutical component, or microorganism. The mixture may include ground cookies that are mixed with the plasticizer, odd chain fatty acids and water to obtain a formable dough or crumbly mass that may be further formed into pieces or pellets and dried to a shelf-stable moisture content.

The present invention will find particular utility in animal feeds. For example, the odd chain fatty acids may be used in conjunction with the feed stocks, e.g., that taught in U.S. Pat. No. 6,777,396, relevant portions incorporated herein by ref-

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erence. The feed for livestock may include a feed for livestock with additives such as nucleic acids, glutamine and glutamic acid. The feed is provided as part of a method for increasing body weight gain efficiency and feed efficiency in livestock.

The present invention also includes an animal feed composition that includes the odd chain fatty acids. It is contemplated that any odd chain fatty acids may be present in the animal feed in any amount effective to provide nutritive fat to the animal. It is contemplated that the odd chain fatty acids content may vary depending upon the animal or upon the intended nutritive qualities of the feed. Generally, the fat is present in the animal feed in an amount of at least 5%, 10%, 17%, 20%, 25%, 30% or even 40% by weight of the animal feed. It is contemplated that two or more odd chain fatty acids sources may be included in the feed.

The animal feed further includes a solid nutritive source, such as a whole grain, whole wheat, whole rice, whole corn, or whole barley. Alternatively, the solid nutritive source may also include nutritive wheat, nutritive rice, nutritive corn or nutritive barley fraction. Other nutritive sources include those derived from soy, oats, sorghum, and the like. The nutritive source may include other nutritive sources, including sources of carbohydrates (such as molasses solids) that are also provided in liquid form. The solid nutritive source may be present in the animal feed in any suitable amount.

Another nutritive source is a protein source, which may be present in any amount effective to provide protein to the animal. Protein may be present in an amount ranging from about 5% to about 40% by weight of the animal feed. For example, young swine are particularly needy of protein, and protein contents in the upper portion of this range (e.g., a protein content of about 36%) are often used in feeds intended for such swine. For feeds of other animals, the protein is present in an amount ranging from about 10% to about 30% by weight of the animal feed, but more often in an amount ranging from about 15% to about 20% by weight of the animal feed.

Animal feeds may also include a source of fiber source. Sources of fiber may include: soybean hulls, rice hulls, corn hulls, cottonseed, wheat hulls, and the like are considered largely non-nutritive (at least in the case of non-ruminant animals). When the animal feed is intended for use by ruminants, the feed very often includes such fiber source in an amount effective to provide fiber to the animal. Different feed formulas for different animals vary greatly in the amount of fiber desired. The fiber source may be prepared in an amount ranging from about 1% to about 25% by weight of the animal feed, the percentage being expressed by the bulk weight of the hulls or other source.

Pelleted High-Fat Animal Feed. The odd chain fatty acids of the present invention may be incorporated into any of a number of animal feeds. Basic types of animal feeds are taught in, e.g., U.S. Pat. No. 6,746,698, relevant portions incorporated herein by reference, which teaches an animal feed, method for preparing animal feed, and method for feeding an animal. Unlike the present invention, however, the fats taught are standard even chain fatty acids and fats. A master batch of horse feed ration containing full fat corn germ from a corn wet-milling process was formulated in a Hobart mixer. The feed was formulated from the following components:

Ground Full-Fat Corn Germ	24.8%
Wheat Midds	21.1%
Ground Whole Corn	17.9%
Soybean Meal	9.5%

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-continued

Odd Chain Fatty Acid	25.0%
Calcium Carbonate	1.1%
Dicalcium Phosphate	1.1%

To an aliquot of the master batch an adhesive may be added, in liquid and/or dry form in an amount sufficient to constitute 5% of the total dry mass of the final pelleted feed. The mixture was then converted into a pelletized horse feed using a pellet mill.

An animal feed was formulated as follows:

Ingredient	Weight %
Corn Germ	23%
Wheat Midds	21.5%
Whole Ground Corn	15%
Soybean Meal	9%
Calcium Carbonate	1%
Dicalcium Phosphate	1%
Odd Chain Fatty Acid	25.0%

The ingredients may be combined to form a mixture and the mixture pelletized. In this formulation, the wheat midds were included to provide a source of fiber, the soybean meal was included to provide a source of protein, and the calcium carbonate and dicalcium phosphate were included as mineral calcium sources. The feed may be evaluated for palatability.

A master batch horse ration formula having the following composition was prepared:

Full-Fat Corn Germ	26%
Wheat Midds	21%
Ground Corn	10%
Soybean Meal	6.5%
Distillers Dried Grain	2%
Calcium Carbonate	1%
Dried Whole Wheat	1%
Dehydrated Alfalfa	1%
Dicalcium Phosphate	1%
Odd Chain Fatty Acid	20.0%

Example 5

Use of Odd Chain Fatty Acids to Treat Mitochondrial Fat Oxidation Defects

Other examples of FODs that derive from errors in metabolism include: Carnitine Palmitoyl Transferase I (CPT I) deficiency, Carnitine-Acylcarnitine Translocase (CACT) deficiency, Carnitine Palmitoyl Transferase I (CPT II) deficiency, very long chain acyl-CoA dehydrogenase (VLCAD) deficiency, Trifunctional Protein (TFP) deficiency, long chain acyl 3-hydroxy CoA dehydrogenase (LCHAD) and short chain acyl CoA dehydrogenase (SCAD) deficiency. Patients having each of these diseases were located and placed on an odd chain fatty acid diet, with informed consent and following under IRB. These patients exhibit one or more of the following symptoms: recurrent hypoglycemia (Reye-like); cardiomyopathy: hypertrophic or dilated; hypotonia and/or delayed development; rhabdomyolysis; muscle weakness and fatigue; peripheral neuropathy; retinopathy; seizures and often sudden death due to, e.g., arrhythmias.

The odd chain fatty acid diet and therapy of the present invention was compared to conventional treatments, which

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include: (1) a low fat, high carbohydrate diet; avoid fasting; frequent or continuous feeding and night time corn starch; or (2) a medium chain triglyceride (MCT oil) for long chain disorders; or (3) insulin and glucose to inhibit lipolysis. However, current treatments fail to treat the hypoglycemia and hepatomegaly (which often persist), the frequency and severity of Rhabdomyolysis; muscle weakness and fatigue are not improved and heart function often remains abnormal. Biochemically, the failure of existing therapies leave patients with impaired Acetyl-CoA production and ketogenesis, there remains an increase Acyl-CoA:CoA (mitochondrion) ratio during illness, a decreased NADH:NAD ratio during illness (mitochondrion), hypoglycemia and hyperammonemia.

TABLE 3

Triheptanoin Patient Demographics.				
FOD	# Patients	Male/ Female	Age @ Entry	*Ethnic
CPT I	2	1/1	4 yr 9 mo-6 yr 11 mo	2 C
CACT	1	0/1	birth	1 H
CPT II	7	2/5	5 yr 6 mo-51 yrs	4 C, 3 J
VLCAD	19	9/10	Neonate-35 yrs	16 C, 1 AA, 1 ME, 1 H
LCHAD	9	4/5	6 mo-23 yrs	6 C, 2 H, 1 J
TFP	5	5/0	32 mo-8 yr 6 mo	5 C
"SCAD"	5	5/0	19 mo-6 yr 9 mo	3 C, 2 J
TOTAL:	48			

*C = caucasian,
H = hispanic,
J = jewish,
AA = afro-american,
ME = middle eastern

FIG. 9 is a graph that shows the effects of triheptanoin on patient creatine kinase activity for selected patient populations. Tri-C7 rapidly suppresses CPK levels in long-chain defects because it provides fuel to muscle via 5-carbon ketone bodies, thereby giving propionate and acetate into the CAC, with an increase in energy.

Table 4 summarized a comparison of patient mortality for patients with FOD of conventional treatment (based on the literature, see *Neonatal onset; Conventional Diet Rx: JIMD 22: 488, 1999), and the odd chain fatty acids of the present invention. There is a demonstrable increase in patient survival across a number of FODs.

TABLE 4

Patient Mortality Comparison Conventional Diet v. C7 Diet		
Disorder	Conventional Diet (41 patients - 51% withdrawal)	C7 Diet (48 patients - 6% withdrawal)
CACT*	5/5 - 100%	1/1 - 100% (rotavirus)
CPT II*	4/5 - 80%	No patients
VLCAD	6/8 - 75%	1/19 - 5%
TFP	4/4 - 100%	1/5 - 0%
LCHAD	2/10 - 20%	0/9 - 0%

The following generalized clinical observations demonstrate the effects of the odd chain fatty acids therapy on the patients. First, sudden weight gain was at first welcome and prevented by reduced carbohydrate intake. The use of emulsifiers and recipes that incorporated the odd chain fatty acids into foodstuffs reduced gastrointestinal (GI) intolerance. As regards chronic (occult) infection a vast improvement was noted and rhabdomyolysis was relieved. In a couple of cases a mild toxicity was noted and resolved by the occasional

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increase in biotin or B12 requirement and was overcome by supplementation in adults, only. Importantly, no “Propionyl Overload” was observed in any patient. Compliance was very good with the minimal withdrawal of 3 (6%) of 51 total patients that withdrew due to complaints of GI discomfort and/or weight gain. Most importantly, the vast majority of patients elected to continue beyond protocol: 22 (85%) of 26.

Regarding protein sparing, the following table summarized the results obtained in a patient with Pompe’s upon

TABLE 5

Plasma Amino Acids (μM) During Initiation of Anaplerotic Therapy: “Protein Sparing” in Adult-Onset Glycogen Storage Disease II (Pompe’s)		Event:						
		Admit	C7 = 1.0 gm/Kg		NPO*		C7 = 1.5 gm/Kg	
			Time:		Time:		Time:	
		Baseline	13 hrs	41 hrs	65 hrs	84 hrs	108 hrs	132 hrs
“Oxaloacetate”:								
ASPARTATE	(0-10)	8	44	112	15	12	50	42
“Urea Cycle”:								
ORNITHINE	(36-118)	35	78	163	59	50	122	95
CITRULLINE	(11-51)	20	1	43	21	20	35	25
ARGININE	(20-122)	50	96	93	71	84	102	97
“Neurotransmitters”:								
**PHENYLALANINE	(35-83)	62	101	156	69	51	90	84
TYROSINE	(30-100)	55	110	62	44	39	56	46
**TRYPTOPHAN	(23-86)	25	121	43	20	<10	40	40
Total Amino Acids:	(1540-4415)	1927	3673	4160	2176	2174	3568	3296

*NPO for gastrostomy - IV glucose only;

**Essential amino acids

Table 6 shows the overall Diet Composition; pCO₂, and Events during Hospitalization of a Pompe’s patient.

DIET	Hospital Day:						
(% Kcal):	0	1	4	5-6	7	23	33
Prot	29	29	25	NPO	25	25	25
CHO	30	30	27		27	24	24
Total	41	41	48		48	51	51
Lipid							
*C7	26	26	35		35	40	40
Total	1500	1500	1700		1700	1900	1900
Kcal:							
pCO ₂	68	43	51		42	48	44
(35-45)							
Event:	Admit			NPO for PEG		Extubated	Discharge

*C7 is included in the “Total Lipid” % Kcal

In conclusion, the odd chain fatty acid diet and therapy of the present invention was found to successfully treat the following FOD disorders: CPT I, CACT, CPT II, VLCAD, TFP, LCHAD and SCAD. The outcome of dietary triheptanoin supplementation included: reduced mortality, cardiomyopathies were resolved and there was a marked attenuation of Rhabdomyolysis. Also, normal glucose homeostasis was observed with the elimination of hepatomegaly. Patients showed improved muscle strength and endurance. In TFP peripheral neuropathy was unchanged and in LCHAD the retinopathy (Choroideremia) was also unchanged.

It will be understood that particular embodiments described herein are shown by way of illustration and not as

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limitations of the invention. The principal features of this invention can be employed in various embodiments without departing from the scope of the invention. Those skilled in the art will recognize, or be able to ascertain using no more than routine experimentation, numerous equivalents to the specific procedures described herein. Such equivalents are considered to be within the scope of this invention and are covered by the claims.

All publications and patent applications mentioned in the specification are indicative of the level of skill of those skilled in the art to which this invention pertains. All publications and patent applications are herein incorporated by reference to the same extent as if each individual publication or patent application was specifically and individually indicated to be incorporated by reference.

In the claims, all transitional phrases such as “comprising,” “including,” “carrying,” “having,” “containing,” “involving,” and the like are to be understood to be open-ended, i.e., to mean including but not limited to. Only the transitional phrases “consisting of” and “consisting essentially of,” respectively, shall be closed or semi-closed transitional phrases.

All of the compositions and/or methods disclosed and claimed herein can be made and executed without undue experimentation in light of the present disclosure. While the compositions and methods of this invention have been described in terms of preferred embodiments, it will be apparent to those of skill in the art that variations may be applied to the compositions and/or methods and in the steps or in the sequence of steps of the method described herein without departing from the concept, spirit and scope of the invention. More specifically, it will be apparent that certain agents which are both chemically and physiologically related may be substituted for the agents described herein while the same or similar results would be achieved. All such similar substitutes and modifications apparent to those skilled in the art are deemed to be within the spirit, scope and concept of the invention as defined by the appended claims.

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What is claimed is:

1. A composition comprising an active agent, wherein the active agent consists essentially of a triheptanoin, and wherein the composition has an acid value of 0.1 or less mg KOH/gr, a hydroxyl value of 2.8 or less mg KOH/gr.

2. The composition of claim 1, wherein the composition is provided in an amount suitable for administration in a range of 1 to 4 grams per kilogram body weight per day.

3. The composition of claim 1, wherein the composition is provided in an amount suitable for administration in a range of 1 to 3 grams per kilogram body weight per day.

4. The composition of claim 1, wherein the composition is provided in an amount suitable for administration in a range of 1 to 2 grams per kilogram body weight per day.

5. The composition of claim 1, wherein the composition is provided in an amount suitable for administration in a range of 0.1 to 2 grams per kilogram body weight per day.

6. The composition of claim 1, wherein the composition is in a dosage unit, wherein the dosage unit comprises at least 15 grams of the triheptanoin.

7. The composition of claim 6, wherein the dosage unit comprises 15 to 75 grams of the triheptanoin.

8. The composition of claim 6, wherein the dosage unit comprises 75 grams of the triheptanoin.

9. The composition of claim 1, wherein the composition is provided in an amount suitable for providing 15% to 40% of the daily dietary calories of a human.

10. The composition of claim 1, wherein the composition is provided in an amount suitable for providing 20% to 35% of the daily dietary calories of a human.

11. The composition of claim 1, wherein the composition is provided in an amount suitable for providing 25% to 35% of the daily dietary calories of a human.

12. The composition of claim 1, wherein the composition is provided in an amount suitable for providing 30% of the daily dietary calories of a human.

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13. The composition of claim 1, wherein the composition is provided in an amount suitable for providing 25% of the daily dietary calories of a human.

14. The composition of claim 1, wherein the composition is a nutritional supplement.

15. The composition of claim 1, wherein the composition is a dietary formulation.

16. The composition of claim 1, wherein the composition is a pharmaceutical composition.

17. The composition of claim 1, wherein the composition is administered parenterally.

18. The composition of claim 1, wherein the composition is administered enterally.

19. The composition of claim 1, wherein the composition is administered orally.

20. The composition of claim 1, wherein the composition is provided in an amount suitable for administration in a range of about 1 gram per kilogram body weight per day for an adult for at least 7 or 14 days.

21. The composition of claim 1, wherein the composition is provided in an amount suitable for administration in a range of about 2 to 4 grams per kilogram body weight per day for at least 7 or 14 days.

22. The composition of claim 1, wherein the composition is provided in an amount suitable for administration in a range of about 1 to 2 grams per kilogram body weight per day for at least 7 or 14 days.

23. The composition of claim 1, wherein the dosage unit is at least 25% of the daily fatty acid dose for an adult.

24. The composition of claim 1, wherein the composition is provided in an amount suitable for administration in a range of about 1 to 4 grams per kilogram body weight per day for infants or 0.1 to 2 grams per kilogram body weight per day for adults.

25. The composition of claim 1, wherein the composition does not contain long-chain or very long-chain fatty acid.

* * * * *

UNITED STATES PATENT AND TRADEMARK OFFICE

(12) CERTIFICATE EXTENDING PATENT TERM
UNDER 35 U.S.C. § 156

(68) PATENT NO. : 8,697,748
(45) ISSUED : April 15, 2014
(75) INVENTOR : Charles R. Roe
(73) PATENT OWNER : Baylor Research Institute
(95) PRODUCT : DOJOLVI® (triheptanoin)

This is to certify that an application under 35 U.S.C. § 156 has been filed in the United States Patent and Trademark Office, requesting extension of the term of U.S. Patent No. 8,697,748 based upon the regulatory review of the product DOJOLVI® (triheptanoin) by the Food and Drug Administration. According to United States Patent and Trademark Office records, the original expiration date of the patent as of the date of issuance of this certificate is October 3, 2025. Because it appears that the requirements of the law have been met, this certificate extends the term of the patent for the period of

(94) 1,303 days

subject to the payment of maintenance fees as provided by law, with all rights pertaining thereto as provided by 35 U.S.C. § 156.

I have caused the seal of the United States Patent and Trademark Office to be affixed this 17th day of October 2023.



Kathi Vidal

Katherine K. Vidal
Under Secretary of Commerce for Intellectual Property and
Director of the United States Patent and Trademark Office

EXHIBIT B

(12) **United States Patent**
Klopp et al.

(10) **Patent No.:** **US 12,551,461 B2**
(45) **Date of Patent:** ***Feb. 17, 2026**

(54) **COMPOSITIONS OF TRIGLYCERIDES AND USES THEREOF**

(71) Applicant: **Ultragenyx Pharmaceutical Inc.,**
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(72) Inventors: **John Klopp,** Novato, CA (US);
Gabrielle Morris, Novato, CA (US);
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(73) Assignee: **Ultragenyx Pharmaceutical Inc.,**
Novato, CA (US)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

This patent is subject to a terminal disclaimer.

(21) Appl. No.: **18/167,551**

(22) Filed: **Feb. 10, 2023**

(65) **Prior Publication Data**

US 2023/0181513 A1 Jun. 15, 2023

Related U.S. Application Data

(63) Continuation of application No. 16/445,557, filed on Jun. 19, 2019, now abandoned, which is a continuation of application No. 14/419,601, filed as application No. PCT/US2014/065693 on Nov. 14, 2014, now abandoned.

(Continued)

(51) **Int. Cl.**
A61K 31/23 (2006.01)
A61K 9/14 (2006.01)
A61K 9/16 (2006.01)
A61K 31/20 (2006.01)

(52) **U.S. Cl.**

CPC **A61K 31/23** (2013.01); **A61K 9/143** (2013.01); **A61K 9/146** (2013.01); **A61K 9/1611** (2013.01); **A61K 9/1652** (2013.01); **A61K 31/20** (2013.01)

(58) **Field of Classification Search**

None
See application file for complete search history.

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Primary Examiner — Isis A Ghali

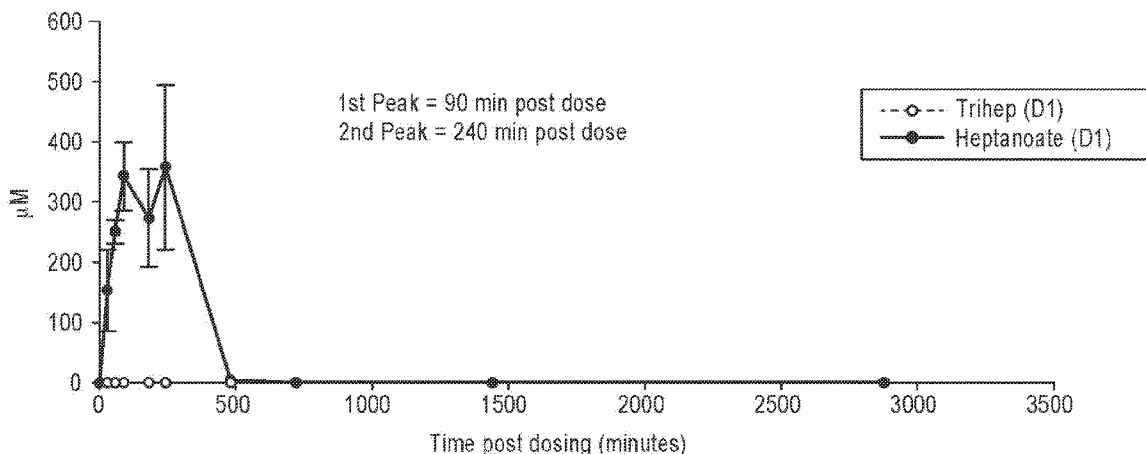
(74) *Attorney, Agent, or Firm* — COOLEY LLP

(57) **ABSTRACT**

The present invention includes solid compositions of triglycerides with one or more fatty acids, such as triheptanoin and glycerol phenylbutyrate, and therapeutic use thereof. The solid compositions can be prepared by spray-drying or other processes.

5 Claims, 40 Drawing Sheets

Triheptanoin and Heptanoate (C7) Metabolite PK Following Triheptanoin Oil Dosing on Day 1 in Male and Female Mini-pigs Combined



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Related U.S. Application Data

- (60) Provisional application No. 61/904,369, filed on Nov. 14, 2013.

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Triheptanoin and Heptanoate (C7) Metabolite PK Following Triheptanoin Oil
Dosing on Day 1 in Male and Female Mini-pigs Combined

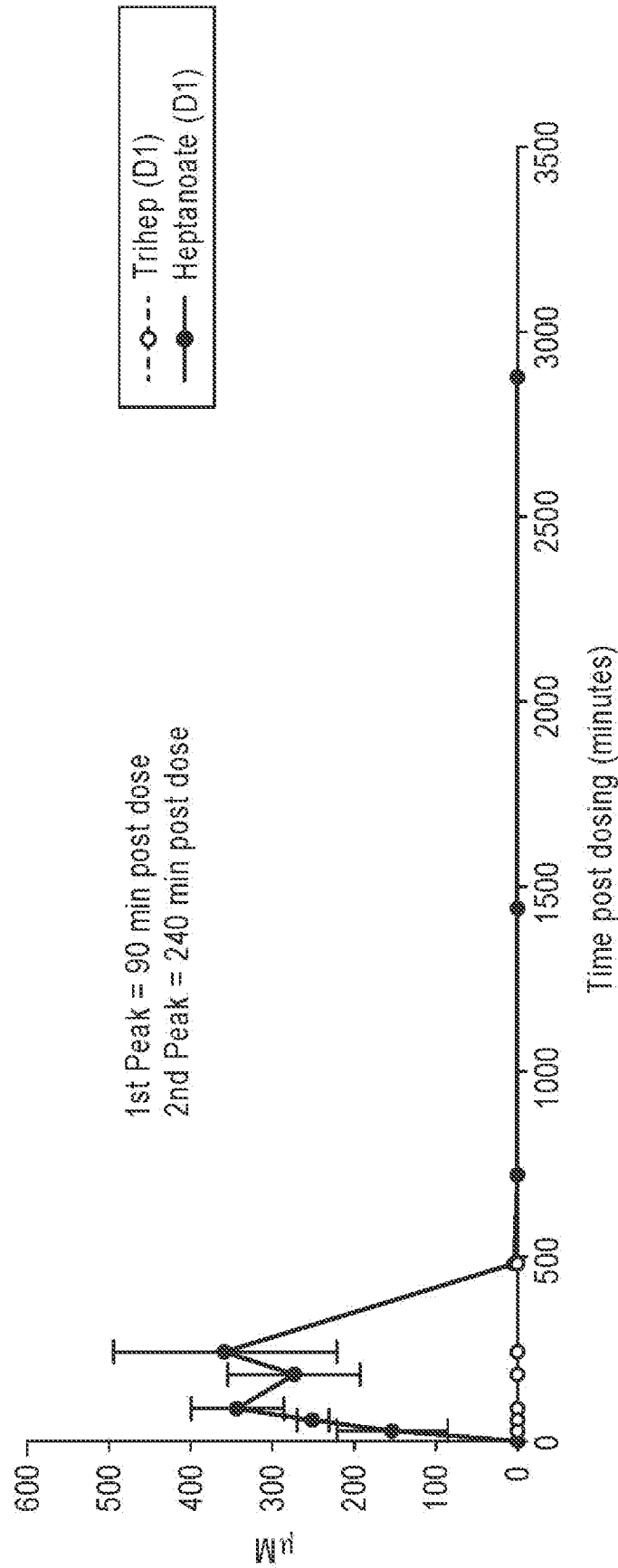


Figure 1

Triheptanoic acid and Heptanoate (C7) Metabolite PK Following Triheptanoic acid Powder-ER
Dosing on Day 1 in Male and Female Mini-pigs Combined

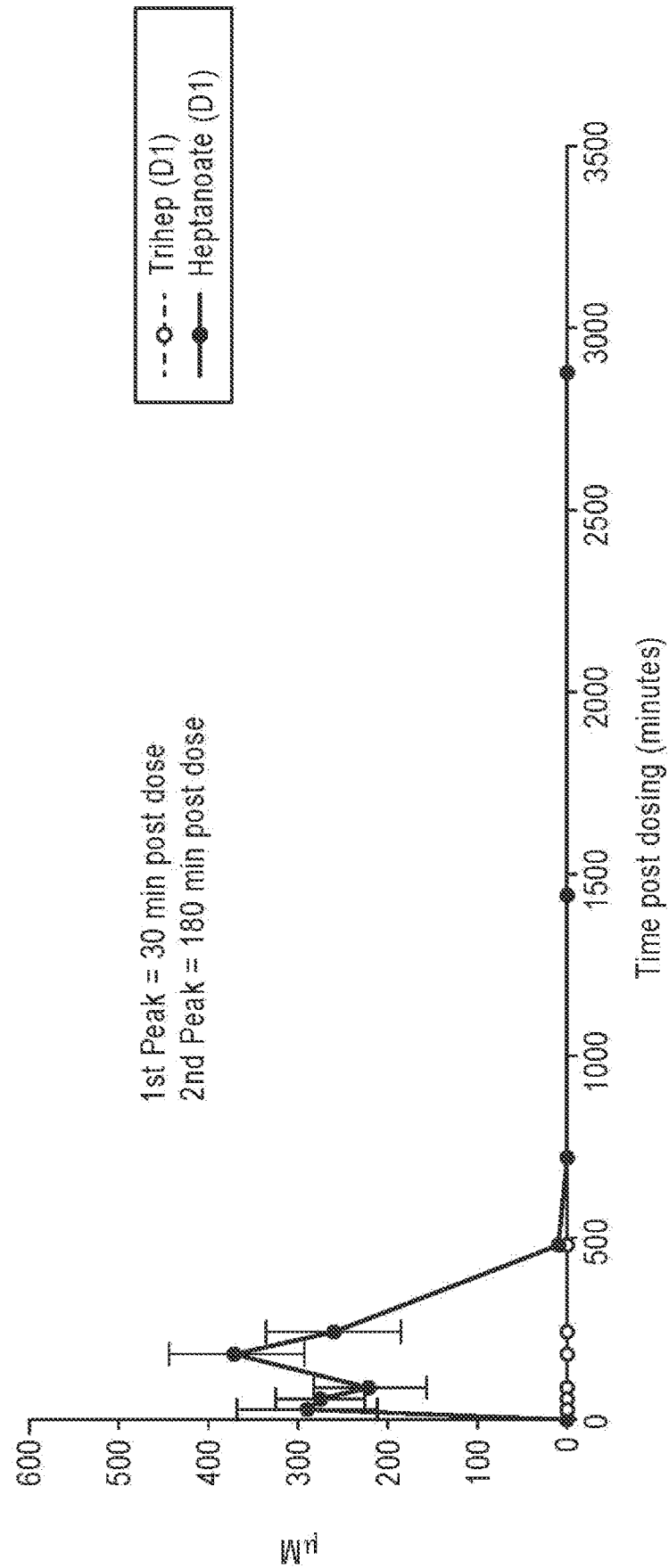


Figure 2

Triheptanoin and Heptanoate (C7) Metabolite PK Following Triheptanoin Powder
Dosing on Day 1 in Male and Female Mini-pigs Combined

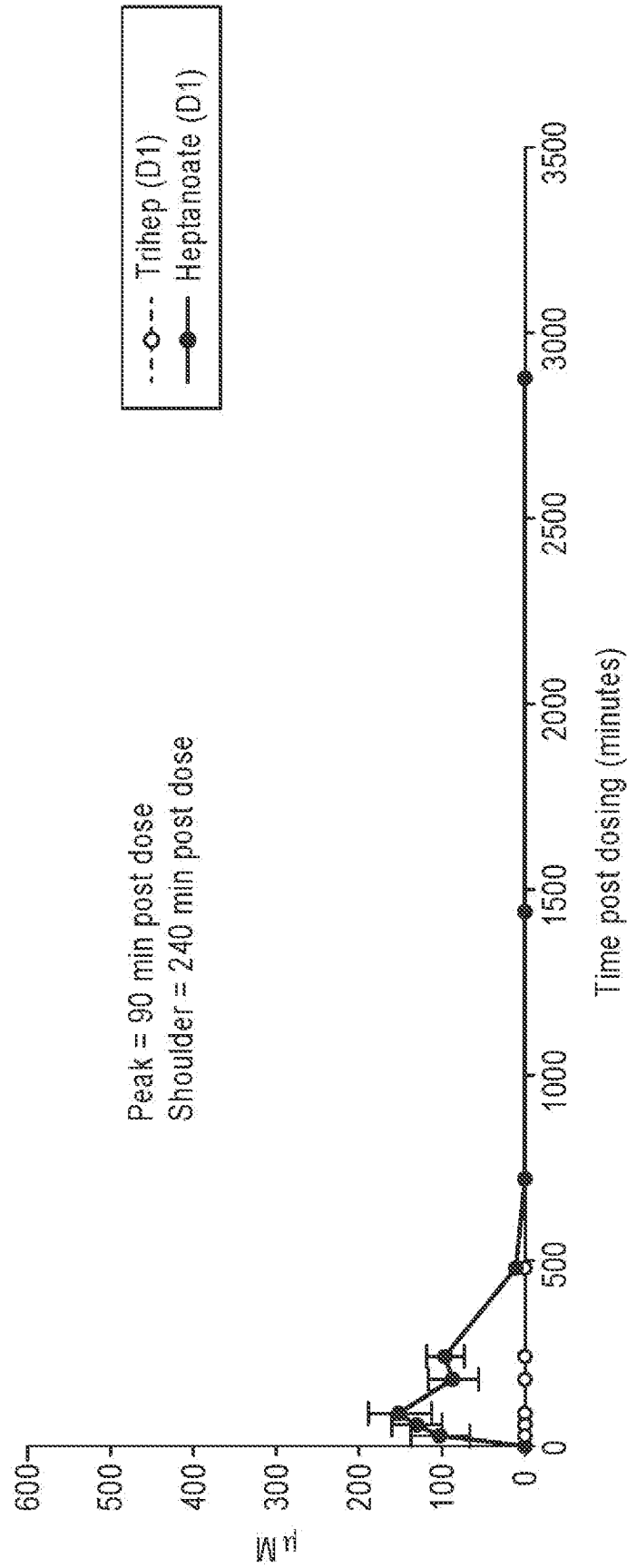
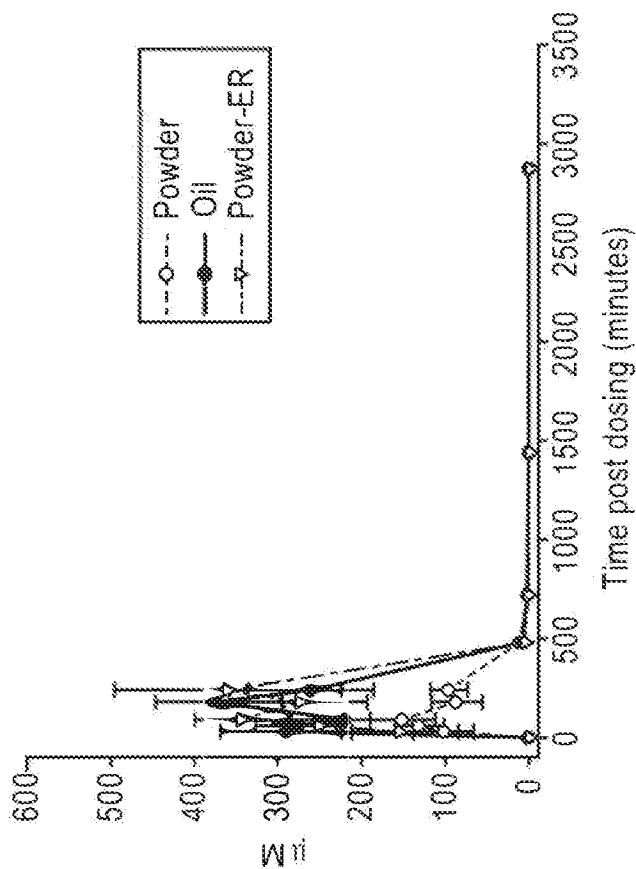


Figure 3

PK up to 48hr timepoint



PK up to 8hr timepoint

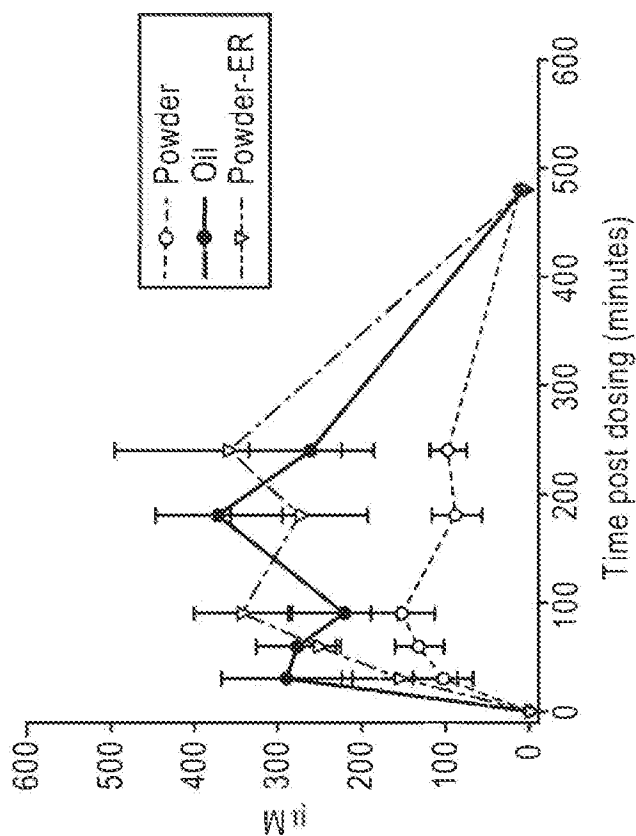


Figure 4

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B-Hydroxybutyric Acid (C4-OH) and B-Hydroxyvaleric Acid (C5-OH) Metabolite PK
Following Triheptanoin Powder-ER Dosing on Day 1 in Male and Female Mini-pigs
Combined

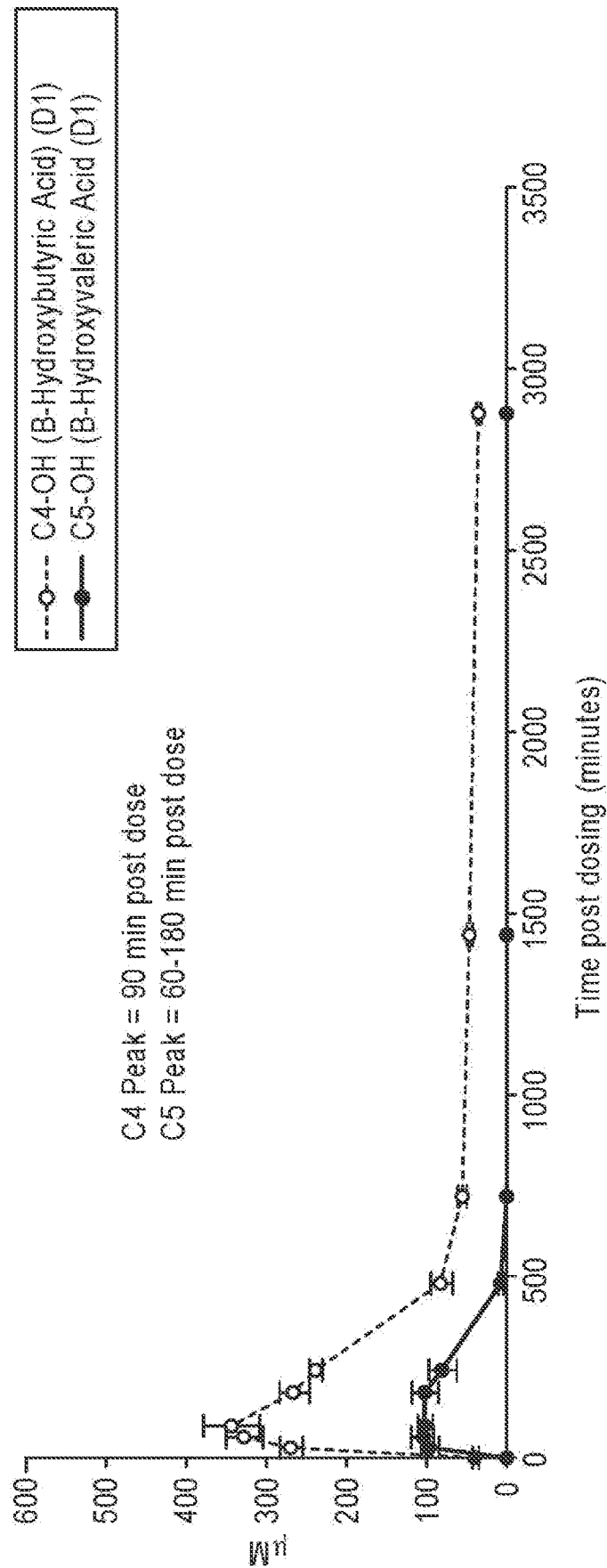


Figure 5

B-Hydroxybutyric Acid (C4-OH) and B-Hydroxyvaleric Acid (C5-OH) Metabolite PK Following Triheptanoin Powder Oil Dosing on Day 1 in Male and Female Mini-pigs Combined

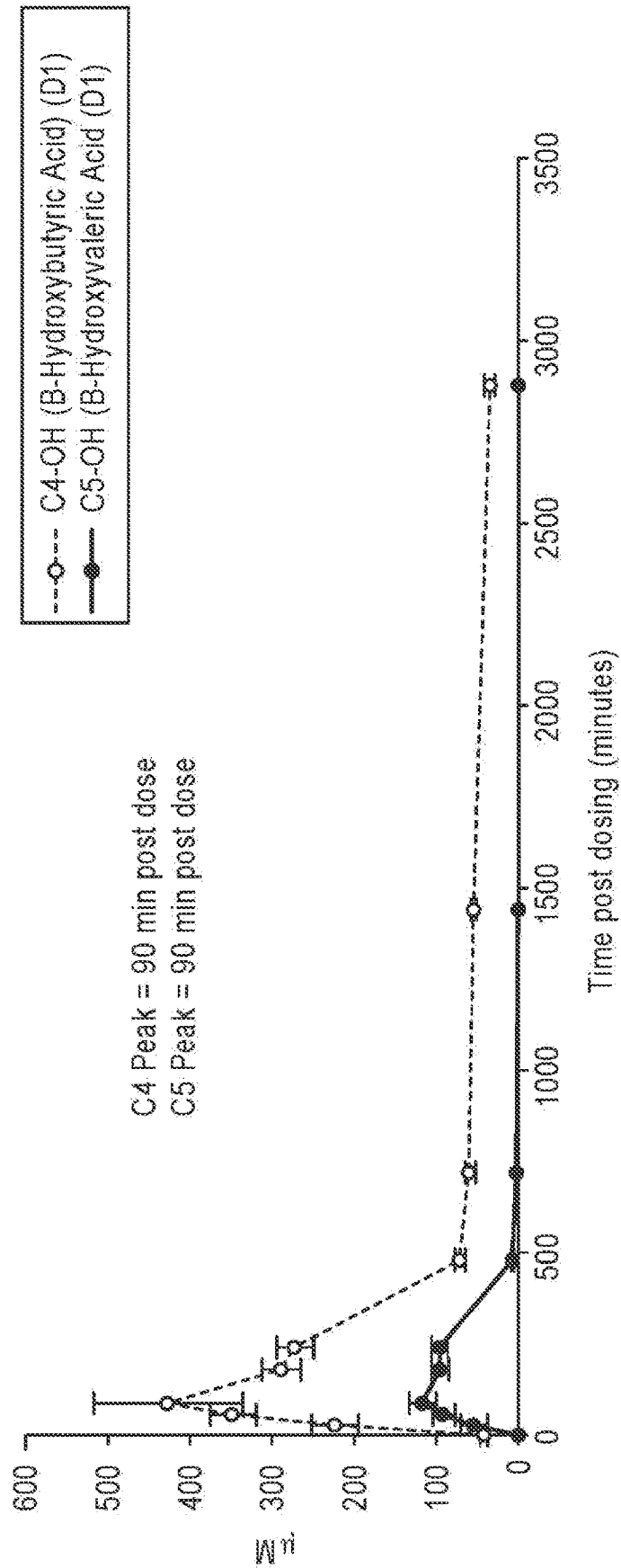


Figure 6

B-Hydroxybutyric Acid (C4-OH) and B-Hydroxyvaleric Acid (C5-OH) Metabolite PK Following Triheptanoin Powder Dosing on Day 1 in Male and Female Mini-pigs Combined

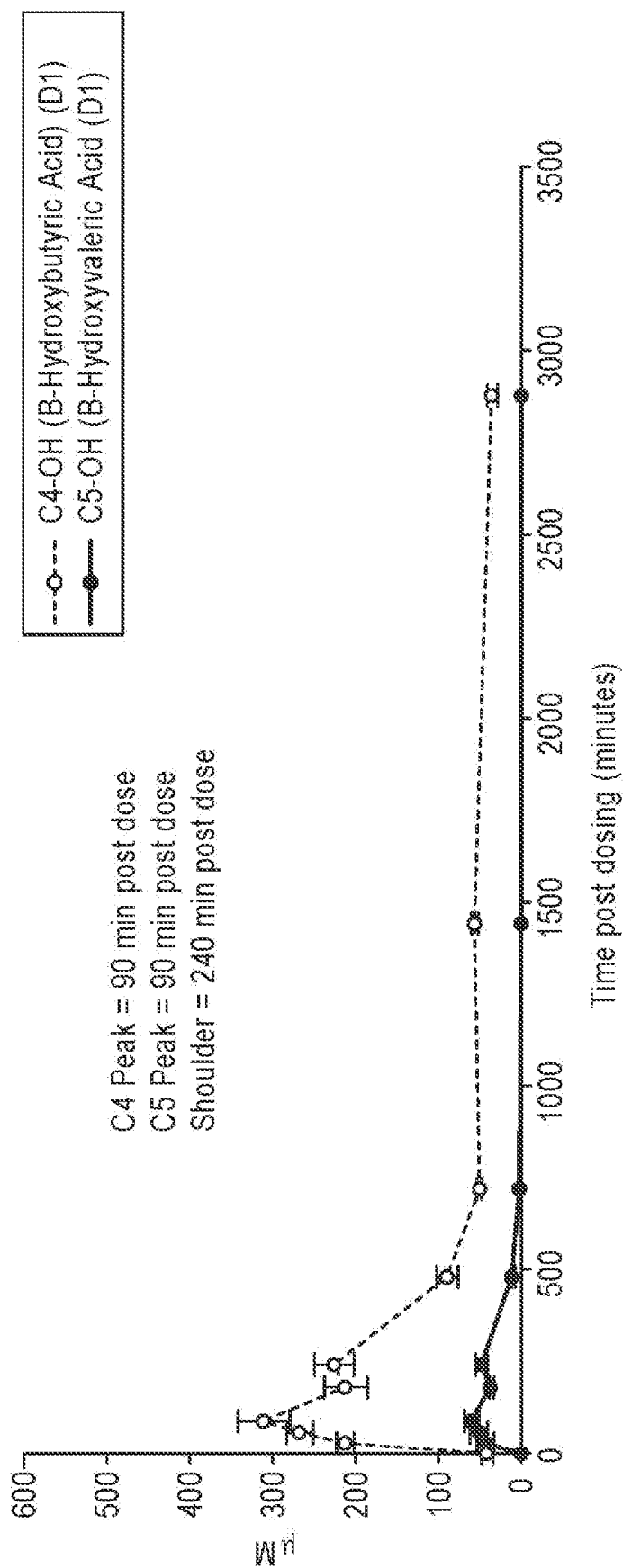


Figure 7

PK up to 48hr timepoint

PK up to 8hr timepoint

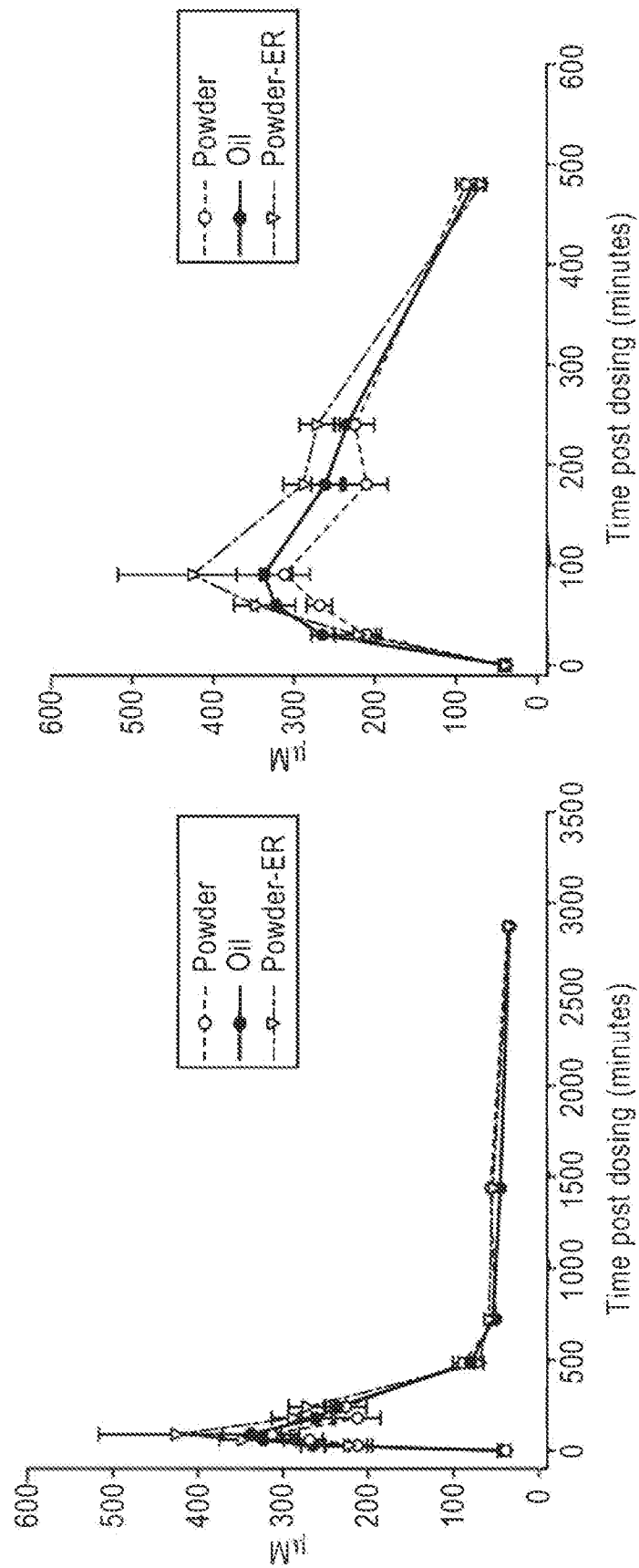
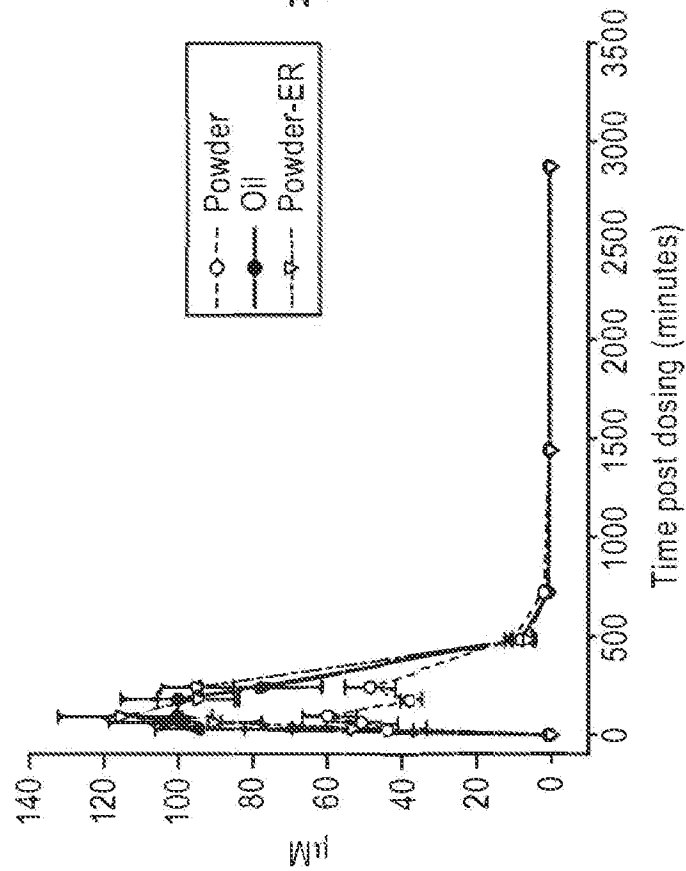


Figure 8

PK up to 48hr timepoint



PK up to 8hr timepoint

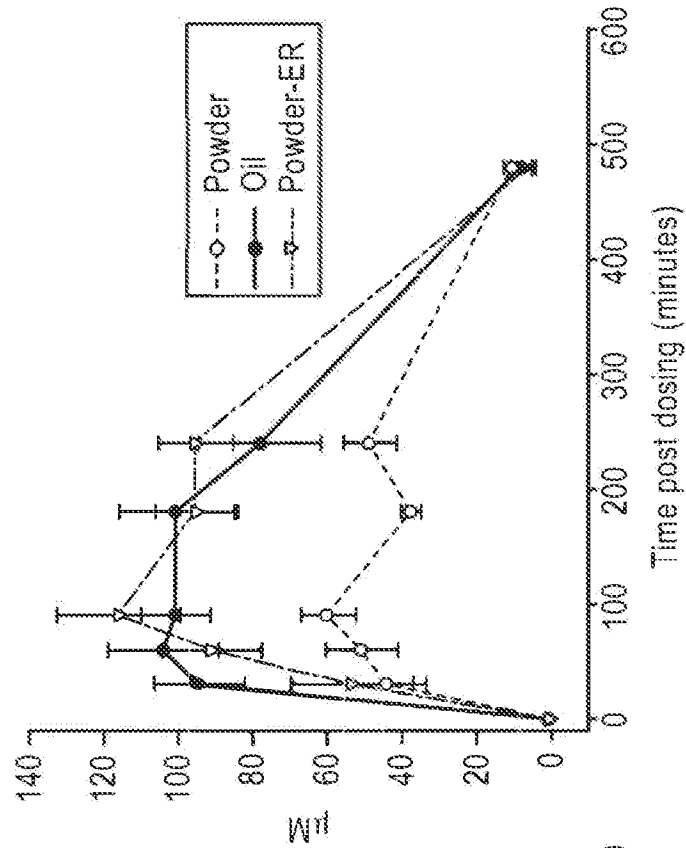


Figure 9

Levels of Aspartate Following Triheptanoin Powder-ER Dosing on Day 1 in Male and Female Mini-pigs Combined

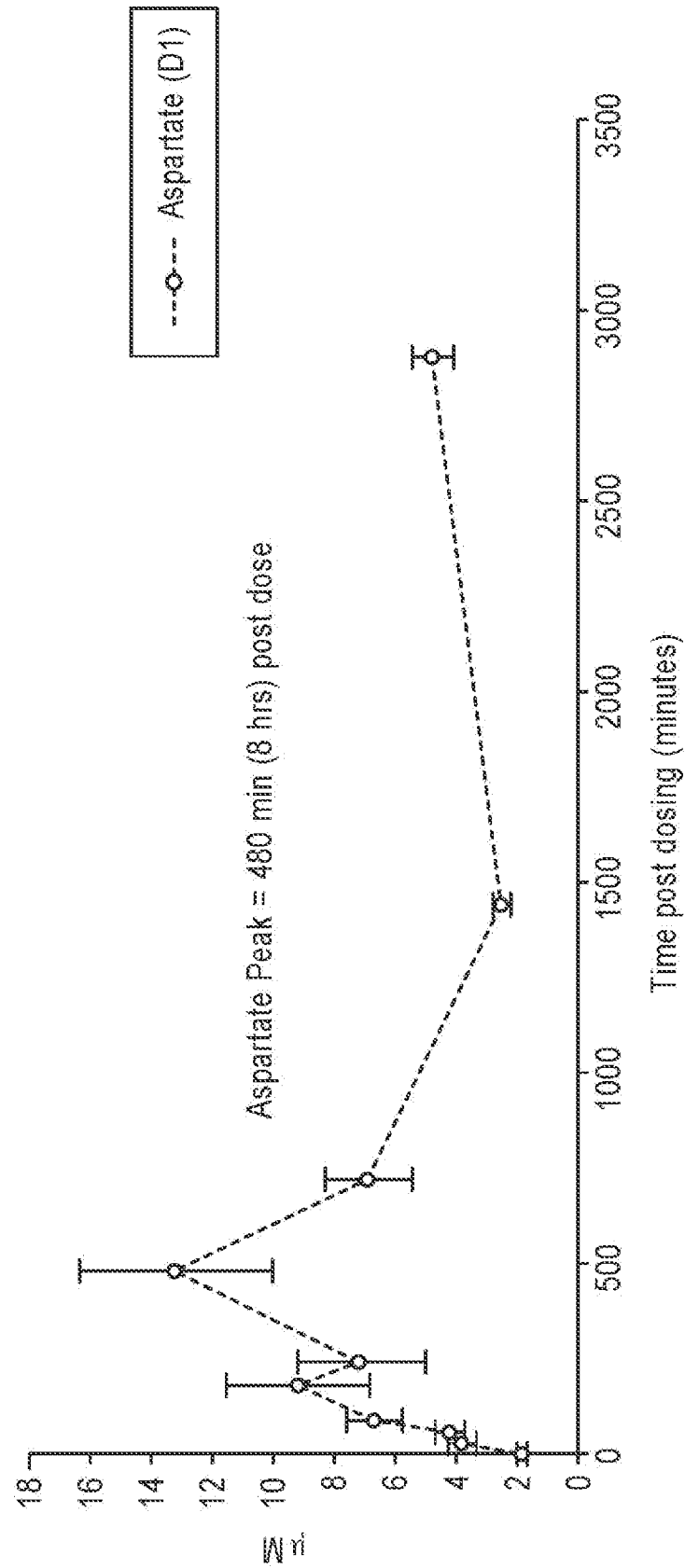


Figure 10

Levels of Aspartate Following Triheptanoic Oil Dosing on Day 1 in Male and Female Mini-pigs Combined

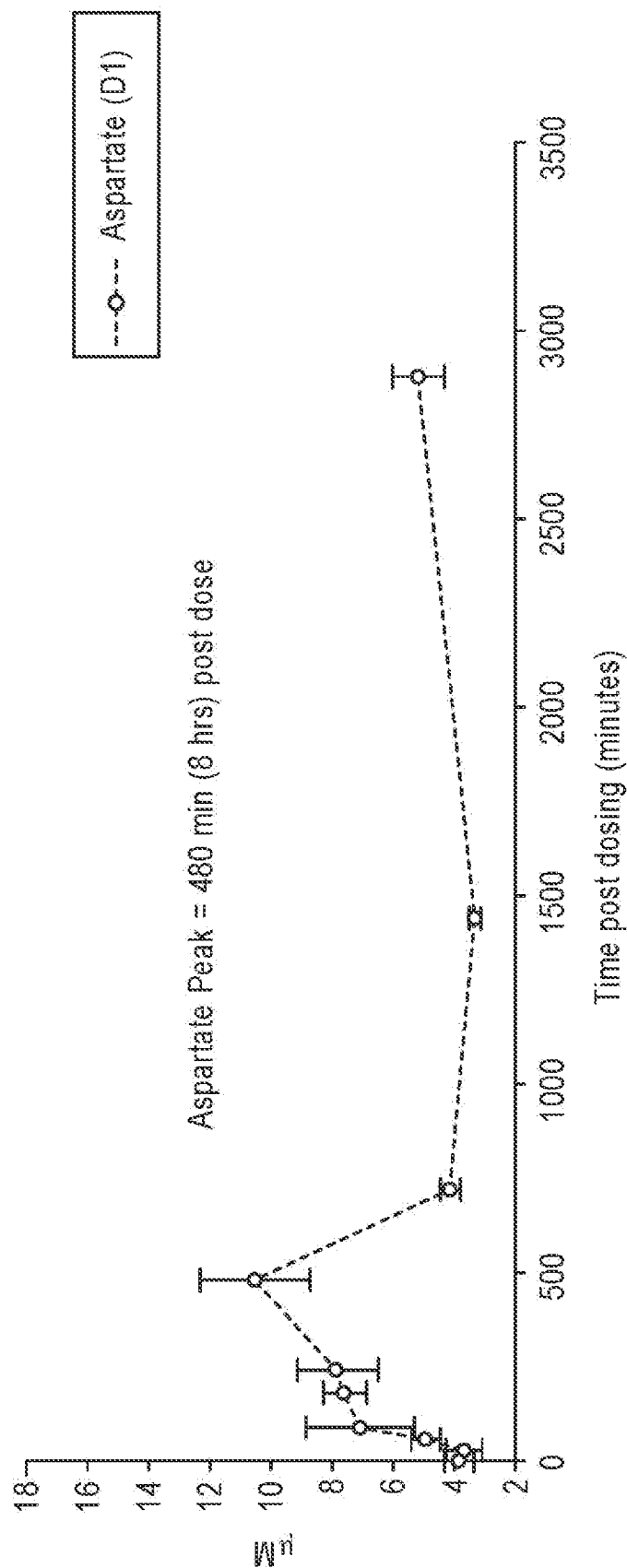


Figure 11

Levels of Aspartate Following Triheptanoic Powder Dosing on Day 1 in Male
and Female Mini-pigs Combined

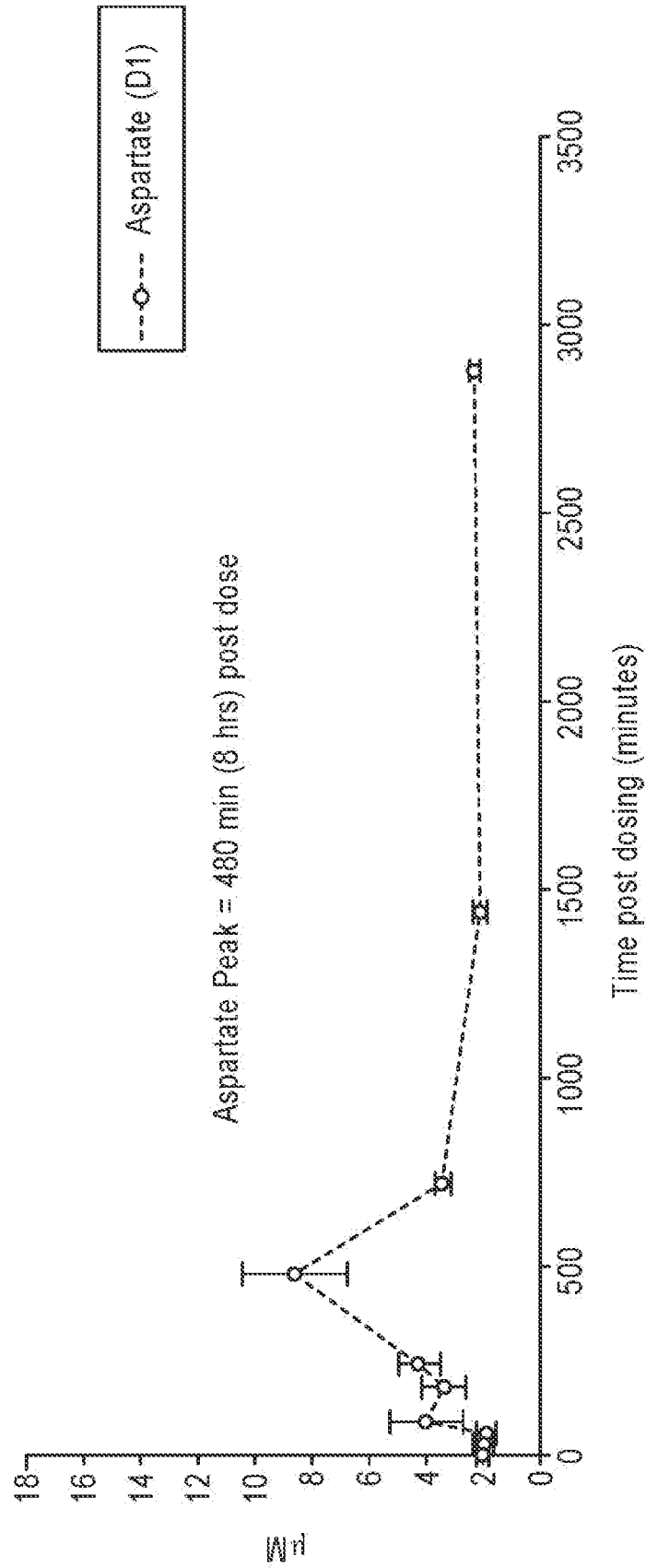


Figure 12

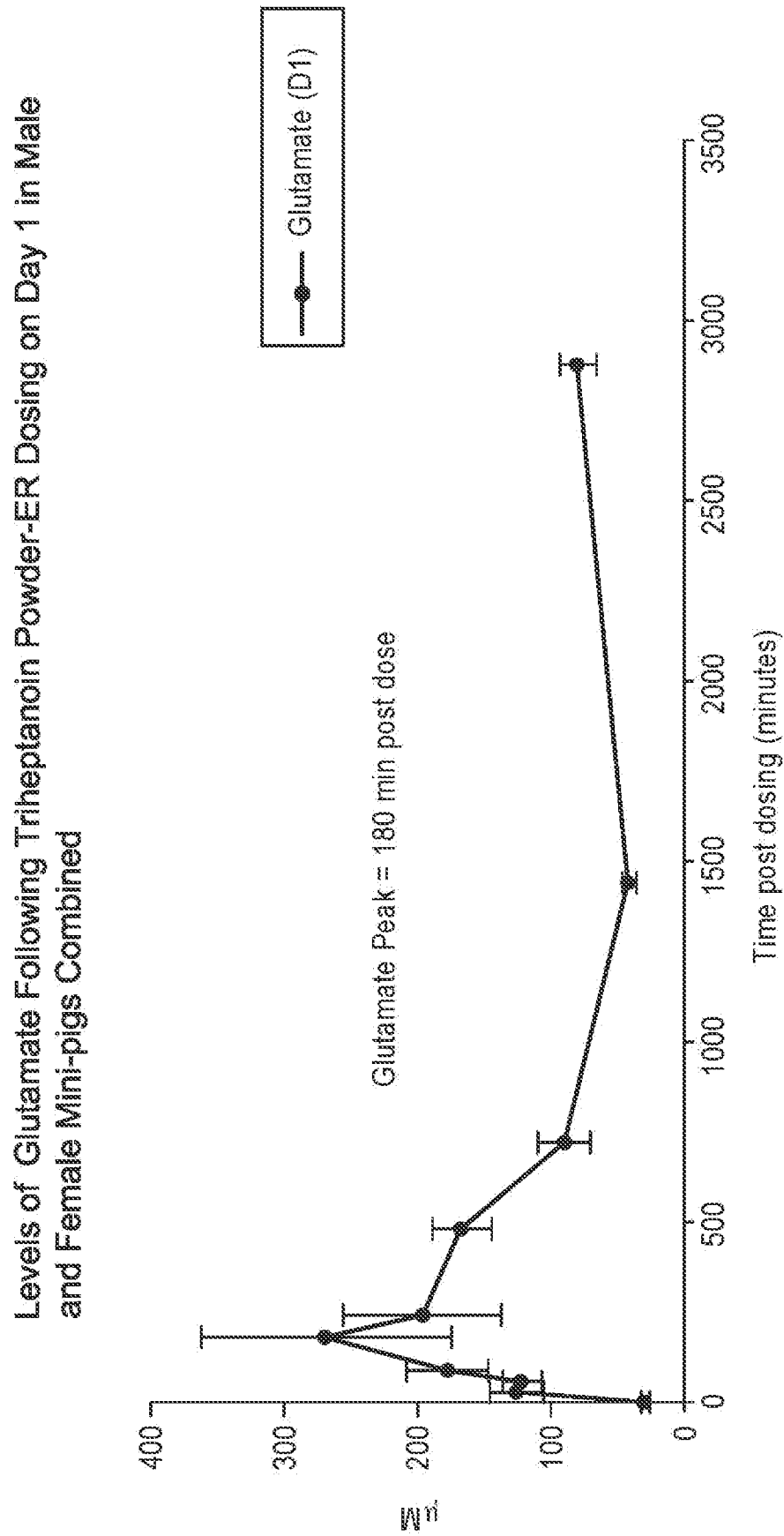


Figure 13

Levels of Glutamate Following Triheptanoin Oil Dosing on Day 1 in Male
and Female Mini-pigs Combined

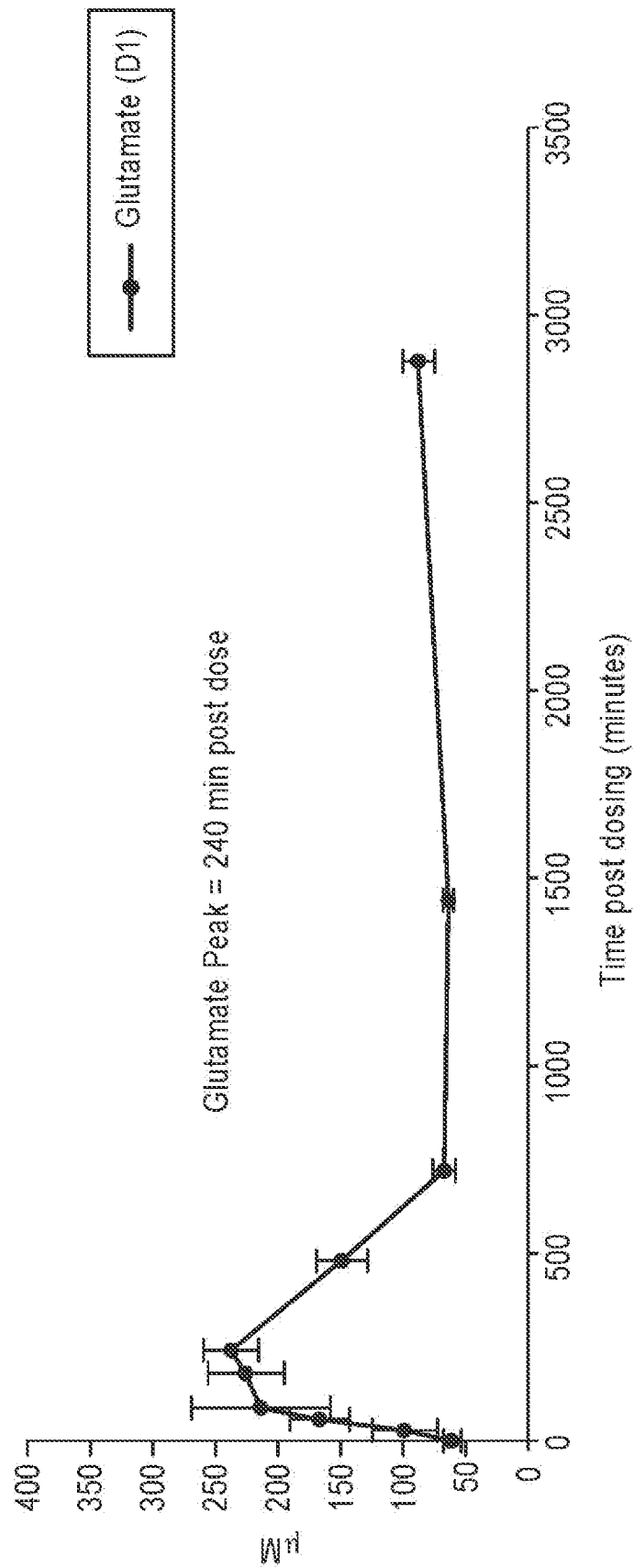


Figure 14

Levels of Glutamate Following Triheptanoin Powder Dosing on Day 1 in Male and Female Mini-pigs Combined

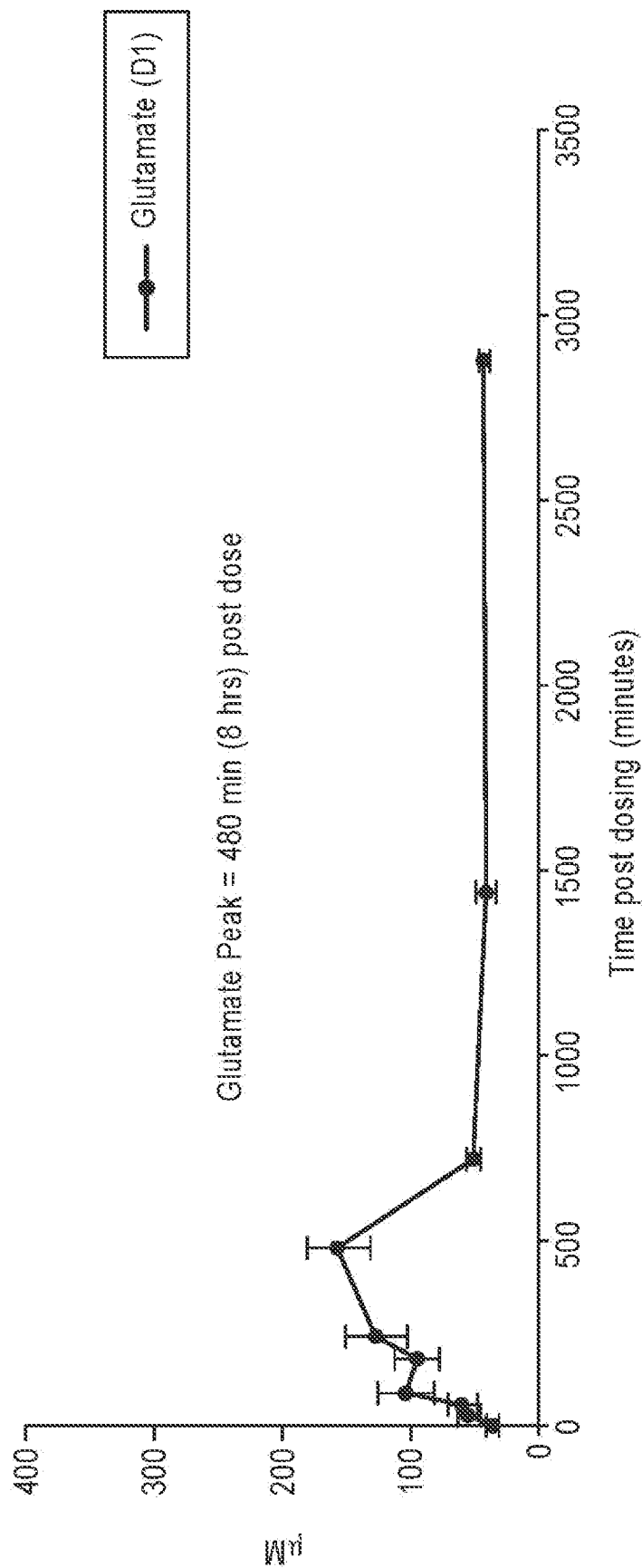


Figure 15

Pimelic Acid, 3-Hydroxypropionate and Propionyl Glycine Metabolite PK Following Triheptanoic Powder-ER Dosing on Day 1 in Male and Female Mini-pigs Combined

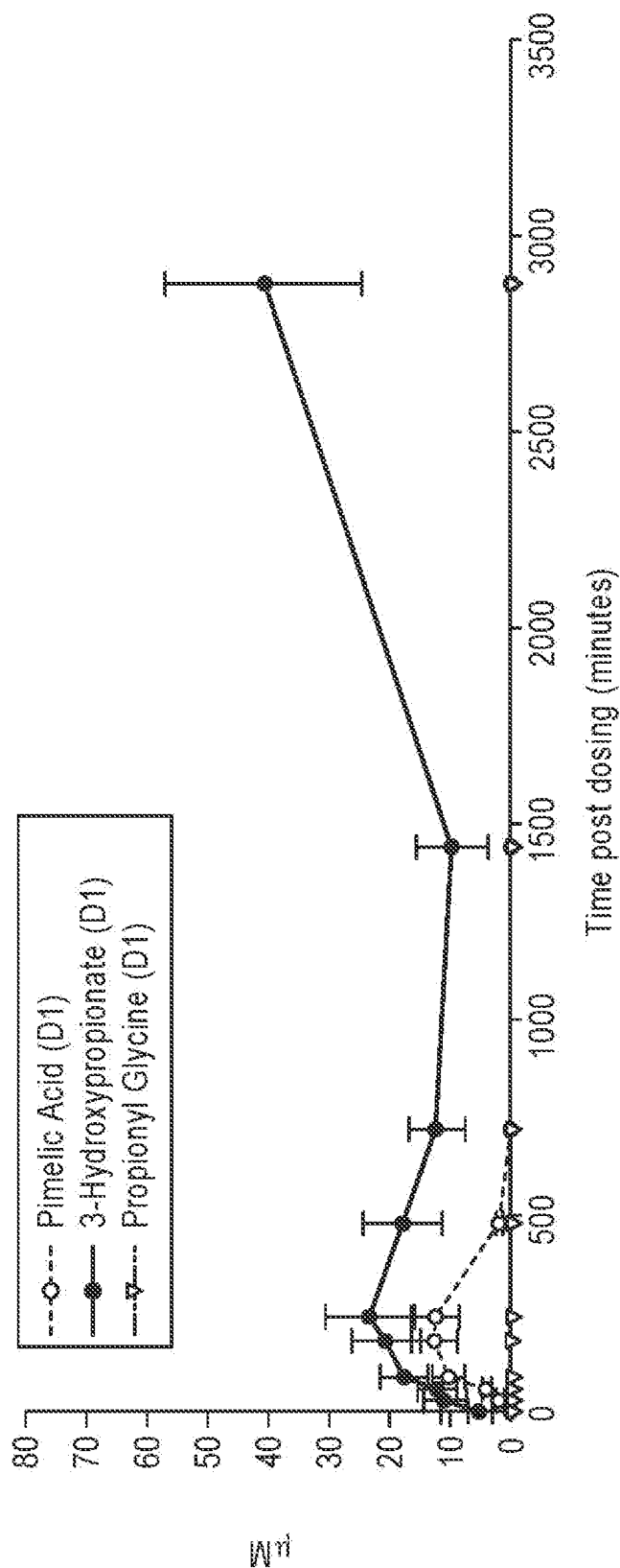


Figure 16

Pimelic Acid, 3-Hydroxypropionate and Propionyl Glycine Metabolite PK Following
Triheptanoin Oil Dosing on Day 1 in Male and Female Mini-pigs Combined

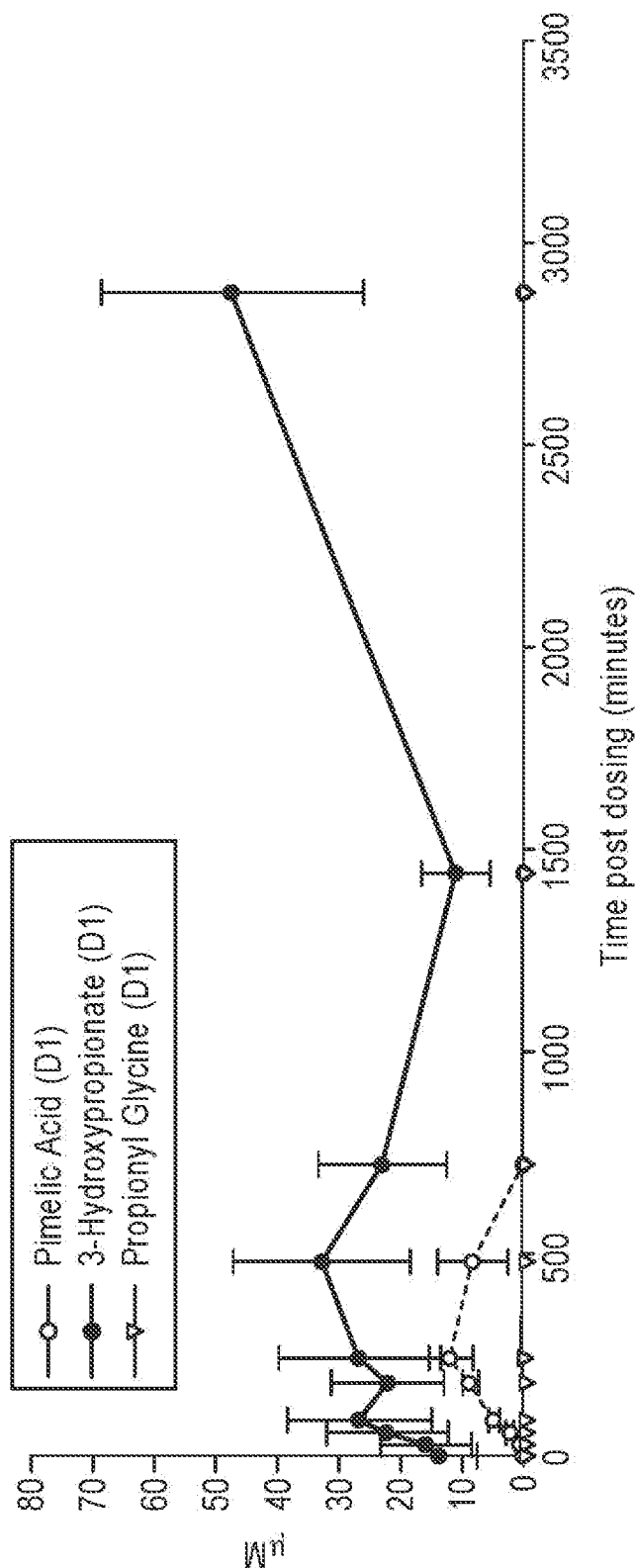


Figure 17

Pimelic Acid, 3-Hydroxypropionate and Propionyl Glycine Metabolite PK Following Triheptanoin Powder Dosing on Day 1 in Male and Female Mini-pigs Combined

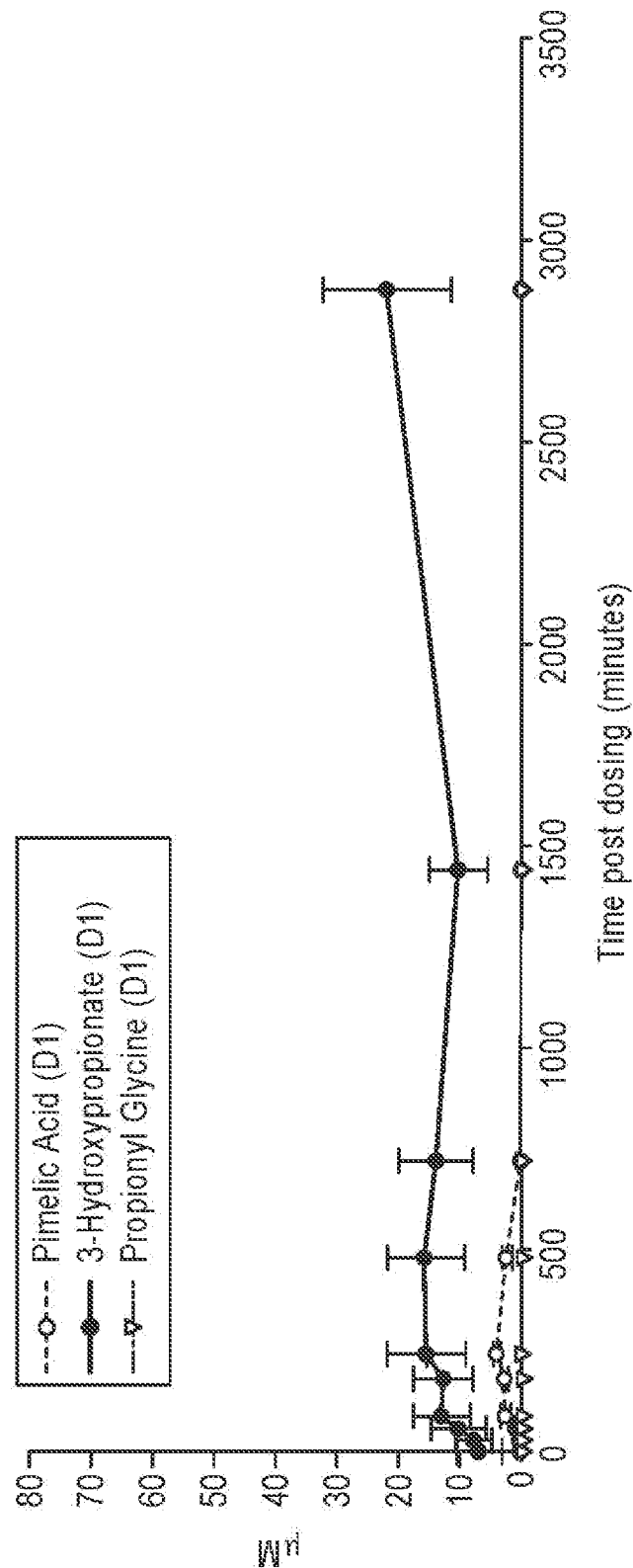


Figure 18

Average Energy and Alternative Metabolites at Each Timepoint Following Triheptanoic Powder-ER Dosing on Day 1 in Male and Female Mini-pigs Combined

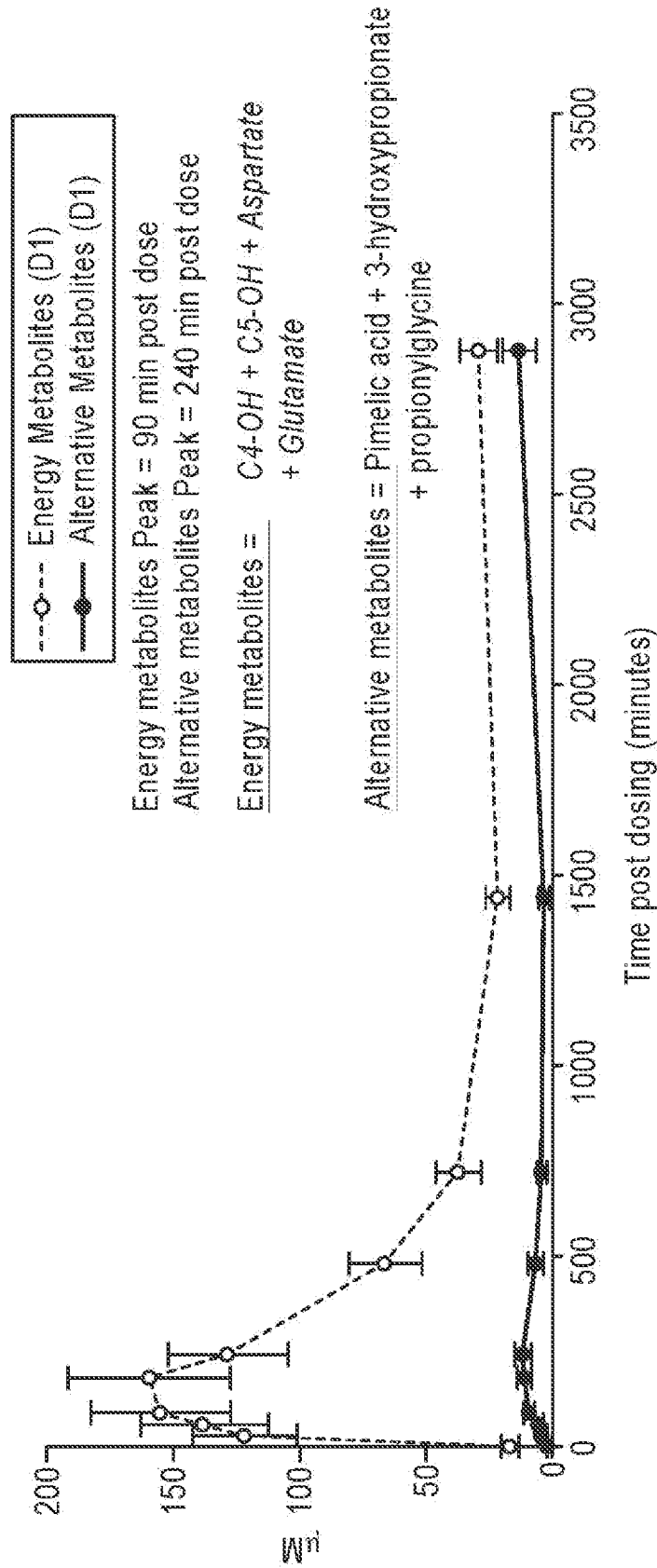


Figure 19

**Average Energy and Alternative Metabolites at Each Timepoint Following
Triheptanoil Oil Dosing on Day 1 in Male and Female Mini-pigs Combined**

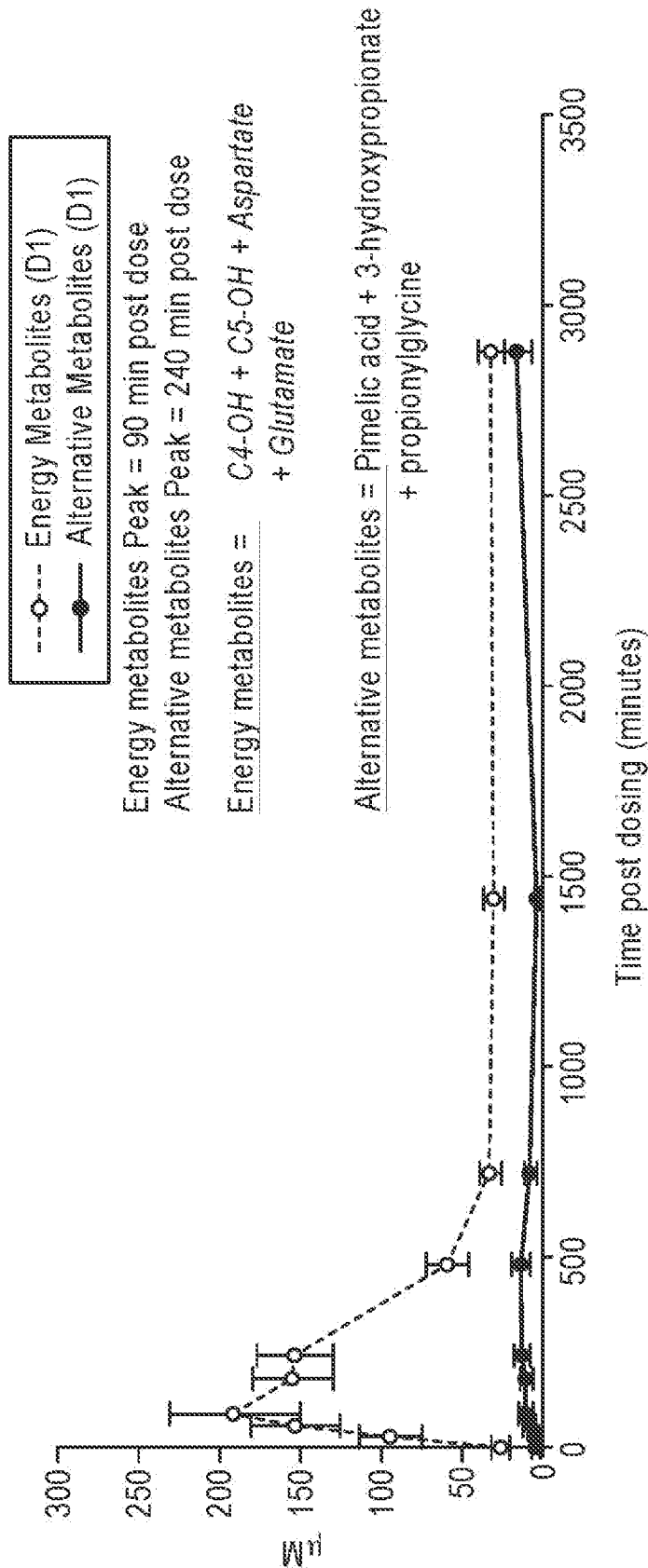


Figure 20

Average Energy and Alternative Metabolites at Each Timepoint Following Triheptanoic Powder Dosing on Day 1 in Male and Female Mini-pigs Combined

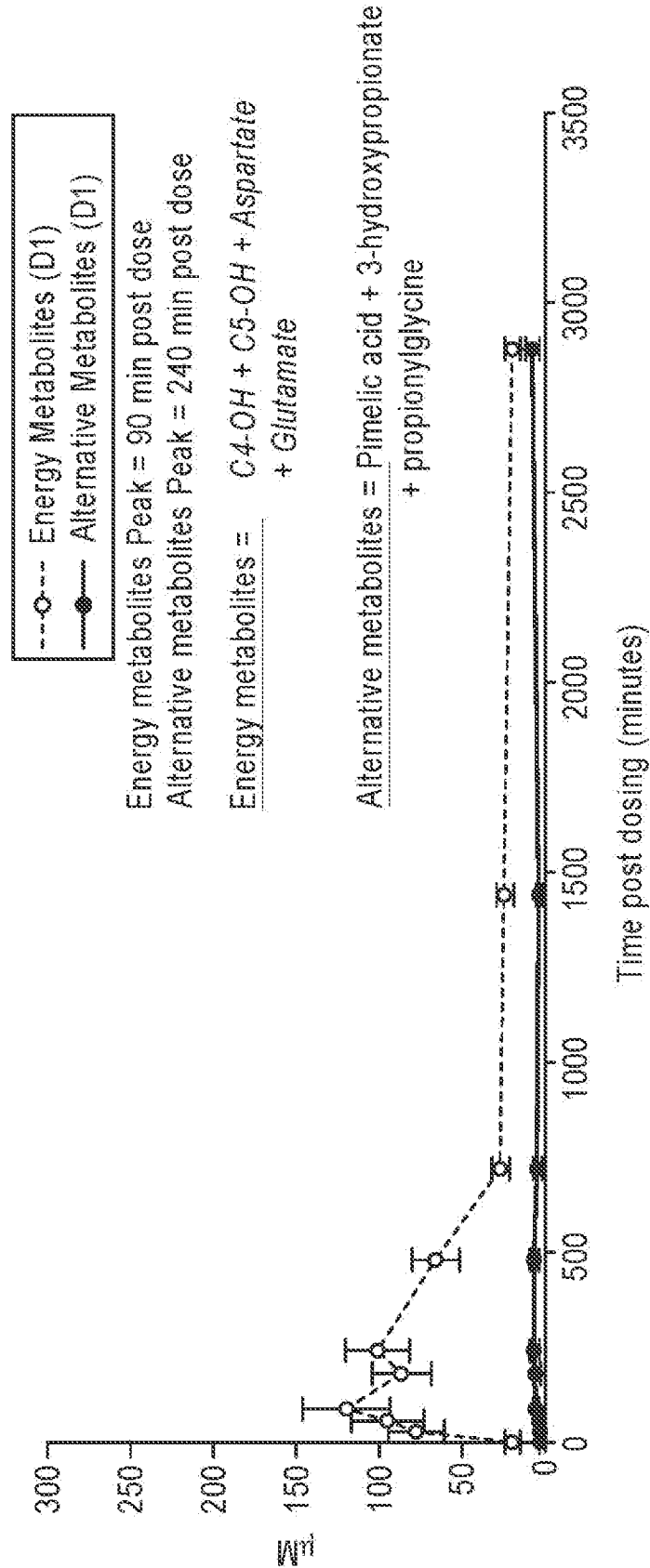


Figure 21

Heptanoic Acid (C7) Metabolite PK Following Triheptanoic Oil, Powder and Powder-ER dosing on Day 1 and Day 7 in Male/Female Combined Mini-pigs

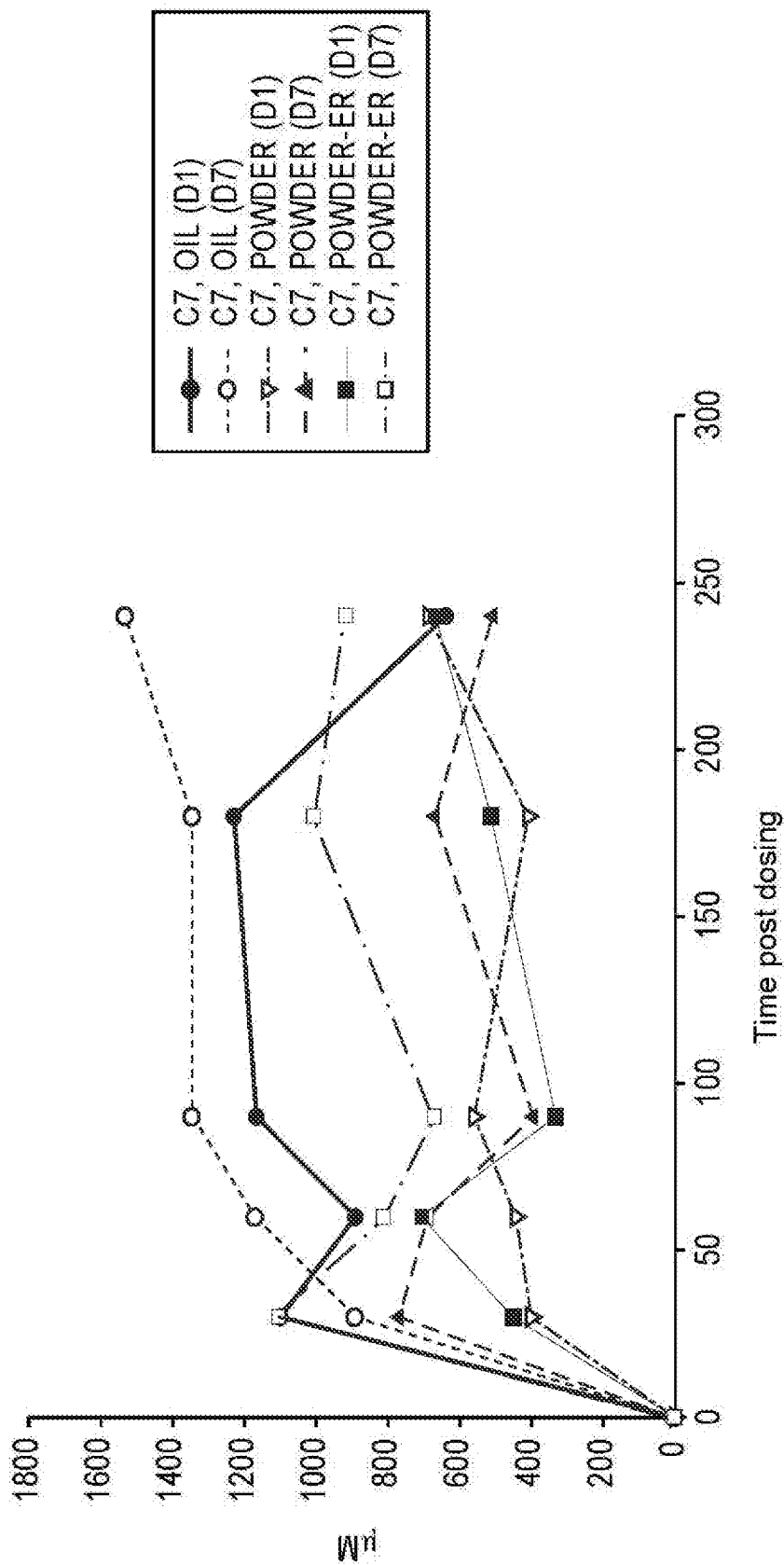


Figure 22

C4 Metabolite PK Following Triheptanoin Oil, Powder and Powder-ER dosing on Day 1 and Day 7 in Male/Female Combined Mini-pigs

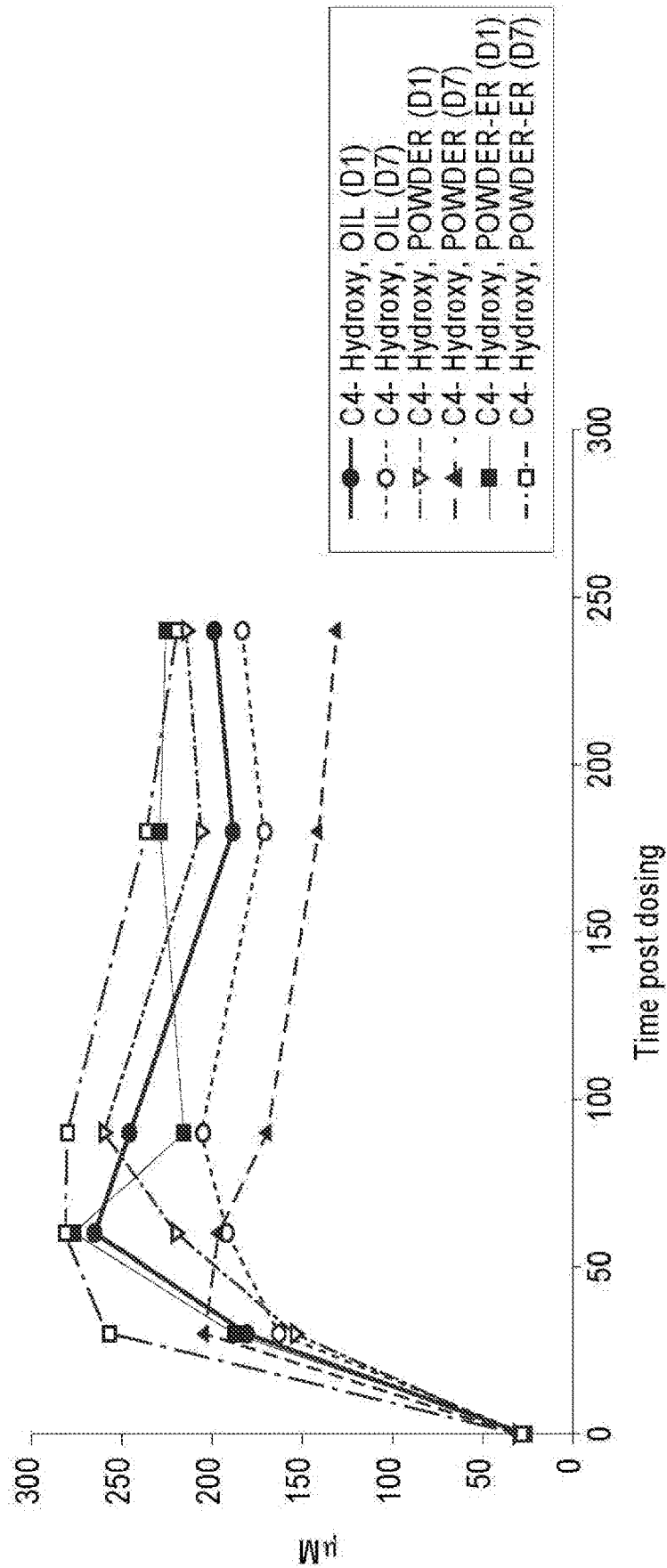


Figure 23

C5 Metabolite PK Following Triheptanoin Oil, Powder and Powder-ER dosing on Day 1 and Day 7 in Male/Female Combined Mini-pigs

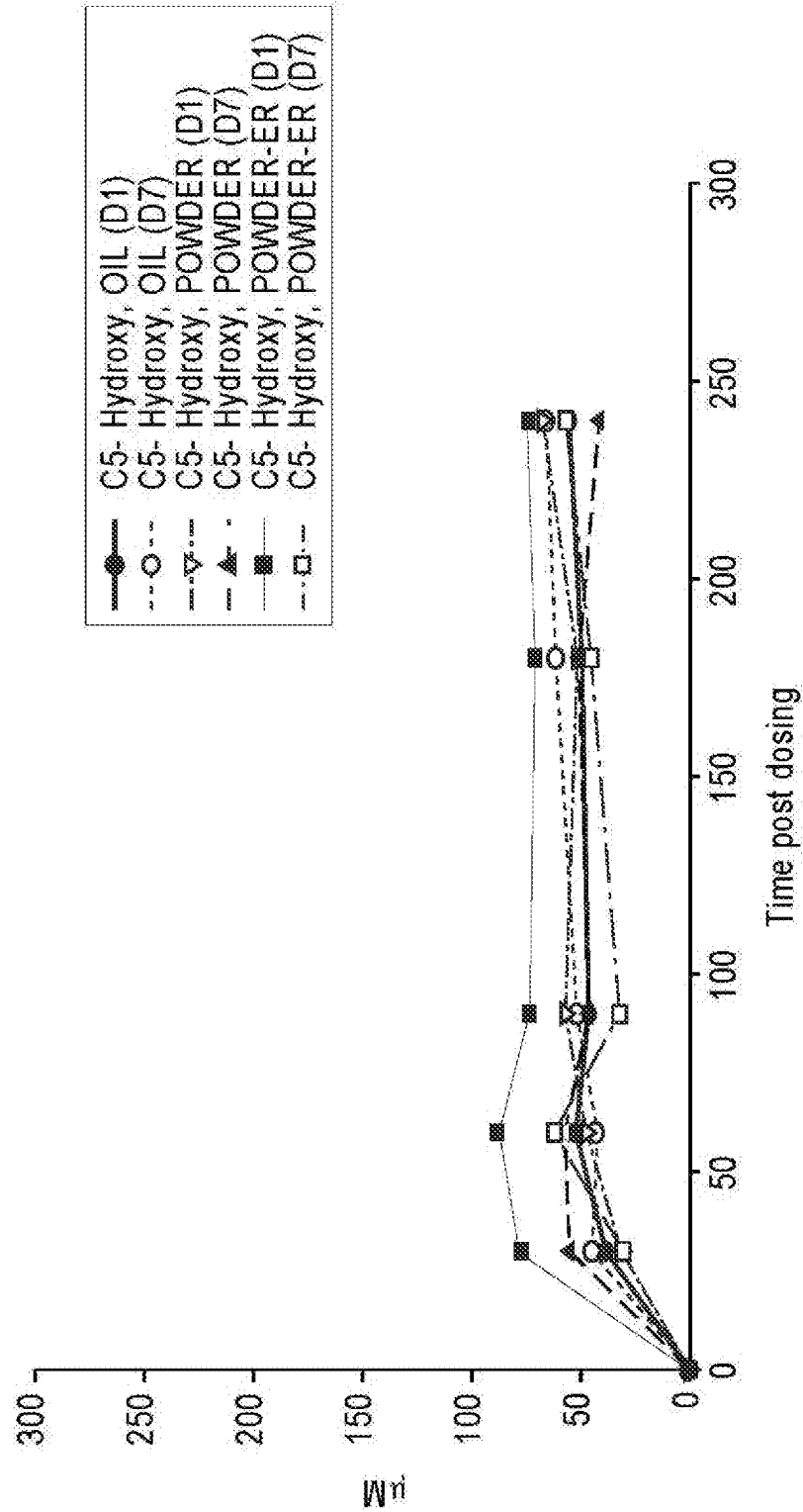


Figure 24

Aspartic Acid Metabolite PK (Normalized to Alanine) Following Triheptanoil Oil, Powder and Powder-ER dosing on Day 1 and Day 7 in Male/Female Combined Mini-pigs

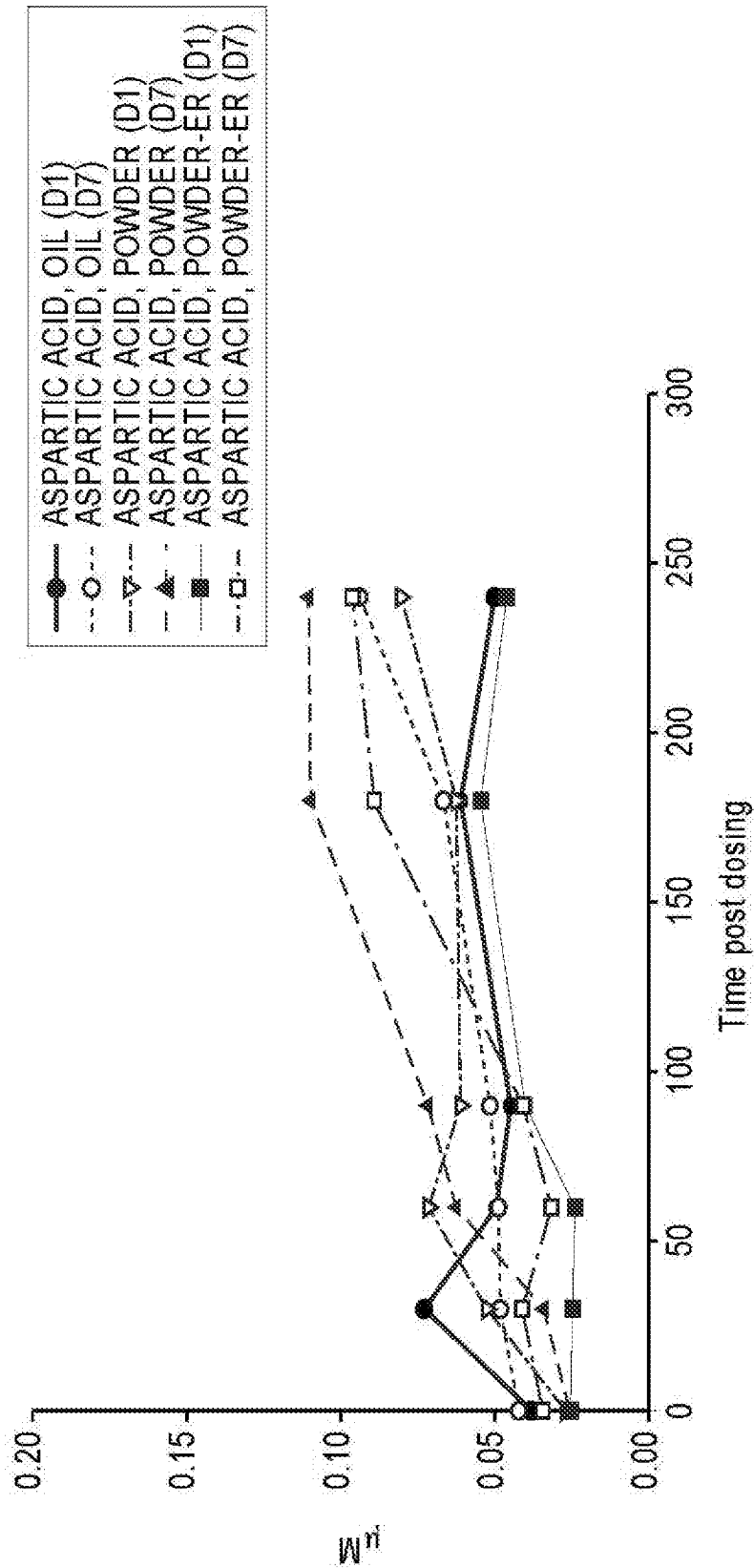


Figure 25

Glutamic Acid Metabolite PK (Relative to Alanine) Following Triheptanoin Oil, Powder and Powder-ER dosing on Day 1 and Day 7 in Male/Female Combined Mini-pigs

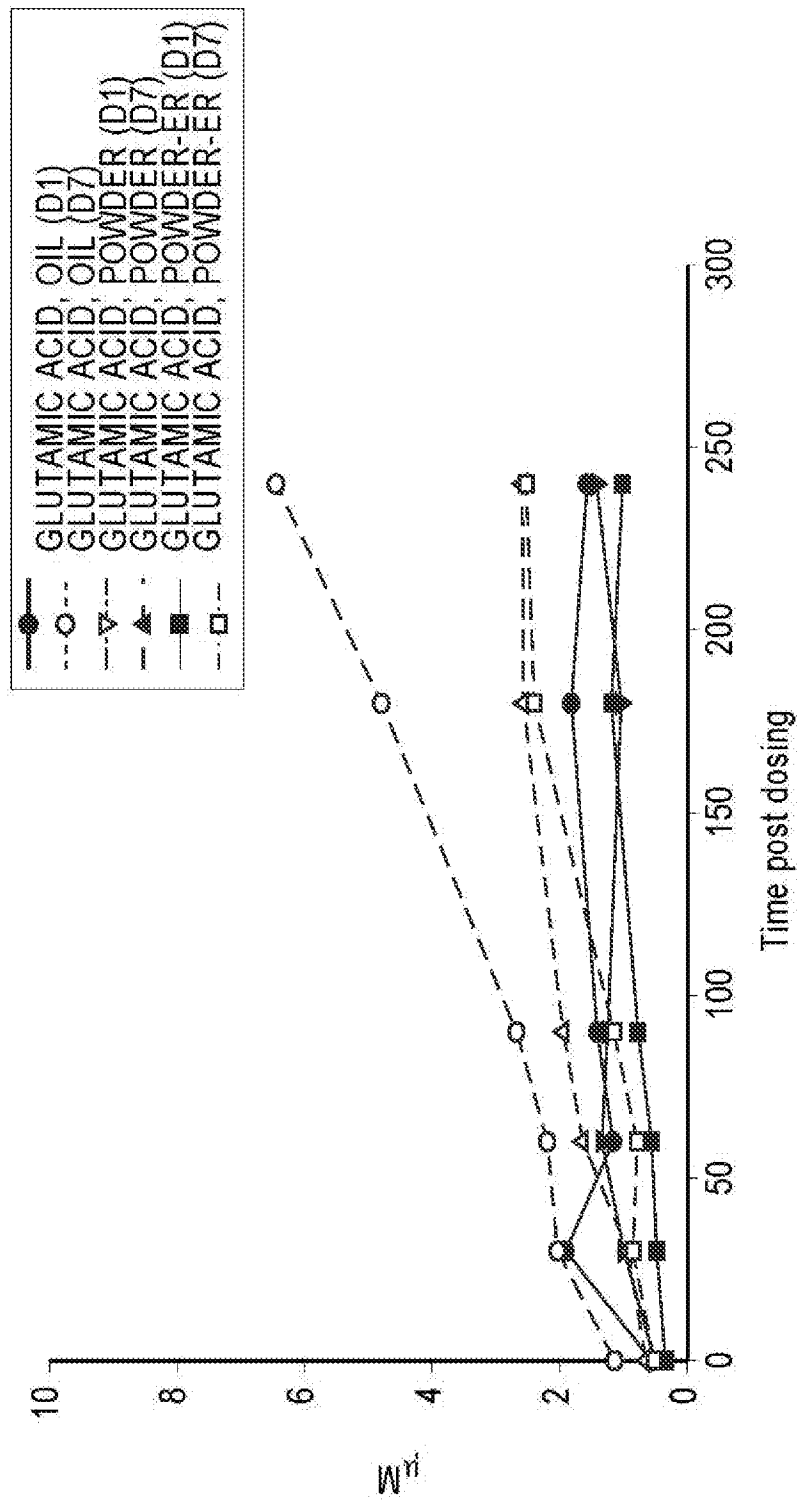


Figure 26

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Pimelic Acid Metabolite PK Following Triheptanoil Oil, Powder and Powder-ER dosing on Day 1 and Day 7 in Male/Female Combined Mini-pigs

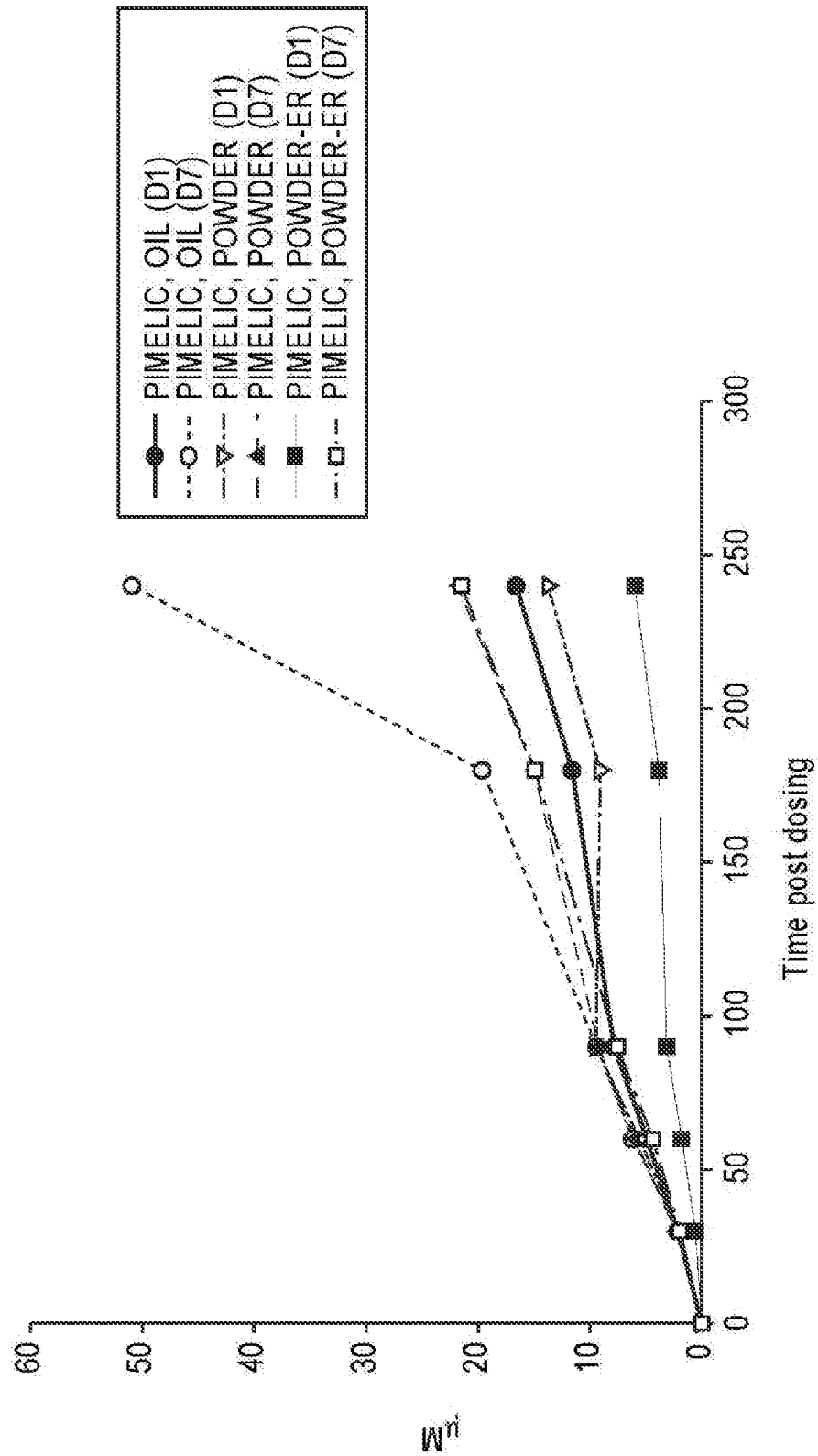


Figure 27

3-Hydroxypropionate Metabolite PK Following Triheptanoil Oil, Powder and Powder-ER dosing on Day 1 and Day 7 in Male/Female Combined Mini-pigs

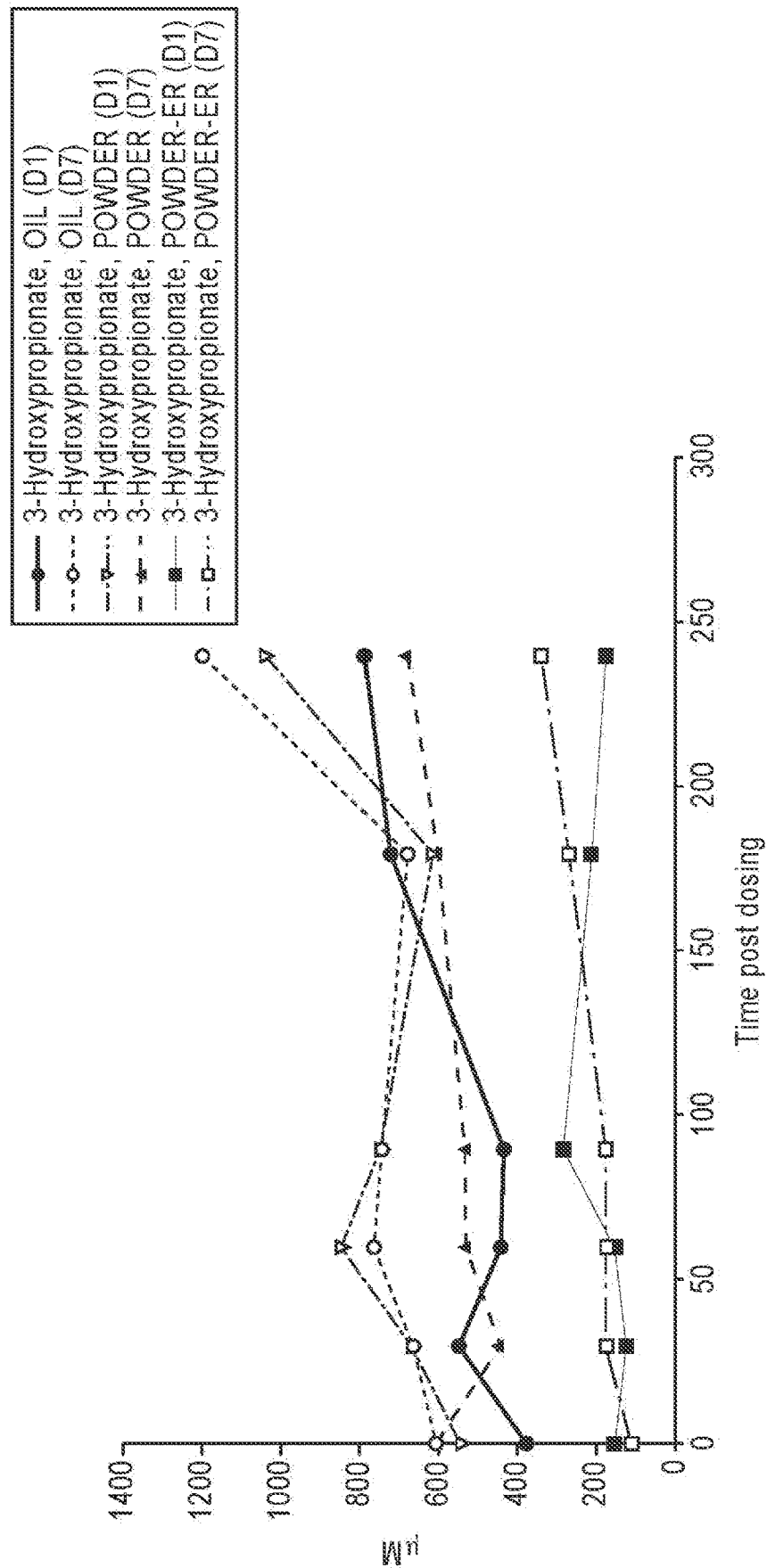


Figure 28

Propionyl Glycine Metabolite PK Following Triheptanoin Oil, Powder and Powder-ER dosing on Day 1 and Day 7 in Male/Female Combined Mini-pigs

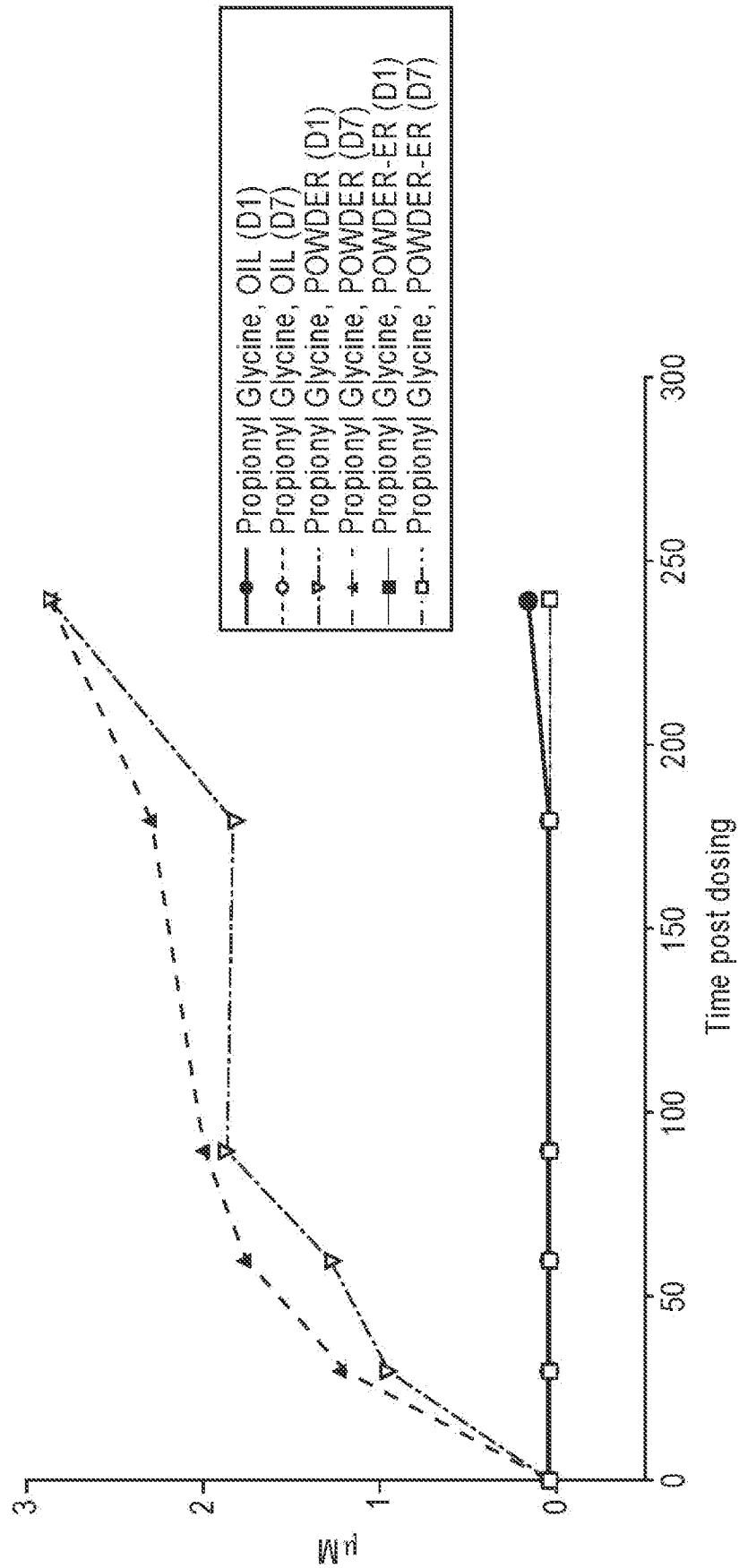


Figure 29

Sum of all Metabolites During PK Timepoints Following Triheptanoin Oil, Powder and Powder-ER dosing on Day 1 and Day 7 in Male/Female Combined Mini-pigs

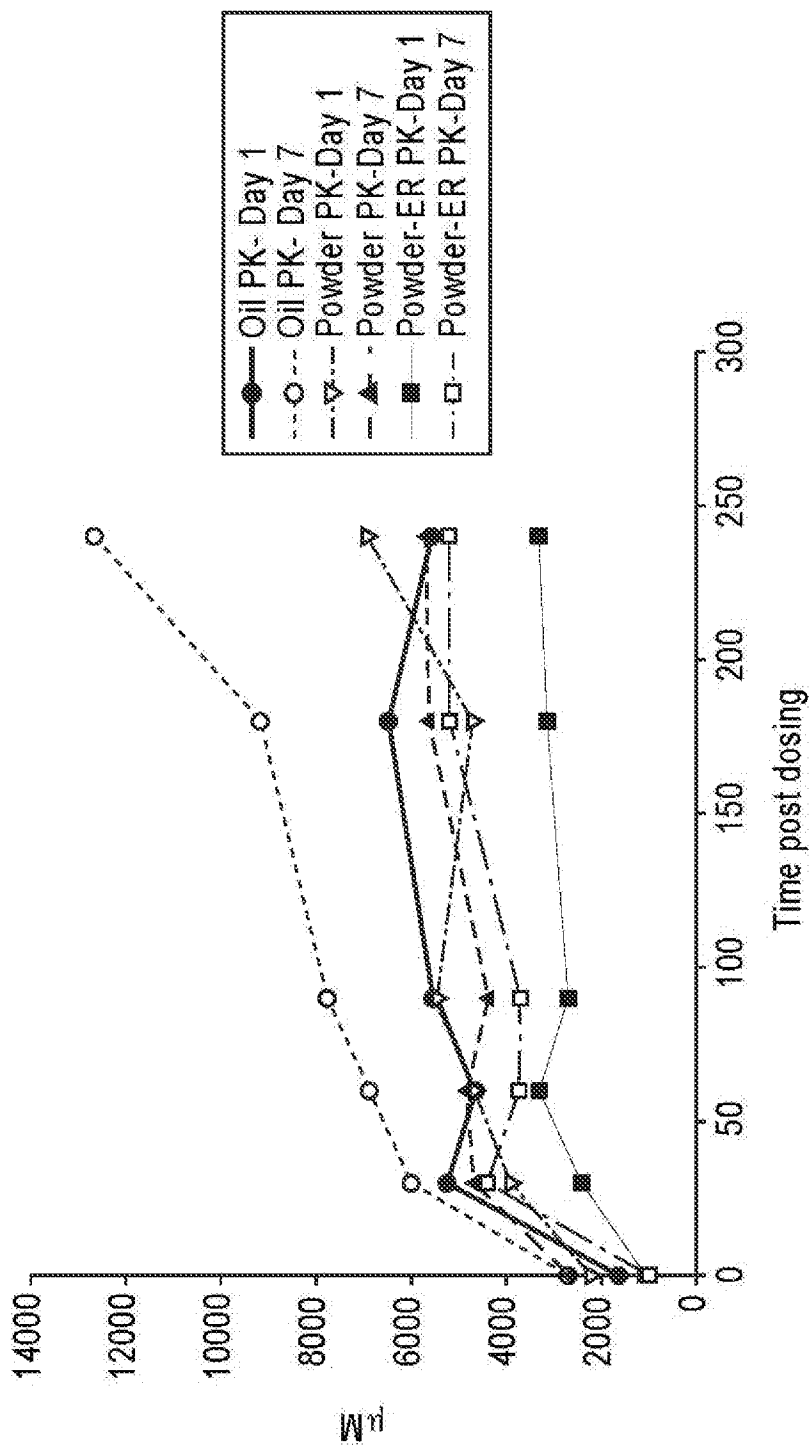


Figure 30

C4-Hydroxy Metabolite PK Following Triheptanoin Powder and Powder-ER Dosing
(Fold Change Relative to Oil Dosing) on Days 1 and 7 in Male and Female Mini-pigs

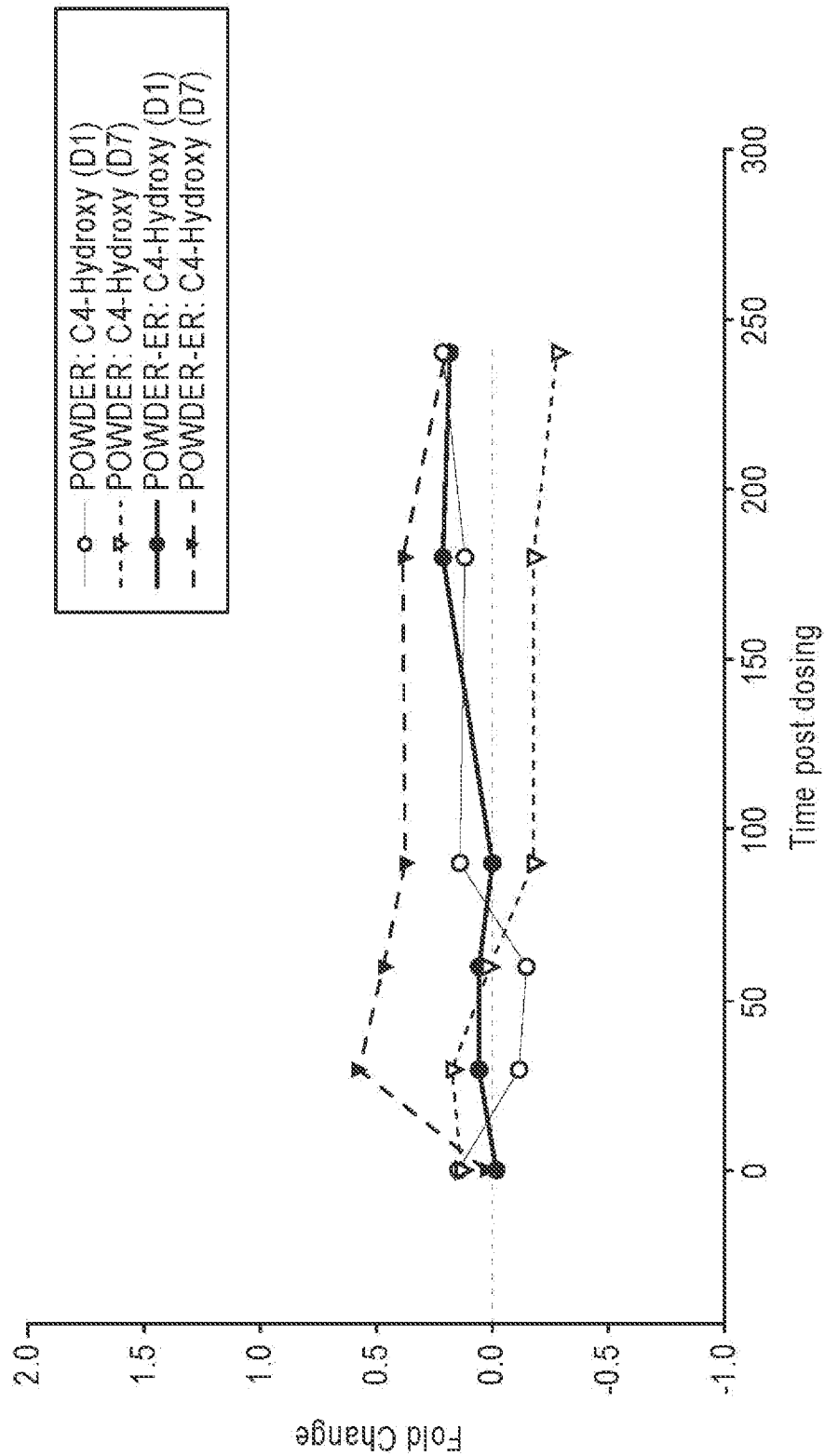


Figure 31

C5-Hydroxy Metabolite PK Following Triheptanoin Powder and Powder-ER Dosing
(Fold Change Relative to Oil Dosing) on Days 1 and 7 in Male and Female Mini-pigs

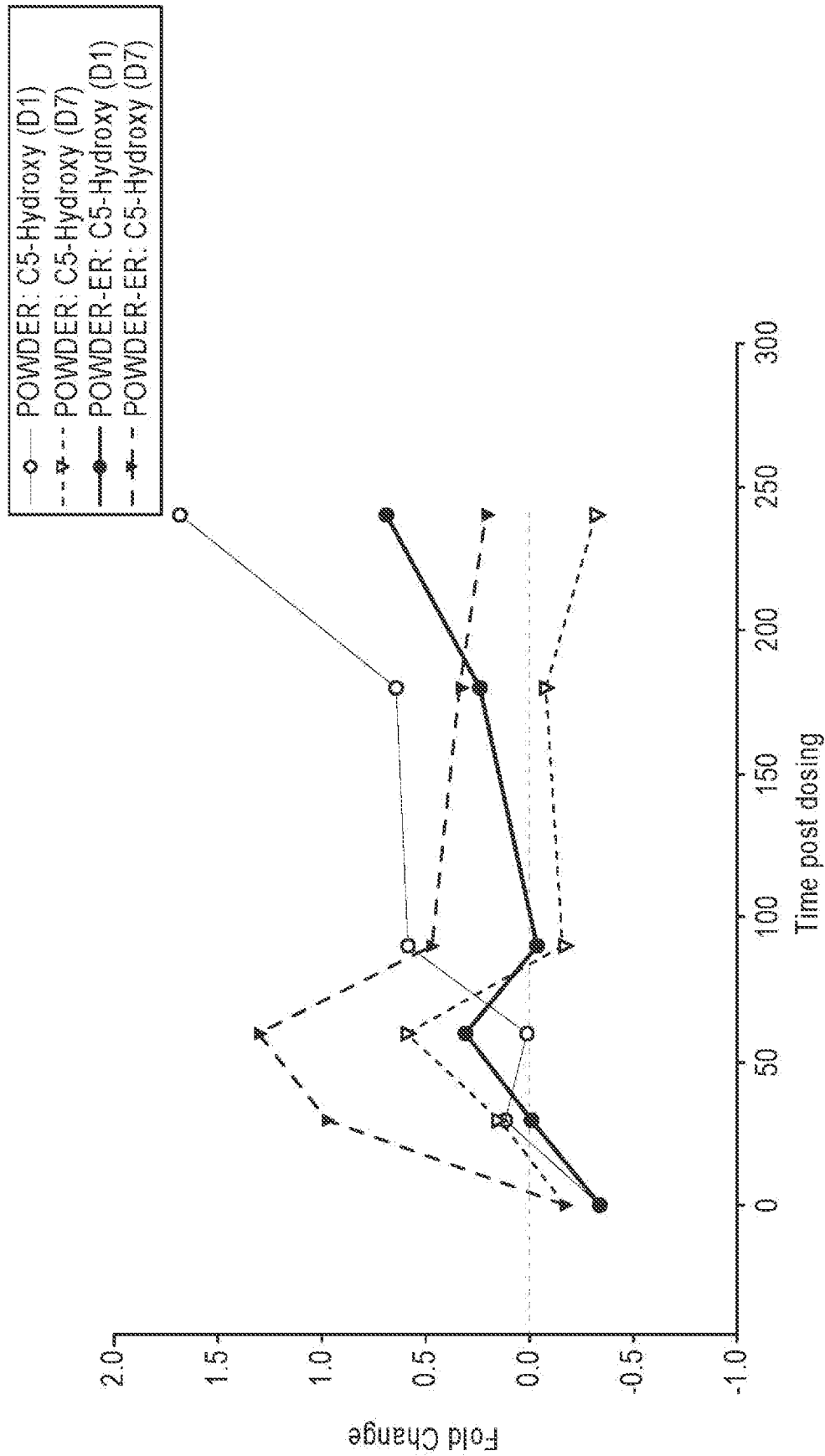


Figure 32

Heptanoic Acid Metabolite PK Following Triheptanoic Powder and Powder-ER Dosing
(Fold Change Relative to Oil Dosing) on Days 1 and 7 in Male and Female Mini-pigs

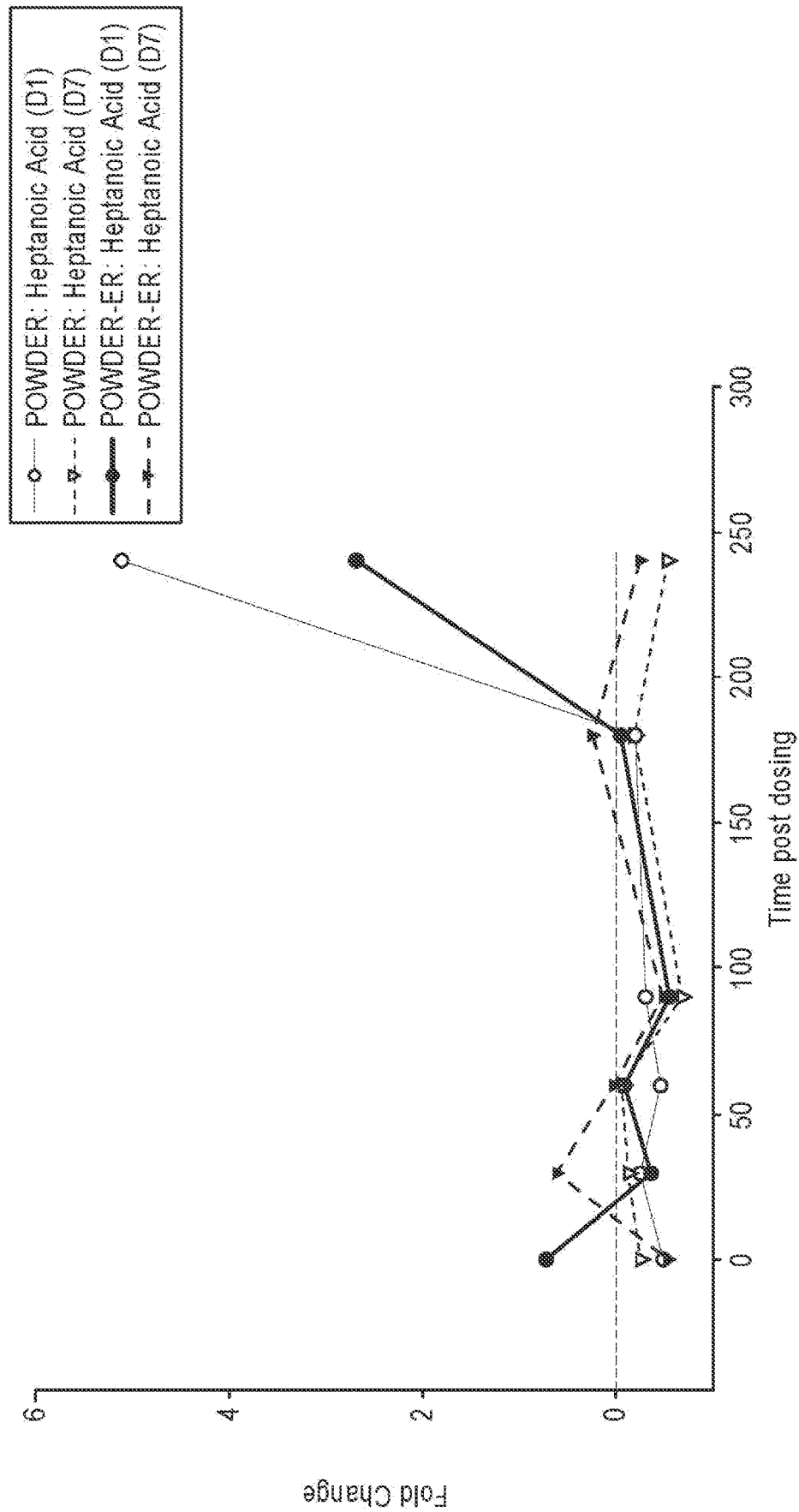


Figure 33

Pimelic Acid Metabolite PK Following Triheptanoin Powder and Powder-ER Dosing
(Fold Change Relative to Oil Dosing) on Days 1 and 7 in Male and Female Mini-pigs

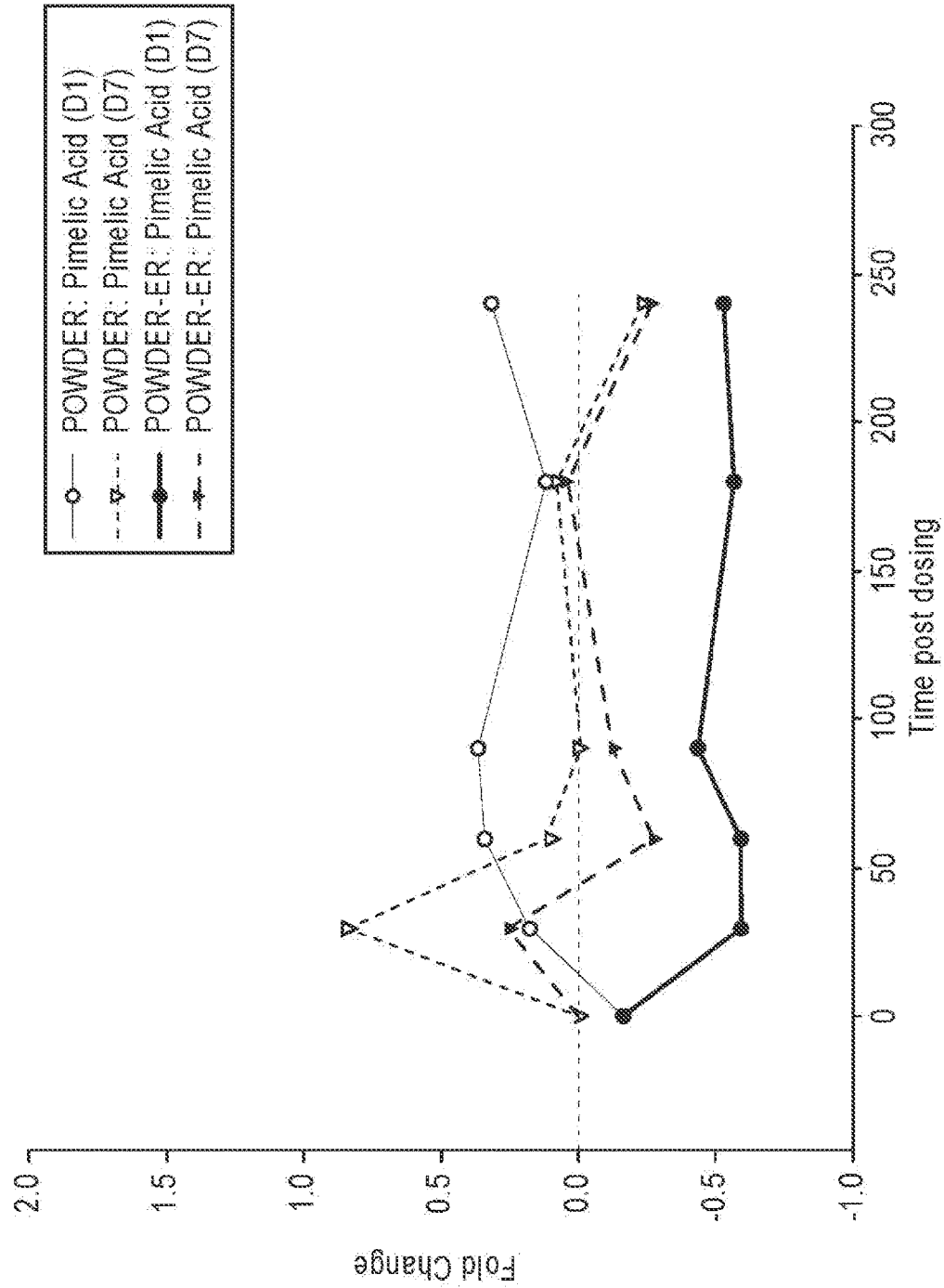


Figure 34

Glutaric Acid Metabolite PK Following Triheptanoin Powder and Powder-ER Dosing
(Fold Change Relative to Oil Dosing) on Days 1 and 7 in Male and Female Mini-pigs

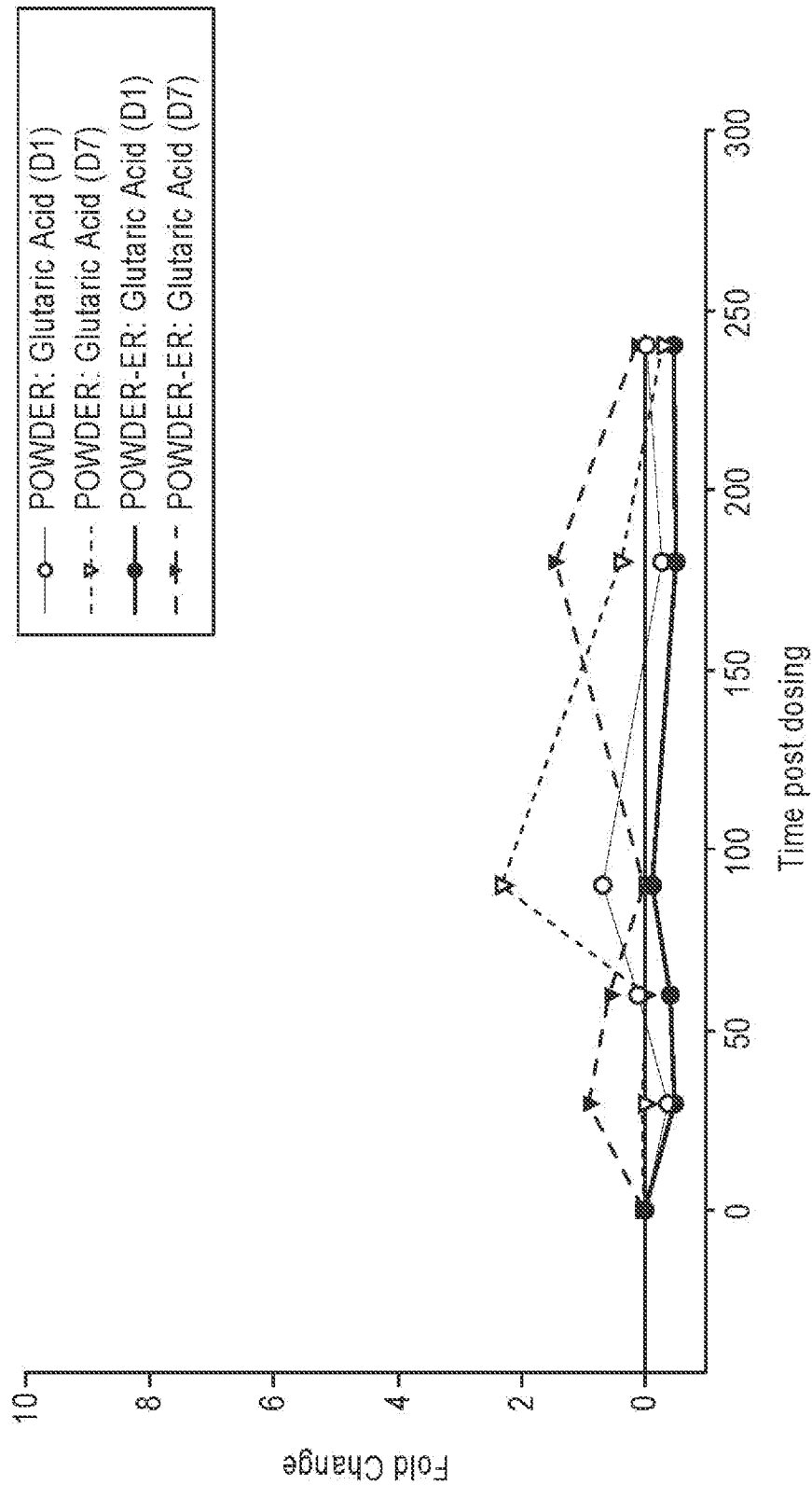


Figure 35

Aspartic Acid Metabolite PK Following Triheptanoil Powder and Powder-ER Dosing
(Fold Change Relative to Oil Dosing) on Days 1 and 7 in Male and Female Mini-pigs

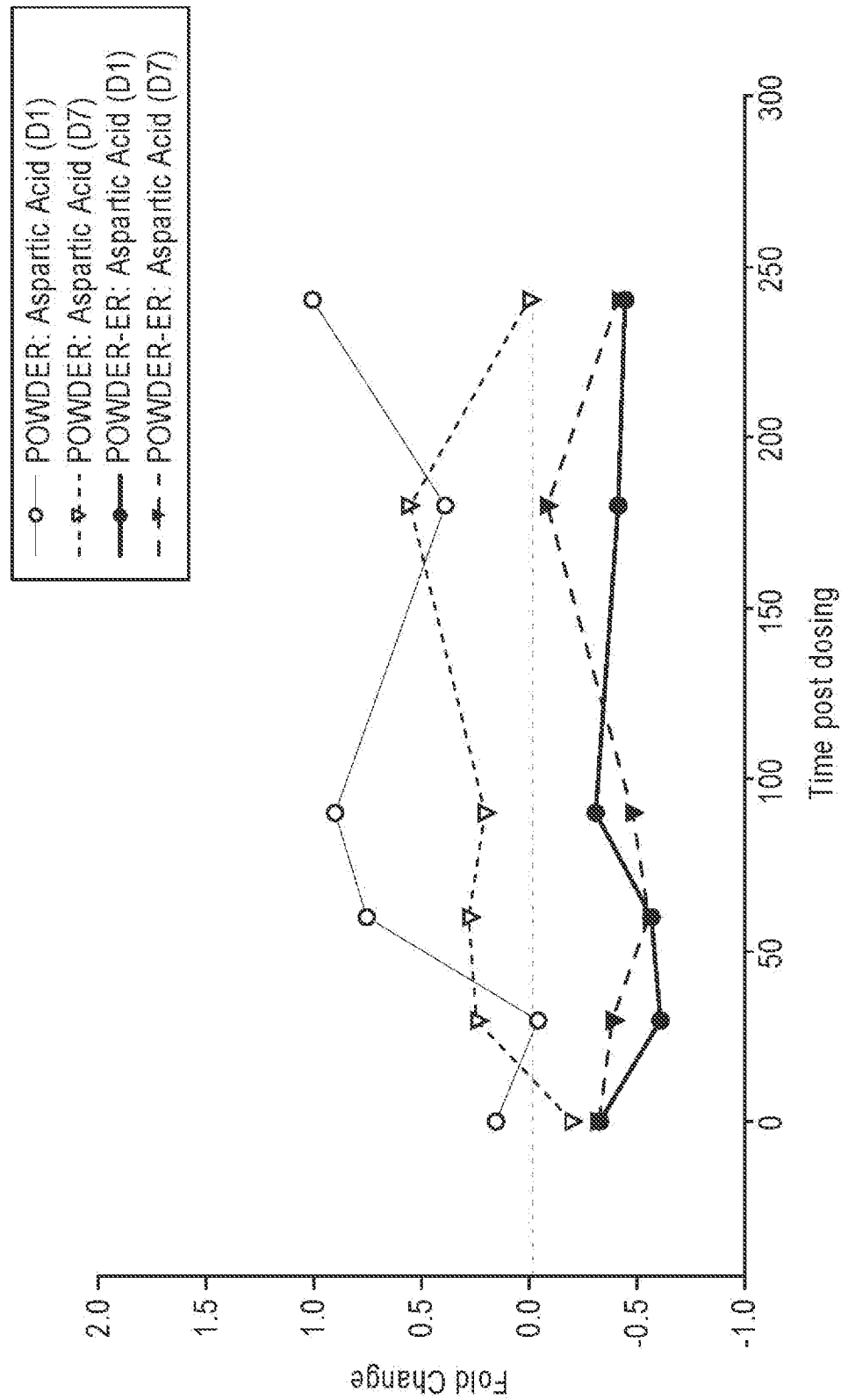


Figure 36

Glutamic Acid Metabolite PK Following Triheptanoin Powder and Powder-ER Dosing
(Fold Change Relative to Oil Dosing) on Days 1 and 7 in Male and Female Mini-pigs

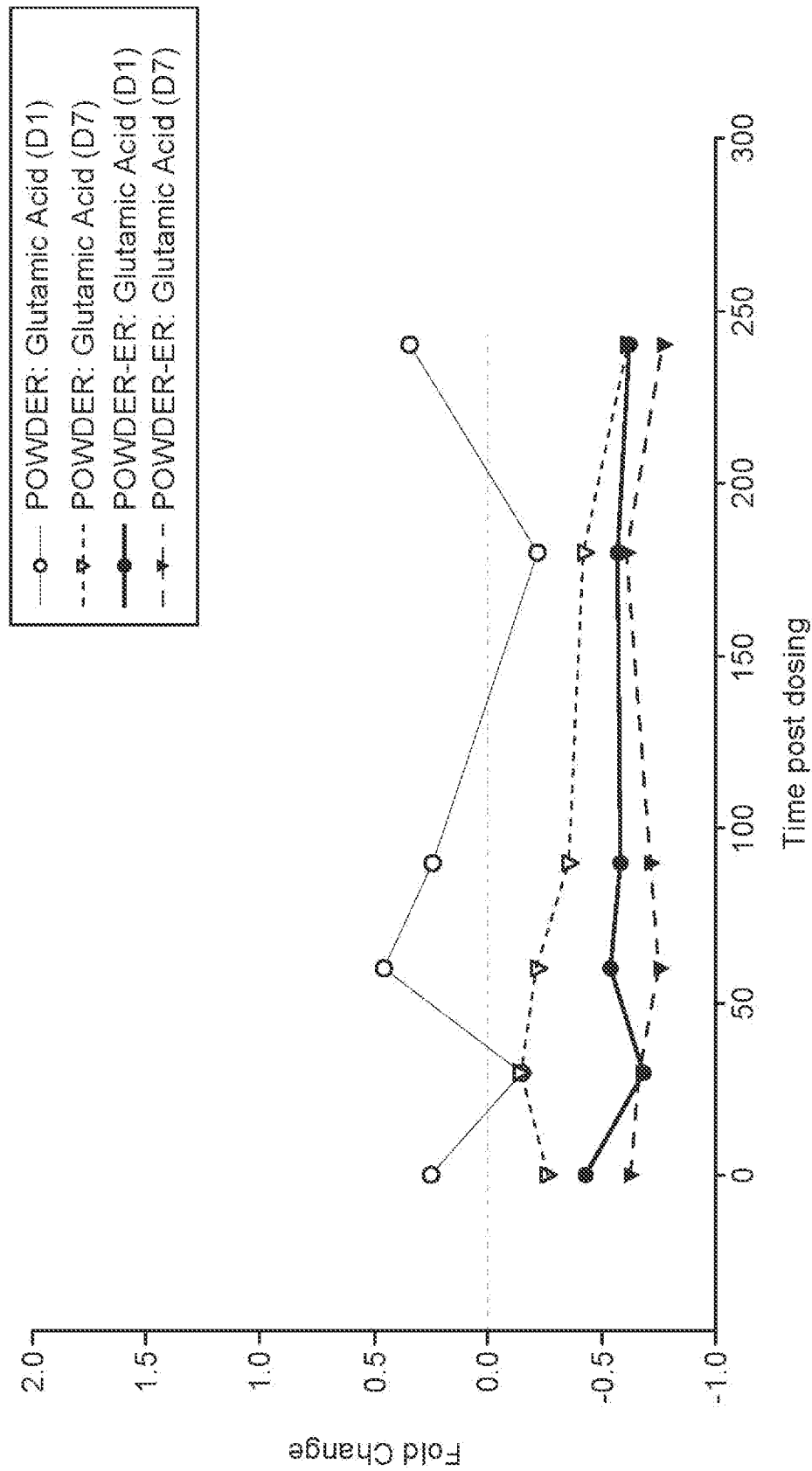


Figure 37

Alanine Metabolite PK Following Triheptanoin Powder and Powder-ER Dosing
(Fold Change Relative to Oil Dosing) on Days 1 and 7 in Male and Female Mini-pigs

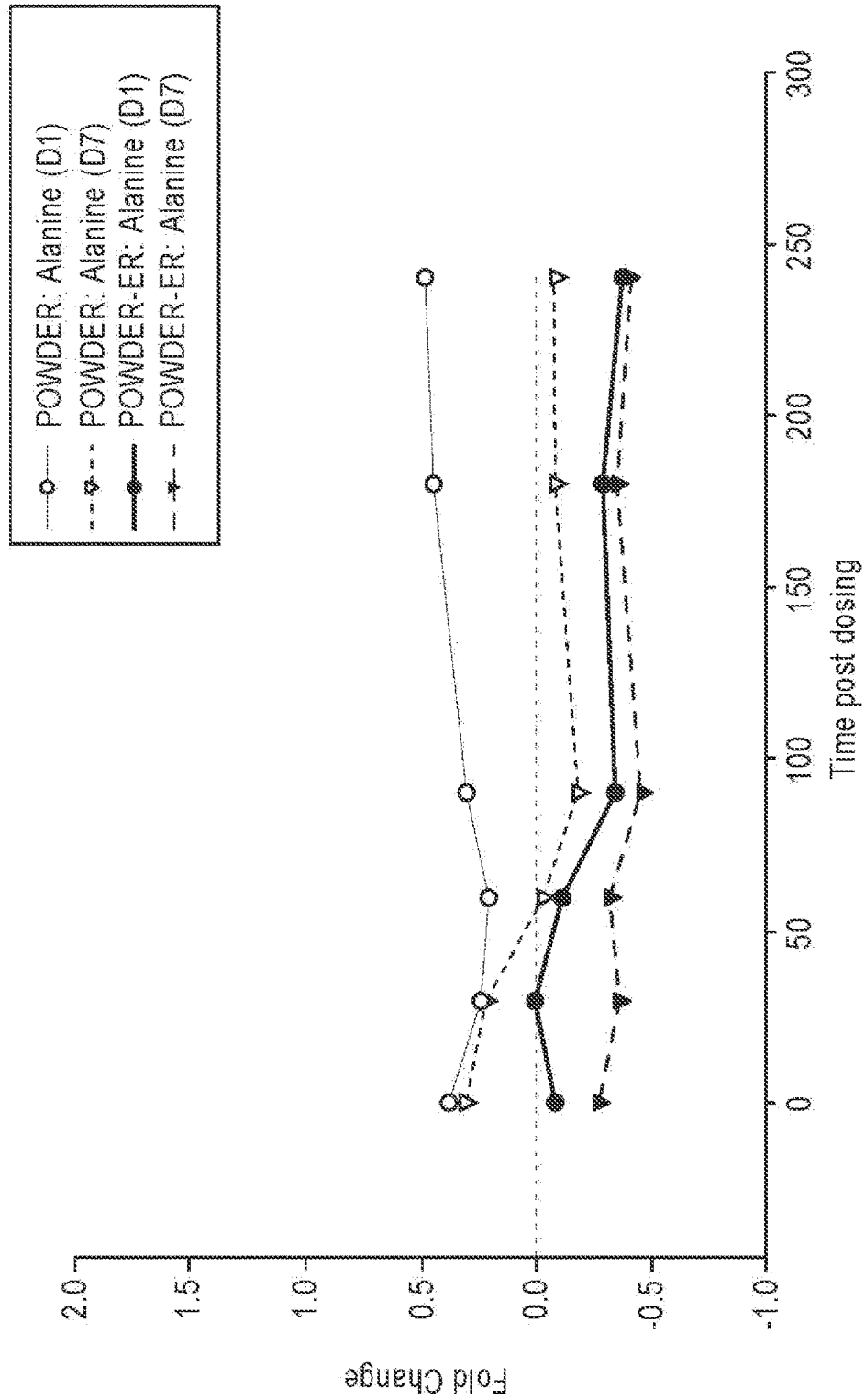


Figure 38

3-Hydroxypropionic acid Metabolite PK Following Triheptanoin Powder and Powder-ER Dosing (Fold Change Relative to Oil Dosing) on Days 1 and 7 in Male and Female Mini-pigs

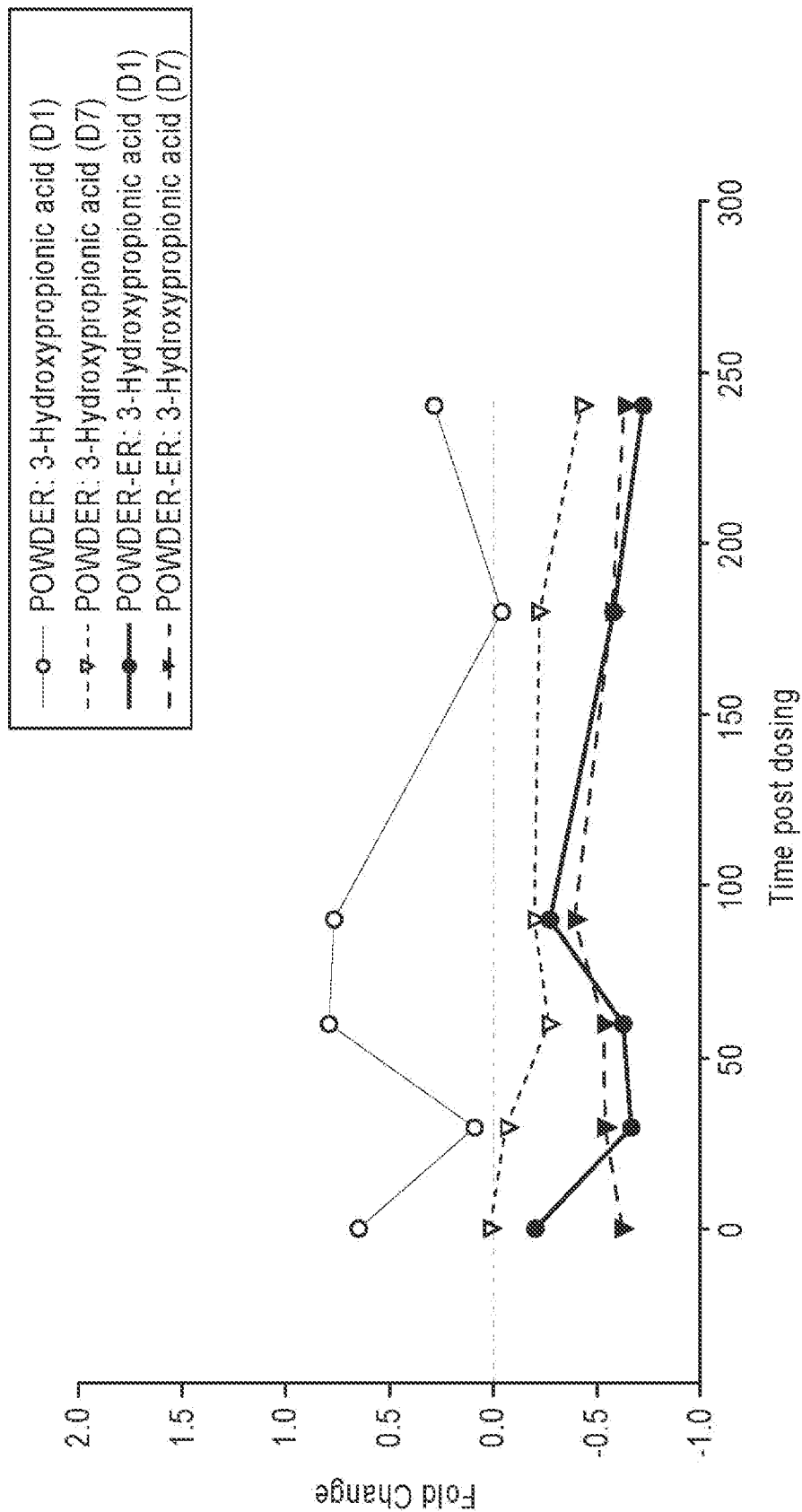


Figure 39

Propionyl Glycine Metabolite PK Following Triheptanoic Powder and Powder-ER
Dosing (Fold Change Relative to Oil Dosing) on Days 1 and 7 in Male and Female Mini-pigs

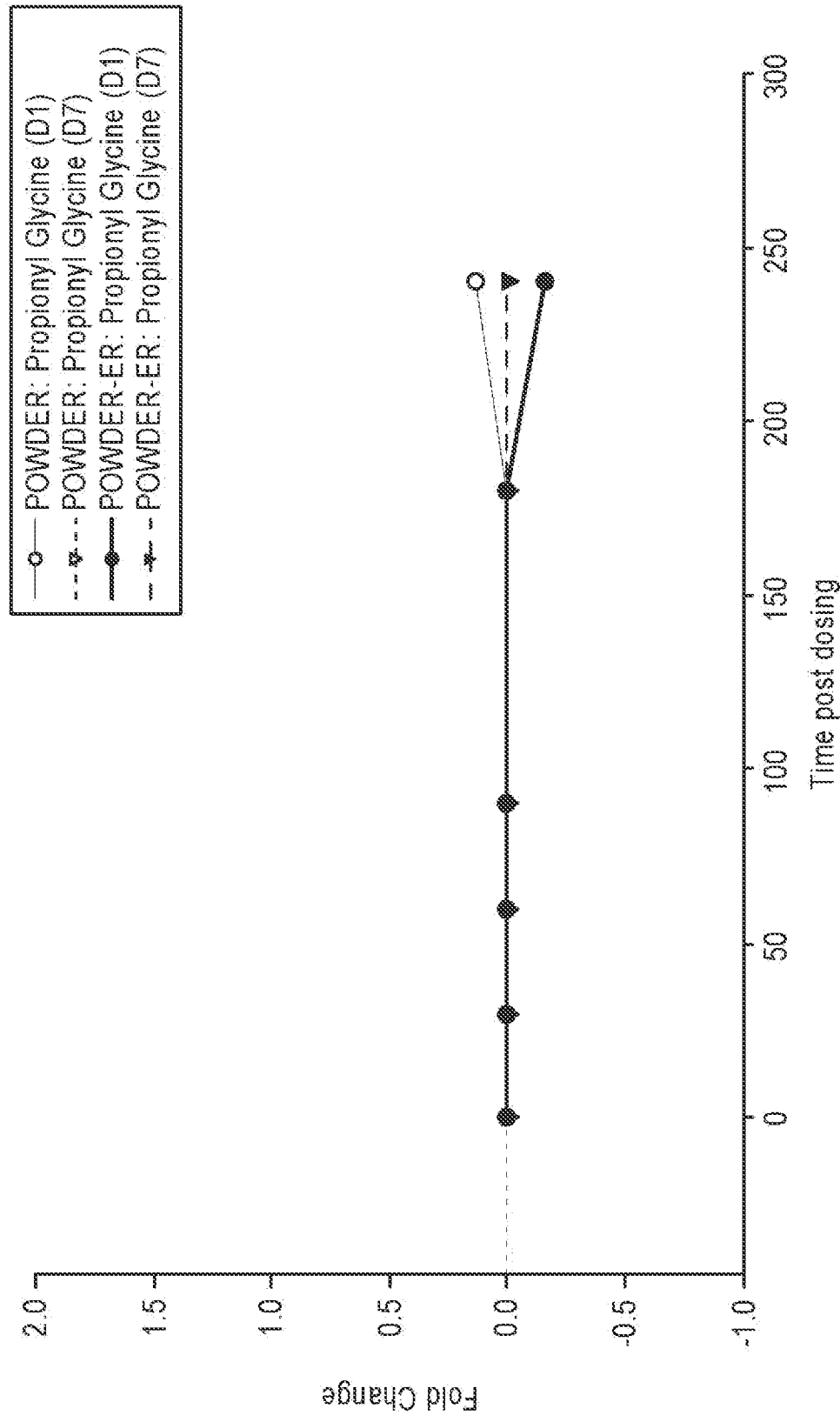


Figure 40

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**COMPOSITIONS OF TRIGLYCERIDES AND
USES THEREOF****CROSS-REFERENCE TO RELATED
APPLICATION**

This application is a Continuation of U.S. application Ser. No. 16/445,557, filed on Jun. 19, 2019, which is a Continuation of U.S. application Ser. No. 14/419,601, filed on Feb. 4, 2015, which is a U.S. National Stage Application of International Application No. PCT/US2014/065693, filed on Nov. 14, 2014, which claims the benefit of U.S. Provisional Patent Application No. 61/904,369 filed Nov. 14, 2013, the contents of each of which are herein incorporated by reference in their entirety for all purposes.

FIELD OF THE INVENTION

The present invention relates to solid compositions of triglycerides with one or more fatty acids, such as triheptanoin and glycerol phenylbutyrate, and the therapeutic use thereof.

BACKGROUND OF THE INVENTION

Current dosage form of the triglycerides drug, e.g., triheptanoin, is a liquid with the consistency of oil. Several problems are associated with the liquid dosage form due to its physical and physiological properties. First, the liquid dosage forms, such as oil, are difficult to be administered due to low miscibility with food or drinks. Second, the oil formulations are difficult to be handled, carried, and dispensed. Furthermore, upon administration, the oil composition is hydrolyzed and released rapidly in the stomach leading to gastric upset, gastric retention, and likely gastrin-related stomach spasm and emesis. In addition, the oil formulations can cause diarrhea by reforming oil droplets and causing a mineral-oil like excess lubrication. Upon repetitive administration of the oil, some patients experience gastric stress which causes vomiting and/or diarrhea. Thus, the tolerability in the oil form can be a dose-limiting toxicity or lead to adverse effects, and the reduced administrable dose would negatively impact the treatment effect for a fatty acid oxidation disorder or deficiency (FAOD); adult polyglucosan body disease; a mitochondrial fat oxidation defect; a glycogen storage disease; a mitochondrial myopathy; glucose transporter type 1 (GLUT1) deficiency syndrome, or other related diseases.

SUMMARY OF THE INVENTION

Among other things, the present invention is directed to a solid composition comprising an ester derived from a polyol and one or more fatty acids as an active ingredient, and a solid substance. To give an example of the ester, triglycerides with one or more fatty acids are suitable for use in accordance with the present invention.

In one aspect, the present invention provides a solid composition comprising triglycerides with one or more odd-numbered carbon fatty acids as an active ingredient and a solid substance; wherein the active ingredient has purity greater than 98% and the one or more odd-numbered carbon fatty acids are selected from the group consisting of C5, C7, C9, C11, C13, C15, and any combinations thereof and the solid composition comprises at least about 50% by weight the triglycerides. The present invention also provides a solid composition comprising a plurality of solid particles, each

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particle comprising triglycerides with one or more odd-numbered carbon fatty acids as an active ingredient adsorbed onto a solid substance; wherein the active ingredient has purity greater than 98% and the one or more odd-numbered carbon fatty acids are selected from the group consisting of C5, C7, C9, C11, C13, C15, and any combinations thereof and the solid composition comprises at least about 50% by weight the triglycerides.

In another aspect, the present invention provides a solid composition comprising triglycerides with one or more phenylalkanoic acids and/or phenylalkenoic acids as an active ingredient and a solid substance; wherein the active ingredient has purity greater than 95% and the solid composition comprises at least about 50% by weight the triglycerides. The present invention also provides a solid composition comprising a plurality of solid particles, each particle comprising triglycerides with one or more phenylalkanoic acids and/or phenylalkenoic acids as an active ingredient adsorbed onto a solid substance; wherein the active ingredient has purity greater than 95% and the solid composition comprises at least about 50% by weight the triglycerides.

In some embodiments, the solid composition comprises at least about 50%, about 55%, about 60%, about 65%, about 70%, about 75%, about 80%, about 85%, or about 90% of the active ingredient by weight of the composition.

In some embodiments, the solid substance includes a solid carrier. The solid carrier can be a fumed silica. Additionally or alternatively, the solid carrier can be selected from the group consisting of SiO₂, TiO₂, Al₂O₃, zeolites, Cab-O-Sil, and combinations thereof.

The solid substance can further comprise one or more sustained release polymers.

In some embodiments, the sustained release polymer is a film-forming, water insoluble polymer. The film-forming, water insoluble polymer can be selected from the group consisting of ethylcellulose, cellulose acetate, cellulose propionate (lower, medium or higher molecular weight), cellulose acetate propionate, cellulose acetate butyrate, cellulose acetate phthalate, cellulose triacetate, poly(methyl methacrylate), poly(ethyl methacrylate), poly(butyl methacrylate), poly(isobutyl methacrylate), poly(hexyl methacrylate), poly(isodecyl methacrylate), poly(lauryl methacrylate), poly(phenyl methacrylate), poly(methyl acrylate), poly(isopropyl acrylate), poly(isobutyl acrylate), poly(octadecyl acrylate), poly(ethylene), poly(ethylene) low density, poly(ethylene) high density, poly(propylene), poly(ethylene oxide), poly(ethylene terephthalate), poly(vinyl isobutyl ether), poly(vinyl acetate), poly(vinyl chloride) or polyurethane, or mixtures thereof. In one embodiment, the film-forming, water insoluble polymer is cellulose acetate.

In some embodiments, the sustained release polymer comprises one or more pH dependent polymers. The pH dependent polymer can be selected from a group consisting of a methyl acrylate-methacrylic acid copolymer, a cellulose acetate succinate, a hydroxy propyl methyl cellulose phthalate, a hydroxy propyl methyl cellulose acetate succinate (hypromellose acetate succinate), a polyvinyl acetate phthalate (PVAP), a methyl methacrylate-methacrylic acid copolymer, alginate and stearic acid, and any combinations thereof.

In some embodiments, the solid composition comprises, by weight of the composition, about 50% to about 80% of the active ingredient; about 10% to about 30% of the solid carrier; and about 10% to about 30% of the sustained release polymer. In some embodiments, the active ingredient, the solid carrier, and the sustained release polymer are in a weight ratio of about 2:1:1.

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The solid composition in accordance with the present invention can be in form of a powder. In some embodiments, the powder comprises particles having an average diameter of less than about 10 micron. The particles can have an average diameter of less than about 9 microns, less than about 8 microns, less than about 7 microns, less than about 6 microns, less than about 5 microns, less than about 4 microns, less than about 3 microns, less than about 2 microns, less than about 1000 nm, less than about 900 nm, less than about 800 nm, less than about 700 nm, less than about 600 nm, less than about 500 nm, less than about 400 nm, less than about 300 nm, less than about 290 nm, less than about 280 nm, less than about 270 nm, less than about 260 nm, less than about 250 nm, less than about 240 nm, less than about 230 nm, less than about 220 nm, less than about 210 nm, less than about 200 nm, less than about 190 nm, less than about 180 nm, less than about 170 nm, less than about 160 nm, less than about 150 nm, less than about 140 nm, less than about 130 nm, less than about 120 nm, less than about 110 nm, less than about 100 nm, less than about 90 nm, less than about 80 nm, less than about 70 nm, less than about 60 nm, less than about 50 nm, less than about 40 nm, less than about 30 nm, less than about 20 nm, less than about 10 nm, less than about 9 nm, less than about 8 nm, less than about 7 nm, less than about 6 nm, or less than about 5 nm.

In some embodiments, the solid substance has a surface area of at least 20 m²/g.

In some embodiments, the solid particles are formed by spray drying a spray suspension comprising the triglycerides with one or more odd-numbered carbon fatty acids as an active ingredient and the solid substance.

The solid composition in accordance with the present invention can include triglycerides with seven carbon fatty acids. In some embodiments, the solid composition further comprises triglycerides with one or more odd-numbered carbon fatty acids selected from the group consisting of C5, C9, C11, C13, C15, and any combinations thereof.

In some embodiments, the triglycerides is a triheptanoin oil comprising greater than about 98% pure glycerol triheptanoate as an active ingredient. The triheptanoin oil can comprise less than 1% w/w of a non-C7 triglyceride or a combination of non-C7 triglycerides.

The solid composition in accordance with the present invention can further include a pharmaceutically acceptable excipient.

In some embodiments, the solid composition includes an active ingredient that has purity greater than 97% after about four weeks of exposure to about 25° C. at about 60% relative humidity when package in a sealed container. In some embodiments, the solid composition has a water content of no more than 1.0% by weight after about four weeks of exposure to about 25° C. at about 60% relative humidity when package in a sealed container. In some embodiments, the solid composition includes an active ingredient that has purity greater than 98% after about four weeks of exposure to about 25° C. at about 60% relative humidity when package in a sealed container. In some embodiments, the solid composition has a water content of no more than 0.5% by weight after about four weeks of exposure to about 25° C. at about 60% relative humidity when package in a sealed container. In some embodiments, the solid composition includes an active ingredient that has purity greater than 99% after about four weeks of exposure to about 25° C. at about 60% relative humidity when package in a sealed container. In some embodiments, the solid composition has a water content of no more than 0.35% by weight after about

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four weeks of exposure to about 25° C. at about 60% relative humidity when package in a sealed container.

In some embodiments, the solid composition includes an active ingredient that has purity greater than 97% after about four weeks of exposure to about 40° C. at about 75% relative humidity when package in a sealed container. In some embodiments, the solid composition has a water content of no more than 1.0% by weight after about four weeks of exposure to 40° C. at about 75% relative humidity when package in a sealed container. In some embodiments, the solid composition includes an active ingredient that has purity greater than 98% after about four weeks of exposure to about 40° C. at about 75% relative humidity when package in a sealed container. In some embodiments, the solid composition has a water content of no more than 0.5% by weight after about four weeks of exposure to 40° C. at about 75% relative humidity when package in a sealed container. In some embodiments, the solid composition includes an active ingredient that has purity greater than 99% after about four weeks of exposure to about 40° C. at about 75% relative humidity when package in a sealed container. In some embodiments, the solid composition has a water content of no more than 0.45% by weight after about four weeks of exposure to 40° C. at about 75% relative humidity when package in a sealed container.

The present invention provides a method of treating a disease, disorder, or condition in a subject comprising orally administering to the subject a therapeutically effective amount of a solid composition in accordance with the present invention, wherein the disease, disorder, or condition is selected from any one or more of the following: a fatty acid oxidation disorder or deficiency; adult polyglucosan body disease; a mitochondrial fat oxidation defect (e.g., relating to carnitine palmitoyl transferase I, carnitine palmitoyl transferase II, carnitine acylcarnitine translocase, very long chain acyl-CoA dehydrogenase, trifunctional protein, long chain hydroxyacyl-CoA dehydrogenase, multiple acyl-CoA dehydrogenase, short chain acyl CoA dehydrogenase, alpha glucosidase, brancher enzyme, debrancher enzyme, myophosphorylase, or phosphofructokinase); a glycogen storage disease (e.g., glycogen storage disease Type II); glucose transporter type 1 (GLUT1) deficiency syndrome; and a mitochondrial myopathy.

Also provided is a method of treating a disease, disorder, or condition in a subject comprising orally administering to the subject a therapeutically effective amount of a solid composition in accordance with the present invention, wherein the disease, disorder, or condition is selected from any one or more of the following: urea cycle disorders (UCD) and hepatic encephalopathy (HE).

In some embodiments, the solid composition is co-administered with a food, drink, or comestible composition.

BRIEF DESCRIPTION OF FIGURES

FIG. 1 is the metabolite pharmacokinetics (PK) profile of triheptanoin and heptanoate following triheptanoin oil dosing on Day 1 in male and female mini-pigs combined.

FIG. 2 is the metabolite PK profile of triheptanoin and heptanoate following triheptanoin powder-ER dosing on Day 1 in male and female mini-pigs combined.

FIG. 3 is the metabolite PK profile of triheptanoin and heptanoate following triheptanoin powder dosing on Day 1 in male and female mini-pigs combined.

FIG. 4 is the metabolite PK profile up to 48 hours (left) or 8 hours (right) of heptanoate following triheptanoin oil, powder and powder-ER dosing on Day 1 in male and female

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mini-pigs combined. Heptanoate (C7) metabolite PK following triheptanoin oil, powder and powder-ER dosing on Day 1 in male and female mini-pigs combined.

FIG. 5 is the metabolite PK profile of B-hydroxybutyric acid (C4-OH) and B-hydroxyvaleric acid (C5-OH) following triheptanoin powder-ER dosing on Day 1 in male and female mini-pigs combined.

FIG. 6 is the metabolite PK profile of B-hydroxybutyric acid (C4-OH) and B-hydroxyvaleric acid (C5-OH) following triheptanoin oil dosing on Day 1 in male and female mini-pigs combined.

FIG. 7 is the metabolite PK profile of B-hydroxybutyric acid (C4-OH) and B-hydroxyvaleric acid (C5-OH) following triheptanoin powder dosing on Day 1 in male and female mini-pigs combined.

FIG. 8 is the metabolite PK profile up to 48 hours (left) or 8 hours (right) of B-hydroxybutyric acid (C4-OH) following triheptanoin oil, powder and powder-ER dosing on Day 1 in male and female mini-pigs combined. B-Hydroxybutyric acid (C4-OH) metabolite PK following triheptanoin oil, powder and powder-ER dosing on Day 1 in male and female mini-pigs combined.

FIG. 9 is the metabolite PK profile up to 48 hours (left) or 8 hours (right) of B-hydroxyvaleric acid (C5-OH) following triheptanoin oil, powder and powder-ER dosing on Day 1 in male and female mini-pigs combined. B-Hydroxyvaleric acid (C5-OH) metabolite PK following triheptanoin oil, powder and powder-ER dosing on Day 1 in male and female mini-pigs combined.

FIG. 10 is the metabolite PK profile of aspartate following triheptanoin powder-ER dosing on Day 1 in male and female mini-pigs combined.

FIG. 11 is the metabolite PK profile of aspartate following triheptanoin oil dosing on Day 1 in male and female mini-pigs combined.

FIG. 12 is the metabolite PK profile of aspartate following triheptanoin powder dosing on Day 1 in male and female mini-pigs combined.

FIG. 13 is the metabolite PK profile of glutamate following triheptanoin powder-ER dosing on Day 1 in male and female mini-pigs combined.

FIG. 14 is the metabolite PK profile of glutamate following triheptanoin oil dosing on Day 1 in male and female mini-pigs combined.

FIG. 15 is the metabolite PK profile of glutamate following triheptanoin powder dosing on Day 1 in male and female mini-pigs combined.

FIG. 16 is the metabolite PK profile of pimelic acid, 3-hydroxypropionate and propionyl glycine following triheptanoin powder-ER dosing on Day 1 in male and female mini-pigs combined.

FIG. 17 is the metabolite PK profile of pimelic acid, 3-hydroxypropionate and propionyl glycine following triheptanoin oil dosing on Day 1 in male and female mini-pigs combined.

FIG. 18 is the metabolite PK profile of pimelic acid, 3-hydroxypropionate and propionyl glycine following triheptanoin powder dosing on Day 1 in male and female mini-pigs combined.

FIG. 19 is the profile of average energy and alternative metabolites following triheptanoin powder-ER dosing on Day 1 in male and female mini-pigs combined.

FIG. 20 is the profile of average energy and alternative metabolites following triheptanoin oil dosing on Day 1 in male and female mini-pigs combined.

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FIG. 21 is the profile of average energy and alternative metabolites following triheptanoin powder dosing on Day 1 in male and female mini-pigs combined.

FIG. 22 is the metabolite PK profile of heptanoic acid following triheptanoin oil dosing on Day 1 and Day 7 in male and female mini-pigs combined.

FIG. 23 is the metabolite PK profile of C4-hydroxy following triheptanoin oil, powder and powder-ER dosing on Day 1 and Day 7 in male and female mini-pigs combined.

FIG. 24 is the metabolite PK profile of C5-hydroxy following triheptanoin oil, powder and powder-ER dosing on Day 1 and Day 7 in male and female mini-pigs combined.

FIG. 25 is the metabolite PK profile of aspartic acid following triheptanoin oil, powder and powder-ER dosing on Day 1 and Day 7 in male and female mini-pigs combined.

FIG. 26 is the metabolite PK profile of glutamic acid following triheptanoin oil, powder and powder-ER dosing on Day 1 and Day 7 in male and female mini-pigs combined.

FIG. 27 is the metabolite PK profile of pimelic acid following triheptanoin oil, powder and powder-ER dosing on Day 1 and Day 7 in male and female mini-pigs combined.

FIG. 28 is the metabolite PK profile of 3-hydroxypropionate following triheptanoin oil, powder and powder-ER dosing on Day 1 and Day 7 in male and female mini-pigs combined.

FIG. 29 is the metabolite PK profile of propionyl glycine following triheptanoin oil, powder and powder-ER dosing on Day 1 and Day 7 in male and female mini-pigs combined.

FIG. 30 is the sum of all metabolite PK profiles following triheptanoin oil, powder and powder-ER dosing on Day 1 and Day 7 in male and female mini-pigs combined.

FIG. 31 is the metabolite PK profile of C4-hydroxy following triheptanoin powder and powder-ER dosing (fold change relative to the oil dosing) on Day 1 and Day 7 in male and female mini-pigs.

FIG. 32 is the metabolite PK profile of C5-hydroxy following triheptanoin powder and powder-ER dosing (fold change relative to the oil dosing) on Day 1 and Day 7 in male and female mini-pigs.

FIG. 33 is the metabolite PK profile of heptanoic acid following triheptanoin powder and powder-ER dosing (fold change relative to the oil dosing) on Day 1 and Day 7 in male and female mini-pigs.

FIG. 34 is the metabolite PK profile of pimelic acid following triheptanoin powder and powder-ER dosing (fold change relative to the oil dosing) on Day 1 and Day 7 in male and female mini-pigs.

FIG. 35 is the metabolite PK profile of glutaric acid following triheptanoin powder and powder-ER dosing (fold change relative to the oil dosing) on Day 1 and Day 7 in male and female mini-pigs.

FIG. 36 is the metabolite PK profile of aspartic acid following triheptanoin powder and powder-ER dosing (fold change relative to the oil dosing) on Day 1 and Day 7 in male and female mini-pigs.

FIG. 37 is the metabolite PK profile of glutamic acid following triheptanoin powder and powder-ER dosing (fold change relative to the oil dosing) on Day 1 and Day 7 in male and female mini-pigs.

FIG. 38 is the metabolite PK profile of alanine following triheptanoin powder and powder-ER dosing (fold change relative to the oil dosing) on Day 1 and Day 7 in male and female mini-pigs.

FIG. 39 is the metabolite PK profile of 3-hydroxypropionate following triheptanoin powder and powder-ER dosing (fold change relative to the oil dosing) on Day 1 and Day 7 in male and female mini-pigs.

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FIG. 40 is the metabolite PK profile of propionyl glycine following triheptanoin powder and powder-ER dosing (fold change relative to the oil dosing) on Day 1 and Day 7 in male and female mini-pigs.

DETAILED DESCRIPTION OF THE INVENTION

A solid composition in accordance with the present invention comprises an ester derived from a polyol and one or more fatty acids as an active ingredient, and a solid substance. The term “polyol” denotes an alcohol containing two or more hydroxyl groups. Examples of polyols include, but are not limited to, diol (e.g., ethylene glycol, propylene glycol, and resorcinol), triol (e.g., glycerol and ethane-1,1,2-triol), tetraol (e.g., pentaerythritol), and sugar alcohols (e.g., maltitol, sorbitol, xylitol, and erythritol). In one embodiment, the polyol is glycerol.

In one embodiment, the present solid compositions are particularly useful as pharmaceutical formulation with both improved physical properties and physiological properties. In one embodiment, the solid compositions are easier to be administered because they are miscible with food, drink, or other comestible compositions with ease. In another embodiment, the solid compositions, such as powder or granule dosage forms, are more portable and easier to handle and dispense during regular daily use or while traveling. In another embodiment, the solid compositions are more stable during storage and easier to be handled and transported during manufacture and commercialization. In another embodiment, the solid compositions may have masked taste and improve dose tolerability and reduce side effects. In another embodiment, the solid compositions, e.g., the powder dosage forms, have delayed release characteristics in the stomach, with limited gastric upset, distress, or spasms. In another embodiment, the solid compositions do not cause diarrhea from the oil leading to excess lubrication of the GI tract through its delayed release and stabilized physical presence allowing better and more complete digestion during passage in the gastrointestinal tract. In another embodiment, the solid compositions comprising a plurality of particles wherein the active ingredient oil is adsorbed on the surface of solid substance thereby enhancing the surface area of the oil on substrate particles. Such enhanced surface area of the oil can enhance absorption efficiency for each dose thereby improving efficacy and reducing diarrhea at the same time. In one embodiment, the solid composition can reduce the gastric stress and other side effects, enhance the therapeutic effect, and improve patient compliance. In another embodiment, the solid composition can reduce gastric and diarrheal tolerability issues, allow higher daily doses of triheptanoin to be achieved and allow for better, and more complete digestion and absorption via the suspended stabilized particles in the GI tract with high surface area. Triglycerides Solid Compositions

In the present invention, “triglyceride” refers to an ester derived from glycerol and one or more fatty acids. The fatty acids can have a carbon chain that is optionally substituted alkyl, optionally substituted alkenyl, or optionally substituted aryl.

“Alkyl,” by itself or as part of another substituent, refers to a saturated branched, straight-chain or cyclic monovalent hydrocarbon radical derived by the removal of one hydrogen atom from a single carbon atom of a parent alkane. The term “alkyl” includes “cycloalkyl” as defined herein below. Typical alkyl groups include, but are not limited to, methyl; ethyl; propyls such as propan-1-yl, propan-2-yl (isopropyl),

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cyclopropan-1-yl, etc.; butan-1-yl, butan-2-yl (sec-butyl), 2-methyl-propan-1-yl (isobutyl), 2-methyl-propan-2-yl (t-butyl), cyclobutan-1-yl, etc.; and the like. In some embodiments, an alkyl group comprises from 1 to 20 carbon atoms (C_1 - C_{20} alkyl). In other embodiments, an alkyl group comprises from 1 to 10 carbon atoms (C_1 - C_{10} alkyl). In still other embodiments, an alkyl group comprises from 1 to 6 carbon atoms (C_1 - C_6 alkyl) or 1 to 4 carbon atoms (C_1 - C_4 alkyl). C_1 - C_6 alkyl is also known as “lower alkyl”.

“Alkenyl,” by itself or as part of another substituent, refers to an unsaturated branched, straight-chain or cyclic monovalent hydrocarbon radical having at least one carbon-carbon double bond derived by the removal of one hydrogen atom from a single carbon atom of a parent alkene. The term “alkenyl” includes “cycloalkenyl” as defined herein below. The group may be in either the cis or trans conformation about the double bond(s). Typical alkenyl groups include, but are not limited to, ethenyl; propenyls such as prop-1-en-1-yl, prop-1-en-2-yl, prop-2-en-1-yl (allyl), prop-2-en-2-yl, cycloprop-1-en-1-yl; cycloprop-2-en-1-yl; butenyls such as but-1-en-1-yl, but-1-en-2-yl, 2-methyl-prop-1-en-1-yl, but-2-en-1-yl, but-2-en-1-yl, but-2-en-2-yl, buta-1,3-dien-1-yl, buta-1,3-dien-2-yl, cyclobut-1-en-1-yl, cyclobut-1-en-3-yl, cyclobuta-1,3-dien-1-yl, etc.; and the like. In some embodiments, an alkenyl group comprises from 2 to 20 carbon atoms (C_2 - C_{20} alkenyl). In other embodiments, an alkenyl group comprises from 2 to 10 carbon atoms (C_2 - C_{10} alkenyl). In still other embodiments, an alkenyl group comprises from 2 to 6 carbon atoms (C_2 - C_6 alkenyl) or 2 to 4 carbon atoms (C_2 - C_4 alkenyl). C_2 - C_6 alkenyl is also known as “lower alkenyl”.

“Aryl,” by itself or as part of another substituent, refers to a monovalent aromatic hydrocarbon group derived by the removal of one hydrogen atom from a single carbon atom of a parent aromatic ring system, as defined herein. Typical aryl groups include, but are not limited to, groups derived from aceanthrylene, acenaphthylene, acephenanthrylene, anthracene, azulene, benzene, chrysene, coronene, fluoranthene, fluorene, hexacene, hexaphene, hexalene, as-indacene, s-indacene, indane, indene, naphthalene, octacene, octaphene, octalene, ovalene, penta-2,4-diene, pentacene, pentalene, pentaphene, perylene, phenalene, phenanthrene, picene, pleiadene, pyrene, pyranthrene, rubicene, triphenylene, trinaphthalene and the like. In some embodiments, an aryl group comprises from 6 to 20 carbon atoms (C_6 - C_{20} aryl). In other embodiments, an aryl group comprises from 6 to 15 carbon atoms (C_6 - C_{15} aryl). In still other embodiments, an aryl group comprises from 6 to 10 carbon atoms (C_6 - C_{10} aryl). In a specific embodiment, an aryl group comprises a phenyl group.

The term “substituted” specifically envisions and allows for one or more substitutions that are common in the art. However, it is generally understood by those skilled in the art that the substituents should be selected so as to not adversely affect the useful characteristics of the compound or adversely interfere with its function. Suitable substituents may include, for example, halogen groups, perfluoroalkyl groups, perfluoroalkoxy groups, alkyl groups, alkenyl groups, alkynyl groups, hydroxy groups, oxo groups, mercapto groups, alkylthio groups, alkoxy groups, aryl or heteroaryl groups, aryloxy or heteroaryloxy groups, arylalkyl or heteroarylalkyl groups, arylalkoxy or heteroarylalkoxy groups, amino groups, alkyl- and dialkylamino groups, carbamoyl groups, alkylcarbonyl groups, carboxyl groups, alkoxy carbonyl groups, alkylaminocarbonyl groups, dialkylamino carbonyl groups, arylcarbonyl groups, aryloxy carbonyl groups, alkylsulfonyl groups, arylsulfonyl groups,

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cycloalkyl groups, cyano groups, C₁-C₆ alkylthio groups, arylthio groups, nitro groups, keto groups, acyl groups, boronate or boronyl groups, phosphate or phosphonyl groups, sulfamyl groups, sulfonyl groups, sulfinyl groups, and combinations thereof. In the case of substituted combinations, such as “substituted arylalkyl,” either the aryl or the alkyl group may be substituted, or both the aryl and the alkyl groups may be substituted with one or more substituents. Additionally, in some cases, suitable substituents may combine to form one or more rings as known to those of skill in the art.

The term “optionally substituted” denotes the presence or absence of the substituent group(s). That is, it means “substituted or unsubstituted”. For example, optionally substituted alkyl includes both unsubstituted alkyl and substituted alkyl. The substituents used to substitute a specified group can be further substituted, typically with one or more of the same or different groups selected from the various groups specified above.

In some embodiments, the triglyceride is an ester derived from glycerol and three fatty acids independently selected from the odd-numbered carbon fatty acids of C5, C7, C9, C11, C13, and C15. In some embodiments, the triglyceride is an ester derived from glycerol and three fatty acids independently selected from phenylalkanoic acids and phenylalkenoic acids.

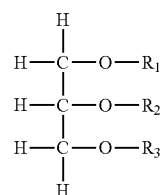
The present invention is based, in part, on the discovery that triglycerides with one or more odd-numbered carbon fatty acids, such as C5, C7, C9, C11, C13, C15, and any combinations thereof, having a purity greater than food grade can be formulated with a solid substance to form a solid composition. In one embodiment, the solid composition contains at least about 50% by weight the triglycerides, which can be in either solid or oil form prior to being formulated with the solid substance. In one embodiment, the triglycerides are an oil prior to being formulated with the solid substance to form the solid composition.

In one embodiment, the present invention provides a solid composition comprising triglycerides with one or more odd-numbered carbon fatty acids as an active ingredient and a solid substance; wherein the active ingredient has purity greater than 98% and the one or more odd-numbered carbon fatty acids are selected from the group consisting of C5, C7, C9, C11, C13, C15, and any combinations thereof and the solid composition comprises at least about 50% by weight the triglycerides. In one embodiment, the triglyceride is an ester derived from glycerol and three fatty acids independently selected from the odd-numbered carbon fatty acids of C5, C7, C9, C11, C13, and C15. In one specific embodiment, the triglyceride is an ester derived from glycerol and three C7 fatty acids, i.e., triheptanoin. In one embodiment, the solid composition comprises triglycerides with seven carbon fatty acids. In another embodiment, the solid composition further comprises triglycerides with one or more odd-numbered carbon fatty acids selected from the group consisting of C5, C9, C11, C13, C15, and any combinations thereof. In another embodiment, the triglycerides with seven carbon fatty acids is a triheptanoin oil comprising greater than about 98% pure glycerol triheptanoate as an active ingredient.

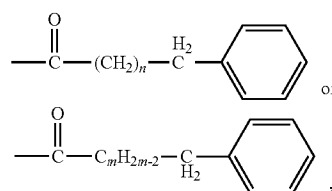
In some embodiment, the present invention provides a solid composition comprising triglycerides with one or more phenylalkanoic acids and/or phenylalkenoic acids as an active ingredient and a solid substance; wherein the active ingredient has purity greater than 95% and the solid composition comprises at least about 50% by weight the triglycerides. In one embodiment, the triglyceride is an ester derived from glycerol and three fatty acids independently

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selected from phenylalkanoic acids and phenylalkenoic acids. In one embodiment, the triglyceride is a compound of formula (I):



wherein R₁, R₂, and R₃ are independently H,



and n is zero or an even number from 2-24 and m is an even number from 2-24, provided that at least one of R₁, R₂, and R₃ is other than H. In one embodiment, n and m are an even number from 2 to 24. In one embodiment, n and m are 0, 2, 4, or 6. In one specific embodiment, the triglyceride is an ester derived from glycerol and three phenylbutyrate, i.e., glycerol phenylbutyrate (e.g., RAVICTI®). Examples of triglycerides with one or more phenylalkanoic acids and/or phenylalkenoic acids are further described in U.S. Pat. No. 5,968,979, the contents of which are hereby incorporated by reference in entirety for all purpose.

In one embodiment, the solid composition comprises at least about 50%, about 55%, about 60%, about 65%, about 70%, about 75%, about 80%, about 85%, or about 90% of the triglycerides. In another embodiment, the solid composition comprises at least about 55%, about 60%, about 65%, about 70%, about 75%, about 80%, or about 85% of the triglycerides. In one specific embodiment, the solid composition comprises about 60% of the triglycerides. In one specific embodiment, the solid composition comprises about 75%, about 80%, or about 85% of the triglycerides.

In some embodiments, the active ingredient in the solid composition has a purity of about or at least about 95, 95.5, 96, 96.5, 97, 97.5, 97.6, 97.7, 97.8, 97.9, 98, 98.1, 98.2, 98.3, 98.4, 98.5, 98.6, 98.7, 98.8, 98.9, 99.0, 99.1, 99.2, 99.3, 99.4, 99.5, 99.6, 99.7, 99.8, or 99.9% or more, or any range derivable therein.

In one embodiment, the solid composition is in a powder form which comprises a plurality of particles, and each particle comprises triglycerides with one or more odd-numbered carbon fatty acids as an active ingredient adsorbed onto a solid substance. In one embodiment, the solid substance has a surface area of at least 20 m²/g.

In one embodiment, the solid substance comprises a solid carrier. The term “solid carrier”, as used herein, denotes any pharmaceutically acceptable solid material, which has a high surface area and does not adversely interact with the present triglycerides. By “high surface area”, it is meant that the solid carrier has a surface area of at least 20 m²/g, or at least 30 m²/g, or at least 50 m²/g, or at least 70 m²/g, or at least 100 m²/g, or at least 150 m²/g, or at least 180 m²/g. In one

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embodiment, the solid carrier can have surface areas of up to 200 m²/g, up to 400 m²/g, or up to 600 m²/g, or more. The surface area of the substrate can be measured using standard procedures, such as low-temperature nitrogen adsorption, based on the Brunauer, Emmett, and Teller (BET) method, which is well known in the art.

In one embodiment, the particles have an average diameter of less than about 100 micron. In one embodiment, the particles have an average diameter of less than about 90 micron, less than about 80 microns, less than about 70 microns, less than about 60 microns, less than about 50 microns, less than about 40 microns, less than about 30 microns, less than about 20 microns, or less than about 10 microns. In another embodiment, the particles have an average diameter of less than about 9 microns, less than about 8 microns, less than about 7 microns, less than about 6 microns, less than about 5 microns, less than about 4 microns, less than about 3 microns, less than about 2 microns, less than about 1000 nm, less than about 900 nm, less than about 800 nm, less than about 700 nm, less than about 600 nm, less than about 500 nm, less than about 400 nm, less than about 300 nm, less than about 290 nm, less than about 280 nm, less than about 270 nm, less than about 260 nm, less than about 250 nm, less than about 240 nm, less than about 230 nm, less than about 220 nm, less than about 210 nm, less than about 200 nm, less than about 190 nm, less than about 180 nm, less than about 170 nm, less than about 160 nm, less than about 150 nm, less than about 140 nm, less than about 130 nm, less than about 120 nm, less than about 110 nm, less than about 100 nm, less than about 90 nm, less than about 80 nm, less than about 70 nm, less than about 60 nm, less than about 50 nm, less than about 40 nm, less than about 30 nm, less than about 20 nm, less than about 10 nm, less than about 9 nm, less than about 8 nm, less than about 7 nm, less than about 6 nm, or less than about 5 nm. The powder may first be formed in the form of small particles ranging in size of from about 5 nm to about 1 micron, or from about 10 nm to about 50 nm, or from about 20 nm to about 100 nm. These particles may in turn form agglomerates ranging in size from about 10 nm to about 100 micron, or 20 nm to about 100 micron, or about 50 nm to about 100 micron.

Examples of the solid carrier include fumed silica and inorganic oxides, such as SiO₂, TiO₂, ZnO₂, ZnO, Al₂O₃, MgAlSilicate, CaSilicate, Al(OH)₃, zeolites, and other inorganic molecular sieves; water insoluble polymers, such as cross-linked cellulose acetate phthalate, cross-linked hydroxypropyl methyl cellulose acetate succinate, cross-linked polyvinyl pyrrolidinone, (also known as cross povidone) microcrystalline cellulose, polyethylene/polyvinyl alcohol copolymer, polyethylene polyvinyl pyrrolidinone copolymer, cross-linked carboxymethyl cellulose, sodium starch glycolate, cross-linked polystyrene divinyl benzene; and activated carbons, including those made by carbonization of polymers such as polyimides, polyacrylonitrile, phenolic resins, cellulose acetate, regenerated cellulose, and rayon. In some specific embodiments, the solid carrier can be Cab-O-Sil, Neusilin, hypromellose acetate succinate (HPMCAS), Casein, or combinations thereof. In one embodiment, the solid carrier is fumed silica. Suitable fumed silica are available from any major producers under various brand names, such as are Evonik (who sells it under the name Aerosil), Cabot Corporation (Cab-O-Sil), Wacker Chemie (HDK), Dow Corning, and OCI (Konasil).

In one embodiment, the solid substance further comprises one or more sustained release polymer. In one embodiment, the one or more sustained release polymer comprises one or

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more pH dependent polymers. In one embodiment, the pH dependent polymer is selected from a group consisting of a methyl acrylate-methacrylic acid copolymer, a cellulose acetate succinate, a hydroxy propyl methyl cellulose phthalate, a hydroxy propyl methyl cellulose acetate succinate (hypromellose acetate succinate), a polyvinyl acetate phthalate (PVAP), a methyl methacrylate-methacrylic acid copolymer, alginate and stearic acid, and any combinations thereof.

In one embodiment, the solid composition may further comprise one or more additional pharmaceutically acceptable excipient, such as fillers, surfactants, and flavorants. Those excipients can be added to improve the powder flowability, dissolvability, and taste.

“Pharmaceutically acceptable” refers to being suitable for use in contact with the tissues of humans and animals without undue toxicity, irritation, allergic response, and the like, commensurate with a reasonable benefit/risk ratio, and effective for their intended use within the scope of sound medical judgment. “Excipient” denotes a diluent, adjuvant, vehicle or carrier with which the active ingredient is administered. Examples of the pharmaceutically acceptable excipient include, are not limited to a filler (or diluent), a binder, a disintegrant, a lubricant, or a combination thereof.

Fillers may be one or more compounds which are capable of providing compactability and good flow. Examples of fillers include microcrystalline cellulose, starch, lactose, sucrose, glucose, mannitol, maltodextrin, sorbitol, dextrose, silicic acid, dibasic calcium phosphate, or a combination comprising at least one of the foregoing fillers. Exemplary lactose forms include lactose monohydrate, NF (Fast Flo), lactose spray-dried monohydrate, and lactose anhydrous. Exemplary microcrystalline celluloses (MCC) include, for example, AVICEL® PH101 and AVICEL® PH102, which are commercially available from FMC Biopolymer, Philadelphia, PA.

Binders may be used to impart cohesive qualities to a formulation, for example, a tablet formulation, and thus ensure that the tablet remains intact after compaction. Examples of binders include starches (for example, Starch 1500® or pregelatinized starch), alginates, gelatin, carboxymethylcellulose, sugars (for example, sucrose, glucose, dextrose, and maltodextrin), polyethylene glycol, waxes, natural and synthetic gums, polyvinylpyrrolidone, and celulosic polymers (for example, microcrystalline cellulose, hydroxypropyl cellulose, hydroxypropyl methylcellulose, methyl cellulose, and hydroxyethyl cellulose) and combinations comprising one or more of the foregoing binders.

Disintegrants are used to facilitate disintegration or “breakup” of a composition, for example, a tablet, after administration. Examples of disintegrants include sodium starch glycolate, sodium croscarmellose (cross-linked carboxy methyl cellulose), crosslinked polyvinylpyrrolidone (PVP-XL), anhydrous calcium hydrogen phosphate, agar-agar, potato or tapioca starch, alginic acid, or a combination comprising one or more of the foregoing disintegrants.

A lubricant may be added to the composition for a minimum period of time to obtain good dispersal. Examples of lubricants include magnesium stearate, calcium stearate, zinc stearate, stearic acid, talc, glyceryl behenate, polyethylene glycol, polyethylene glycol, polyethylene oxide, sodium lauryl sulfate, magnesium lauryl sulfate, sodium oleate, sodium stearyl fumarate, DL-leucine, colloidal silica, or a combination comprising one or more of the foregoing lubricants.

If desired, the solid composition may optionally comprise small amounts of nontoxic auxiliary substances such as

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wetting or emulsifying agents, or pH buffering agents, for example, sodium acetate, sorbitan monolaurate, triethanolamine sodium acetate, triethanolamine oleate, sodium lauryl sulfate, dioctyl sodium sulfosuccinate, and polyoxyethylene sorbitan fatty acid esters.

In another embodiment of the present invention, the solid composition comprises one or more sustained release polymer. In one embodiment, the solid substance comprises one or more sustained release polymer. By “sustained release polymer”, it is meant any polymer which can control the release of the active ingredient from the composition in such a way to obtain the desired release profile. In one embodiment, the sustained release polymer is a hydrophilic polymer. The term “hydrophilic polymer” refers to a polymer having a strong affinity for water and tending to dissolve in, mix with, or be wetted by water. Examples of the hydrophilic polymer include, but are not limited to polyethylene oxide, hydroxypropyl cellulose, hydroxypropyl methyl cellulose, hydroxyethyl cellulose, sodium carboxymethylcellulose, calcium carboxymethyl cellulose, methyl cellulose, polyacrylic acid, maltodextrin, pre-gelatinized starch, guar gum, sodium alginate, polyvinyl alcohol, chitosan, locust bean gum, amylase, any other water-swelling polymer, and a combination thereof. In another embodiment, the sustained release polymer is a film-forming, water insoluble polymer. Examples of the film-forming, water insoluble polymer include, but are not limited to ethylcellulose, cellulose acetate, cellulose propionate (lower, medium or higher molecular weight), cellulose acetate propionate, cellulose acetate butyrate, cellulose acetate phthalate, cellulose triacetate, poly(methyl methacrylate), poly(ethyl methacrylate), poly(butyl methacrylate), poly(isobutyl methacrylate), poly(hexyl methacrylate), poly(isodecyl methacrylate), poly(lauryl methacrylate), poly(phenyl methacrylate), poly(methyl acrylate), poly(isopropyl acrylate), poly(isobutyl acrylate), poly(octadecyl acrylate), poly(ethylene), poly(ethylene) low density, poly(ethylene) high density, poly(propylene), poly(ethylene oxide), poly(ethylene terephthalate), poly(vinyl isobutyl ether), poly(vinyl acetate), poly(vinyl chloride) or polyurethane, or any other water insoluble polymer, or mixtures thereof. In another embodiment, the sustained release polymer is a film-forming, water soluble polymer. Examples of the film-forming, water soluble polymer include, but are not limited to polyvinyl alcohol, polyvinylpyrrolidone, methyl cellulose, hydroxypropyl cellulose, hydroxypropylmethyl cellulose and polyethylene glycol, Pluronic F108, Pluronic F127, Pluronic F68 or mixtures thereof. In one embodiment, the present solid composition comprises one or more the film-forming, water insoluble polymers. In one specific embodiment, the present solid composition comprises cellulose acetate.

In another embodiment, the sustained release polymer comprises one or more pH dependent polymers. By “pH dependent polymers”, it is meant polymers which are not soluble at the highly acidic pH but soluble in neutral to basic pH environment. Examples of pH dependent polymers include, but are not limited to, methyl acrylate-methacrylic acid copolymers, cellulose acetate succinates, hydroxy propyl methyl cellulose phthalates, hydroxy propyl methyl cellulose acetate succinates (hypromellose acetate succinates), polyvinyl acetate phthalates (PVAP), methyl methacrylate-methacrylic acid copolymers, alginate and stearic acid, and any combinations thereof. Furthermore, materials, such as fatty acids, waxes, shellac, plastics, and plant fibers, may also be suitable for pH dependent polymers.

The solid compositions as described herein can be used alone for various purposes, such as in treatment methods as

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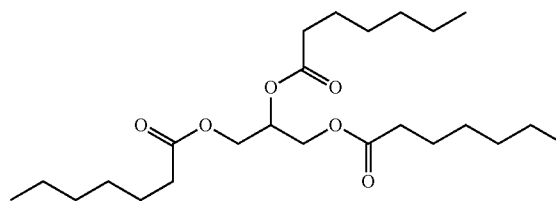
described herein. In this regard, the solid compositions can be pharmaceutically acceptable.

The present solid compositions can be formed by a spray-drying process. In one embodiment, the solid particles are formed by spray drying a spray suspension comprising the triglycerides with one or more odd-numbered carbon fatty acids as an active ingredient and the solid substance. The spray suspension contains the triglycerides dissolved in a solvent and the solid substance suspended therein. Such spray suspension is spray-dried to form a solid powder by rapidly removing the solvent. The formed powder particles may have the triglycerides adsorbed on the surface of the solid substance or the triglycerides and the solid substance interspersed with one another to form a matrix. In another embodiment, the solid composition further comprises a sustained release polymer. For example, the solid particles may be formed by spray drying a spray suspension comprising the triglycerides and the sustained release polymer dissolved in a solvent with the solid substance suspended therein. The spray suspension is spray-dried to form a solid powder by rapidly removing the solvent.

In one embodiment, some powder formulations (with 75%, 80%, 85%, and 90% loading of triheptanoin on solid carrier) were prepared. The triheptanoin oil as described herein was first dissolved in acetone and then mixed with the solid carrier, such as Cab-O-Sil, to form a suspension. The sustained release polymer, e.g., Eudragit, was optionally added. The resulting suspension was spray dried to obtain a powder. The spray dried powder is then dried further to reduce the amount of solvent (acetone) to a desirable level.

Triheptanoin Oil

In one embodiment, the triglyceride in the present solid composition is a triheptanoin oil. Triheptanoin oil as described herein comprises triglycerides with seven carbon fatty acids. Triheptanoin, also known as glycerol triheptanoate, glycerol triheptanoate, glyceryl triheptanoate, 1,3-di(heptanoyloxy)propan-2-yl heptanoate, triheptanthin, 1, 2, 3-tri(heptanoyl)glycerol, and propane-1,2,3-triyl triheptanoate, is a triglyceride of the seven carbon (C7) fatty acid heptanoate (CAS Registry No. 620-67-7) and has the following structure:



Triheptanoin oil as described herein is of a purity or grade greater than food grade triheptanoin oil. In some embodiments, the triheptanoin oil is considered ultrapure pharmaceutical grade triheptanoin oil.

In one embodiment, the present triheptanoin oil has the general properties listed in Table 1:

Physical and chemical property	Triheptanoin with Pharmaceutical Grade Purity
Form	Liquid
Color	Light yellow
Cloud point	<0° C.
Flash point	ca. 220° C.

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-continued

Physical and chemical property	Triheptanoin with Pharmaceutical Grade Purity
Vapor pressure	<0.01 hPa; 20° C.
Relative Density	ca. 0.96 g/cm ³ ; 20° C.
Water solubility	<0.01 g/l; 20° C.
Partition coefficient (n-octanol/water)	Log Pow: >3.0; glycerides of saturated C ₇ fatty acid
Viscosity, dynamic	ca. 20 mPa · s at 20° C.
Refractive Index	1.4440 to 1.4465

Typically, the total concentration of impurities in triheptanoin oil is less than the total concentration of impurities in food grade triheptanoin oil. In some embodiments, food grade triheptanoin oil can have a purity of about or at most about 95, 95.5, 96, 96.5, 97, or 97.5%, or any range derivable therein.

In some embodiments, total impurities in triheptanoin oil amount to less than 5% w/w. In some embodiments, total impurities in triheptanoin oil amount to less than 5, 4.5, 4, 3.5, 3, 2.5, 2, 1.5, 1, or 0.5% w/w or less, or any range derivable therein.

In one embodiment, triheptanoin oil comprises less than 2% w/w of triglycerides esterified with acids other than C₇ acids (e.g., C₂, C₃, C₄, C₅, C₆, C₈, or C₉ acids, or others, or combinations thereof (“non-C₇ triglycerides”)). In some embodiments, triheptanoin oil comprises less than 2, 1.9, 1.8, 1.7, 1.6, 1.5, 1.4, 1.3, 1.2, 1.1, 1, 0.9, 0.8, 0.7, 0.6, 0.5, 0.4, 0.3, 0.2 or 0.1% w/w or less of non-C₇ triglycerides, or any range derivable therein.

In one embodiment, triheptanoin oil comprises glycerol triheptanoate of a purity of greater than 97.5%. In some embodiments, triheptanoin oil comprises glycerol triheptanoate of a purity of about or at least about 97.5, 97.6, 97.7, 97.8, 97.9, 98, 98.1, 98.2, 98.3, 98.4, 98.5, 98.6, 98.7, 98.8, 98.9, 99.0, 99.1, 99.2, 99.3, 99.4, 99.5, 99.6, 99.7, 99.8, or 99.9% or more, or any range derivable therein.

In one embodiment, triheptanoin oil comprises less than 0.01% w/w ash. In some embodiments, triheptanoin oil comprises less than about 0.01, 0.009, 0.008, 0.007, 0.006, 0.005, 0.004, 0.003, 0.002, or 0.001% w/w ash or less, or any range derivable therein.

In one embodiment, triheptanoin oil comprises less than 0.04% w/w water. In some embodiments, triheptanoin oil comprises less than about 0.04, 0.035, 0.03, 0.025, 0.02, 0.015, 0.01, 0.005, or 0.001% w/w water or less, or any range derivable therein.

In one embodiment, triheptanoin oil comprises less than 2.5% w/w glycerol. In some embodiments, triheptanoin oil comprises less than 2.5, 2.4, 2.3, 2.2, 2.1, 2.0, 1.9, 1.8, 1.7, 1.6, 1.5, 1.3, or 1.2% w/w glycerol or less, or any range derivable therein.

In one embodiment, triheptanoin oil comprises less than 1.5% w/w monoheptanoate. In some embodiments, triheptanoin oil comprises less than 1.5, 1.4, 1.3, 1.2, 1.1, 1.0, 0.9, 0.8, 0.7, 0.6% w/w monoheptanoate or less, or any range derivable therein.

In one embodiment, triheptanoin oil comprises less than 3.0% w/w diheptanoate. In some embodiments, triheptanoin oil comprises less than 3.0, 2.9, 2.8, 2.7, 2.6, 2.5, 2.4, 2.3, 2.2, 2.1, 1.9, 1.8, or 1.7% w/w diheptanoate or less, or any range derivable therein.

In one embodiment, triheptanoin oil comprises less than 2.5% w/w Hexano-Diheptanoate. In some embodiments, triheptanoin oil comprises less than 2.5, 2.4, 2.3, 2.2, 2.1,

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2.0, 1.9, 1.8, 1.7, 1.6, 1.5, 1.3, or 1.2% w/w Hexano-Diheptanoate or less, or any range derivable therein.

In one embodiment, triheptanoin oil comprises less than 2.5% w/w Hexanoic acid. In some embodiments, triheptanoin oil comprises less than 2.5, 2.4, 2.3, 2.2, 2.1, 2.0, 1.9, 1.8, 1.7, 1.6, 1.5, 1.3, or 1.2% w/w glycerol or less, or any range derivable therein.

In some embodiments, the triheptanoin oil as described herein is a colorless oil of low viscosity and is odorless and tasteless. Triheptanoin oil as described herein is typically stored at room temperature and protected from light, and its stability can exceed 24 months under such conditions.

In one embodiment, the present invention provides a solid composition comprises triheptanoin as the active ingredient; fumed silica; and cellulose acetate. In a further embodiment, the solid composition a plurality of solid particles, each particle comprising triheptanoin adsorbed onto a solid substance comprising fumed silica and cellulose acetate. In a further embodiment, the solid composition comprises, by weight of the composition, about 50% to about 80% of triheptanoin; about 10% to about 30% of fumed silica; and about 10% to about 30% of cellulose acetate. In another embodiment, the solid composition comprises, by weight of the composition, about 60% of triheptanoin; about 20% of fumed silica; and about 20% of cellulose acetate.

In one specific embodiment, the present triheptanoin oil has the characteristics listed in Table 2:

Parameter	Specification	Method
IDENTITY		
Triheptanoate	Comparable to reference	Ph. Eur. 2.2.28
QUALITY		
Appearance	Clear substance and substance is not more intensely colored than reference solution Y ₃	Ph. Eur. 2.2.1, 2.2.2. method I
Relative density	0.95-0.98	Ph. Eur. 2.2.5
Refractive index	1.4440-1.4465	Ph. Eur. 2.2.6
Viscosity	15 mPa · s-23 mPa · s	Ph. Eur. 2.2.9
PURITY		
Composition of fatty acids		
Heptanoic acid	>99%	Ph. Eur. 2.4.22 method C
Hexanoic acid	Max. 0.8%	
All other, each	<0.03%	
Saponification value	360 to 410 mg KOH/g	Ph. Eur. 2.5.6
Acid value	Max. 0.2 mg KOH/g	Ph. Eur. 2.5.1
Hydroxyl value	Max. 10 mg KOH/g	Ph. Eur. 2.5.3 method A
Peroxide value	Max. 1.0 mg KOH/g	Ph. Eur. 2.5.5 method A
Water	Max. 0.2%	Ph. Eur. 2.5.12
Total ash	Max. 0.1%	Ph. Eur. 2.4.16
Heavy metals	Max. 10 ppm	Ph. Eur. 2.4.8
POTENCY/STRENGTH		
Triheptanoate	90% to 110%	Ph. Eur. 2.2.28

In one specific embodiment, the present triheptanoin oil has the characteristics listed in Table 3:

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Parameter	Specification	Method
IDENTITY		
Triheptanoate	Comparable to reference	Ph. Eur. 2.2.28
IR (infrared)	Comparable to reference	Ph. Eur. 2.2.24
QUALITY		
Appearance		
Clarity and Opalescence	Clear substance	Ph. Eur. 2.2.1
Color	Substance is not more intensely colored than reference solution Y ₃	Ph. Eur. 2.2.2, method I
Composition of fatty acids		
Heptanoic acid	≥99.0%	Ph. Eur. 2.4.22 method C
Hexanoic acid	≤1.0%	
Individual Unidentified Impurities	≤0.10%	
Impurities		
Glycerol	≤1.0%	Ph. Eur. 2.2.28
Monoheptanoate	≤0.5%	
Diheptanoate	≤1.5%	
Hexano-Diheptanoate	≤1.0%	
Individual Unidentified Impurities	≤0.5%	
Total Impurities	≤5.0%	
Elemental Impurities		
As	≤0.02 ppm	Ph. Eur. 2.2.57 or 2.2.58
Cd	≤0.19 ppm	
Pb	≤0.08 ppm	
Hg	≤0.12 ppm	
Acid value	Max. 0.2 mg KOH/g	Ph. Eur. 2.5.1
Water	Max. 0.2%	Ph. Eur. 2.5.12
Total ash	Max. 0.1%	Ph. Eur. 2.4.16
ASSAY		
Triheptanoate	95% to 103%	Ph. Eur. 2.2.28

In certain specific embodiments, the triheptanoin oil has the characteristics listed in Table 4:

Parameter	Example A	Example B
Triheptanoate	Comparable to reference	Comparable to reference
Triheptanoate Impurities:	99.0%	98.2%
Glycerol	not detectable	not detectable
Monoheptanoate	not detectable	not detectable
Diheptanoate	0.66%	0.62%
Hexano-Diheptanoate	0.39%	0.40%
Individual Unidentified Impurities	0.20%	0.18%
	(RRT* 1.258)	(RRT 1.258)
Total Impurities	1.25%	1.20%
Elemental Impurities		
As	not detected	not detected
Cd	not detected	not detected

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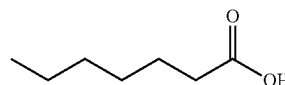
-continued

Parameter	Example A	Example B
Pb	not detected	not detected
Hg	not detected	not detected

*RRT: relative retention time.

Triheptanoin Oil Precursors and Manufacture of Triheptanoin Oil

Also provided herein are precursors to triheptanoin oil. In some embodiments, a precursor is n-heptanoic acid. n-Heptanoic acid has the following structure:



n-Heptanoic acid as described herein is of a purity or grade greater than food grade n-heptanoic acid. In some embodiments, purity of food grade n-heptanoic acid is between 90% and 98%. In some embodiments, purity of food grade n-heptanoic acid is between 95% and 98%. In some embodiments, purity of food grade n-heptanoic acid is between 95% and 97.5%. In some embodiments, purity of food grade n-heptanoic acid has a maximum purity of 98, 97.9, 97.6, 97.5, 97.4, 97.3, 97.2, 97.1, 97.0, 96.9, 96.8, 96.7, 96.6, 96.5, 96.4, 96.3%, 96.2, 96.1, 96.0, 95.9, 95.8, 95.7, 95.6, 95.5, 95.4, 95.3, 95.2, or 95.1%.

In some embodiments, n-heptanoic acid as described herein is considered ultrapure pharmaceutical grade n-heptanoic acid.

Typically, the total concentration of impurities in n-heptanoic acid is less than the total concentration of impurities in food grade n-heptanoic acid. In some embodiments, total impurities in n-heptanoic acid amount to less than 4.0% w/w. In some embodiments, total impurities amount to less than about 4.0, 3.9, 3.8, 3.7, 3.6, 3.5, 3.4, 3.3, 3.2, 3.1, 3.0, 2.9, 2.8, 2.7, 2.6, 2.5, 2.4, 2.3, 2.2, 2.1, 2.0, 1.9, 1.8, 1.7, 1.6, 1.5, 1.4, 1.3, 1.2, 1.1, 1.0, 0.9, 0.8, 0.7, 0.6, 0.5, 0.4, 0.3, 0.2, 0.1, 0.09, 0.08, 0.07, 0.06, 0.05, 0.04, 0.03, 0.02, or 0.01% w/w or less, or any range derivable therein.

In one embodiment, n-heptanoic acid comprises less than 2% w/w of triglycerides esterified with acids other than C7 acids (e.g., C2, C3, C4, C5, C6, C8, or C9 acids, or others, or combinations thereof ("non-C7 triglycerides")). In some embodiments, n-heptanoic acid comprises less than 2, 1.9, 1.8, 1.7, 1.6, 1.5, 1.4, 1.3, 1.2, 1.1, 1.0, 0.9, 0.8, 0.7, 0.6, 0.5, 0.4, 0.3, 0.2 or 0.1% w/w or less of non-C7 triglycerides, or any range derivable therein.

In one embodiment, an n-heptanoic acid composition comprises a sum of C7 carboxylic acids having greater than 97% purity. In some embodiments, the purity is greater than 97, 97.1, 97.2, 97.3, 97.4, 97.5, 97.6, 97.7, 97.8, 97.9, 98, 98.1, 98.2, 98.3, 98.4, 98.5, 98.6, 98.7, 98.8, 98.9, 99, 99.1, 99.2, 99.3, 99.4, 99.5, 99.6, 99.7, 99.8 or 99.9%, or more, or any range derivable therein. Compositions, such as pharmaceutical compositions, comprising n-heptanoic acid are also provided. Further, any composition comprising n-heptanoic acid can optionally be further defined as a pharmaceutical composition.

Triheptanoin oil can be prepared from n-heptanoic acid as described herein or a composition comprising n-heptanoic acid. For example, n-heptanoic acid can be esterified with glycerol to produce triheptanoin oil, such as in the presence of a basic catalyst and heat.

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In one embodiment, an n-heptanoic acid composition comprises less than 3.0% w/w of 2-methylhexanoic acid. In some embodiments, the composition comprises less than 3.0, 2.9, 2.8, 2.7, 2.6, 2.5, 2.4, 2.3, 2.2, 2.1, 2.0, 1.9, 1.8, 1.7, 1.6, 1.5, 1.4, 1.3, 1.2, 1.1, 1.0, 0.9, 0.8, 0.7, 0.6, 0.5, 0.4, 0.3, 0.2, or 0.1% w/w 2-methylhexanoic acid, or less, or any range derivable therein.

In one embodiment, an n-heptanoic acid composition comprises less than 0.1% w/w water. In some embodiments, a composition comprises less than 0.1, 0.09, 0.08, 0.07, 0.06, 0.05, 0.04, 0.03, 0.02, or 0.01% w/w of water or less, or any range derivable therein.

In one embodiment, an n-heptanoic acid composition comprises a carbonyl content of less than 0.2% w/w. Carbonyl content measurements are well-known in the art. In some embodiments, the carbonyl content is less than 0.15, 0.1, 0.05, or 0.01% w/w, or less, or any range derivable therein.

In one embodiment, an n-heptanoic acid composition comprises an iodine number of less than 0.5%. Iodine number measurements are well-known in the art. In some embodiments, the iodine number is less than 0.5, 0.4, 0.3, 0.2, or 0.1% or less, or any range derivable therein.

In one embodiment, an n-heptanoic acid composition comprises less than 0.2 mg/kg iron. In some embodiments, a composition comprises less than 0.2, 0.15, 0.1, 0.05, or 0.01 mg/kg iron, or any range derivable therein.

In one embodiment, a composition comprising n-heptanoic acid has a platinum/cobalt color number of less than 0.25. Determination of the platinum/cobalt color number is well-known in the art. In some embodiments, the platinum/cobalt color number is less than 0.25, 0.2, 0.15, 0.1, or 0.05 or less, or any range derivable therein.

Therapeutic Treatments

Provided herein in some embodiments are methods of using the present solid composition for treatment. The solid composition can be administered to a subject in need thereof in an amount effective to treat the subject.

For example, a subject may suffer from any one or more of the following: a fatty acid oxidation disorder or deficiency; adult polyglucosan body disease; a mitochondrial fat oxidation defect (e.g., relating to carnitine palmitoyl transferase I, carnitine palmitoyl transferase II, carnitine acylcarnitine translocase, very long chain acyl-CoA dehydrogenase, trifunctional protein, long chain hydroxyacyl-CoA dehydrogenase, multiple acyl-CoA dehydrogenase, short chain acyl CoA dehydrogenase, alpha glucosidase, brancher enzyme, debrancher enzyme, myophosphorylase, or phosphofructokinase); a glycogen storage disease (e.g., glycogen storage disease Type II); glucose transporter type 1 (GLUT1) deficiency syndrome; or a mitochondrial myopathy. In some embodiments, triheptanoin oil as described herein or a pharmaceutical composition comprising triheptanoin oil can be used to treat any condition described in U.S. Pat. No. 8,106,093.

In another embodiment, a subject may suffer from urea cycle disorders (UCD), which include several inherited deficiencies of enzymes or transporters necessary for the synthesis of urea from ammonia, including enzymes involved in the urea cycle; or hepatic encephalopathy (HE), which a spectrum of neurologic signs and symptoms believed to result from hyperammonemia, which frequently occur in subjects with cirrhosis or certain other types of liver disease. In some embodiments, glycerol phenylbutyrate as described herein or a pharmaceutical composition comprising glycerol phenylbutyrate can be used to treat these conditions. Further details of methods of treatment using

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glycerol phenylbutyrate can be found in U.S. Pat. Nos. 8,404,215 and 8,642,012, the contents of which are hereby incorporated by reference in entirety for all purpose.

In one embodiment, the solid compositions can be administered to a patient directly in various dosage forms (e.g., powders, granules, tablets, capsules, and the like) without mixing with food, drink, or other comestible compositions. In another embodiment, the solid compositions, such as powders or granules, can be taken together with food, drink, or other comestible compositions by mixing the solid compositions into the food, drink, or comestible compositions. The food, drink, or comestible compositions can be in any form including liquid, solid, semi-solid, foamy material, paste, gel, cream, lotion, and combinations thereof. The solid compositions can provide therapeutic effects while minimize gastric upset and gastrin release, and co-administration with food, drink, or other comestible compositions can enhance such tolerability benefit. In addition, co-administration of food, drink, or other comestible compositions with the solid composition can provide delayed and/or extended release of the active ingredients and thus allow less frequent dosing, longer term of energy exposure, and potentially higher efficiency in absorption from the GI tract. The delayed and/or extended release characteristic of the solid composition can also be independently achieved by including one or more sustained release polymers in the solid compositions as discussed above.

As used herein, the term “effective” (e.g., “an effective amount”) means adequate to accomplish a desired, expected, or intended result. An effective amount can be a therapeutically effective amount. A “therapeutically effective amount” refers to the amount of an active ingredient that, when administered to a subject, is sufficient to effect such treatment of a particular disease or condition. The “therapeutically effective amount” will vary depending on, e.g., the disease or condition, the severity of the disease or condition, and the age, weight, etc., of the subject to be treated.

In general, “treating” or “treatment” of any condition, disease or disorder refers, in some embodiments, to ameliorating the condition, disease or disorder (i.e., arresting or reducing the development of the disease or at least one of the clinical symptoms thereof). In some embodiments “treating” or “treatment” refers to ameliorating at least one physical parameter, which may not be discernible by the subject. In some embodiments, “treating” or “treatment” refers to inhibiting the condition, disease or disorder, either physically, (e.g., stabilization of a discernible symptom), physiologically, (e.g., stabilization of a physical parameter) or both. In some embodiments, “treating” or “treatment” refers to delaying the onset of a condition, disease, or disorder.

EXAMPLES

Example 1

Table 5 shows the high purity of exemplary triheptanoin oil samples and solid samples comprising triheptanoin oil in accordance with the present invention.

Further, the stability of an exemplary solid composition (solid sample 2 in Table 5) comprising triheptanoin oil was tested. Measurements were conducted after storage at 25° C./60% RH (relative humidity) in double LDPE (low density polyethylene) bags in a 60 cc induction sealed HDPE (high density polyethylene) bottle and results are shown in Table 6. Measurements were also conducted after storage at 40° C./75% RH in double LDPE bags in a 60 cc induction sealed HDPE bottle and results are shown in Table 7.

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TABLE 5

Initial Purity Test Results						
Impurities	Specification	Parameter				
		OIL Sample 1	OIL Sample 2	SOLID Sample 1	SOLID Sample 2	SOLID Sample 3
Glycerol	≤1.0%	ND	ND	ND	ND	ND
Monoheptanoate	≤0.5%	ND	ND	ND	ND	ND
Diheptanoate	≤1.5%	1.2%	0.7%	1.1%	1.1%	1.1%
Hexano-Diheptanoate	≤1.0%	0.4%	0.4%	0.4%	0.4%	0.4%
Individual Unidentified Impurities:	≤0.5%	RRT 0.91: 0.06%	RRT 1.26: 0.20%	RRT 0.91: 0.06%	RRT 0.91: 0.06%	RRT 0.91: 0.06%
RRT* (%)		RRT 1.05: 0.16%		RRT 1.05: 0.16%	RRT 1.05: 0.16%	RRT 1.05: 0.16%
		RRT 1.06: 0.17%		RRT 1.06: 0.17%	RRT 1.06: 0.17%	RRT 1.06: 0.17%
		RRT 1.07: 0.11%		RRT 1.07: 0.11%	RRT 1.07: 0.11%	RRT 1.07: 0.11%
		RRT 1.25: 0.06%		RRT 1.25: 0.06%	RRT 1.25: 0.06%	RRT 1.25: 0.06%
		RRT 1.26: 0.23%		RRT 1.26: 0.23%	RRT 1.26: 0.23%	RRT 1.26: 0.24%
		RRT 1.30: 0.10%		RRT 1.30: 0.10%	RRT 1.30: 0.10%	RRT 1.30: 0.10%
		RRT 1.36: 0.11%		RRT 1.36: 0.11%	RRT 1.36: 0.11%	RRT 1.36: 0.11%
Total Impurities	≤5.0%	2.6%	1.3%	2.6%	2.6%	2.6%

RRT = relative retention time.

ND = not detected.

TABLE 6

Stability Study Results for an exemplary solid composition stored at 25° C./60% RH:				
Parameter	Specification	Time points (months)		
		Freshly Made (Five months before)	0	1
Appearance	White to off-white powder	Conforms	Conforms	Conforms
Impurities				
Glycerol	≤1.0%	ND	ND	ND
Monoheptanoate	≤0.5%	ND	ND	ND
Diheptanoate	≤1.5%	1.1%	1.1%	1.1%
Hexano-Diheptanoate	≤1.0%	0.4%	0.4%	0.4%
Individual Unidentified Impurities: RRT (%)	≤0.5%	RRT 0.91: 0.06%	RRT 0.91: 0.06%	RRT 0.91: 0.05%
		RRT 1.05: 0.16%	RRT 1.05: 0.16%	RRT 1.05: 0.16%
		RRT 1.06: 0.17%	RRT 1.06: 0.17%	RRT 1.06: 0.17%
		RRT 1.07: 0.11%	RRT 1.07: 0.11%	RRT 1.07: 0.11%
		RRT 1.25: 0.06%	RRT 1.25: 0.06%	RRT 1.25: 0.06%
		RRT 1.26: 0.23%	RRT 1.26: 0.24%	RRT 1.26: 0.23%
		RRT 1.30: 0.10%	RRT 1.30: 0.10%	RRT 1.30: 0.09%
		RRT 1.36: 0.11%	RRT 1.36: 0.11%	RRT 1.36: 0.09%
Total Impurities	≤5.0%	2.6%	2.6%	2.5%
Water	NMT 5%	NS	0.29%	0.35%
Triheptanoate	90%-110%	99%	101%	100%
Acetone	≤200 ppm	<100 ppm	NS	NS

RRT = relative retention time.

NMT = not more than.

ND = not detected.

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TABLE 7

Stability Study Results for an exemplary solid composition stored at 40° C./75% RH:				
Parameter	Specification	Time points (months)		
		Freshly Made (Five months before)	0	1
Appearance	White to off-white powder	Conforms	Conforms	Conforms
Impurities				
Glycerol	≤1.0%	ND	ND	ND
Monoheptanoate	≤0.5%	ND	ND	ND
Diheptanoate	≤1.5%	1.1%	1.1%	1.1%
Hexano-Diheptanoate	≤1.0%	0.4%	0.4%	0.4%
Individual Unidentified Impurities: RRT (%)	≤0.5%	RRT 0.91: 0.06% RRT 1.05: 0.16% RRT 1.06: 0.17% RRT 1.07: 0.11% RRT 1.25: 0.06% RRT 1.26: 0.23% RRT 1.30: 0.10% RRT 1.36: 0.11%	RRT 0.91: 0.06% RRT 1.05: 0.16% RRT 1.06: 0.17% RRT 1.07: 0.11% RRT 1.25: 0.06% RRT 1.26: 0.24% RRT 1.30: 0.10% RRT 1.36: 0.11%	RRT 0.91: 0.05% RRT 1.05: 0.16% RRT 1.06: 0.17% RRT 1.07: 0.11% RRT 1.25: 0.06% RRT 1.26: 0.23% RRT 1.30: 0.09% RRT 1.36: 0.10%
Total Impurities	≤5.0%	2.6%	2.6%	2.5%
Water	NMT 5%	NS	0.29%	0.45%
Triheptanoate	90%-110%	99%	101%	98%
Acetone	≤200 ppm	<100 ppm	NS	NS

RRT = relative retention time.

NMT = not more than.

ND = not detected.

Example 2

Single dose study of each arm with full pharmacokinetics (PK) profile through 48 hours post dose was performed to determine when metabolites return to baseline (one week washout between each arm) as well as if acute release of gastrin and cholecystokinin (CCK) hormones cause spasmodic stomach contractions. Blood samples were collected from each arm 0-90 min plus anytime outside this window if gastric distress observed.

More specifically, animals (n=3/sex) were fasted prior to dose and fed 4 hrs post dose. Blood samples for PK analysis of triheptanoin and metabolites were collected pre-dose through 48 hrs post dose. As seen in the multiple dose study (Example 3), there were no major differences between males and females thus data was combined for all metabolites.

In FIGS. 1-21, animals were administered a single oral gavage dose level of a triheptanoin oil sample (i.e., the oil) or exemplary solid samples comprising triheptanoin oil in accordance with the present invention. The two exemplary solid compositions comprise 1) 60%:20%:20% of triheptanoin oil:Cab-O-Sil:cellulose acetate (i.e., the Powder-ER); and 2) 80%:20% of triheptanoin oil:Cab-O-Sil (i.e., the Powder), respectively.

Triheptanoin and Heptanoic Acid

Triheptanoin is metabolized to heptanoic acid. The metabolite PK profiles for the three samples are shown in FIGS. 1-4. Bio-modal peaks were observed for heptanoic acid on three arms. It is contemplated that: 1) the first peak may be when the triheptanoin hits the stomach and the second peak may be when the triheptanoin hits the duodenum; 2) fat soluble materials can be dissolved in the stomach but free heptanoin is absorbed in the duodenum (higher 2nd peak for the powders); and 3) the cellulose acetate in the Powder-ER formulation delays delivery by ~one hour.

Energy Metabolites: Ketones

As seen in FIGS. 5-9, we observed comparable levels of ketones with oil treatment and powder-ER treatment. Delivery of C4 ketones were faster with powders but also blunted by the powders. The ratio C4 ketone:C5 ketone indicates the state of feeding; as this ratio increases it indicates that triheptanoin is being used as a source of calories.

Energy Metabolites: Aspartate

The levels of aspartate of the three samples are shown in FIGS. 10-12. We observed sustained release with the Powder-ER. It also showed a more favorable effect on gluconeogenic precursors because of the delay in release. The levels of aspartate of the Oil and the powder return to baseline at 8 hrs, while that of the powder-ER returns to baseline at 24 hrs post dose.

Energy Metabolites: Glutamate

The levels of glutamate of the three samples are shown in FIGS. 13-15. When glutamate increases, the trend indicates that triheptanoin protects against hypoglycemia. Triheptanoin increases the chemicals that can be made into glucose. Livers in minipigs store gluconeogenic precursors in the form of glutamate based on presence in the blood.

Alternative Metabolites

The metabolite PK profiles of alternative metabolites including pimelic acid, 3-hydroxypropionate and propionyl glycine are shown in FIGS. 16-18. Pimelic acid and 3-hydroxypropionate production was not increased with the Powder-ER when compared to the oil. The levels were lower for the powder. 3-hydroxypropionate is a normal acid made by gut bacteria, but very dependent on the individual animal.

We have increased co-enzyme A activity by giving triheptanoin. In addition, the levels of average energy and alternative metabolites are shown in FIGS. 19-21.

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Example 3

Multiple dose study was performed on the oil, the Powder and the Powder-ER as used in Example 2 and the results are shown below.

Powder Arm:

Day 1: Animal #6501 vomited ~1 hour post dose

Day 3: Animal #6502 vomited

Day 3: Animal #6510 gastric distressed observed

Oil Arm:

Day 1: Animal #6501 vomited ~3 hours post dose

Day 1: Animal #6516 vomited ~3 hours post dose

Day 5: Animal #6509 gastric distressed observed

Powder ER Arm:

No vomiting or gastric distress observed

Heptanoic Acid Across Dose Groups

Overall, results as expected, the triheptanoin releases heptanoic acid. Referring to FIG. 22, triheptanoin was metabolized primarily in liver to C7 fatty acids and ketone bodies which distribute via circulation to other tissues to provide an energy source. There were no triheptanoin values in oil arm. All triheptanoin oil was converted to C7.

Trace amounts of triheptanoin were observed in powder arm. The powder matrix may have caused a delay in the lipase action, this delay is slowing the release of C7. The Powder ER arm released slightly higher levels of heptanoic acid than the powder arm but not as high as the oil.

Energy Metabolites

As seen in FIGS. 23 and 24, beta hydroxy levels rose 10-fold above resting state while ketone bodies did not increase from Day 1 to Day 7, which indicated no chance for ketosis.

Referring to FIGS. 25 and 26, we observed glutamic acid and aspartic acid increased and the Powder had higher levels of both metabolites than the powder ER. Glutamic acid and aspartic acid are gluconeogenic amino acids, and can represent a novel source of energy.

FIGS. 27-29 shows PK profiles of additional metabolites: A) pimelic acid, B) 3-hydroxypropionate, and C) propionylglycine following dosing on Day 1 and Day 7. The oil had the highest amount of pimelic acid compared to the powders. 3-Hydroxypropionate was increasing in all 3 groups but the powder-ER had the smallest increase and the lowest levels. Propionylglycine was essentially zero for the oil and the powder-ER but the powder dosing caused an increase. Overall, the Powder ER produced the least amount of excess metabolites.

The PK profiles of all metabolites are shown in FIG. 30. The metabolites include Triheptanoin, C4 (Hydroxy and Keto), C5 (Hydroxy and Keto), Heptanoic acid, Pimelic acid, Glutaric acid, Aspartic acid, Glutamic acid, Alanine and 3-Hydroxypropionic acid.

The results of the powder and the powder-ER as compared to that of the oil are shown in FIGS. 31-40. The Powder ER raised the C5-hydroxy levels the most of all the treatments, which indicates it's the most effective at delivering fatty acids to the cells. We observed differences between Days 1 and Day 7 for the powders. Day 7 levels were lower than Day 1, spike at 240 min on Day 1, which indicates that PK sampling can be carried out longer. The Powder-ER dosing resulted in lowest levels of pimelic acid. The Powder had a higher release of aspartic acid than the Powder-ER. The Powder had higher levels of glutamic acid and alanine than the Powder-ER. The Powder also had higher levels of 3-Hydroxypropionate than the Powder-ER.

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Following long-standing patent law convention, the terms "a", "an", and "the" refer to "one or more" when used in this application, including the claims.

The use of the term "or" in the claims is used to mean "and/or" unless explicitly indicated to refer to alternatives only or the alternatives are mutually exclusive. It is specifically contemplated that any listing of items using the term "or" means that any of those listed items may also be specifically excluded from the related embodiment.

Throughout this application, the term "about" is used to indicate that a value includes the standard deviation of error for the device or method being employed to determine the value.

Unless defined otherwise, all technical and scientific terms used herein have the same meaning as commonly understood to one of ordinary skill in the art to which the present application belongs. Although any methods and materials similar or equivalent to those described herein can be used in the practice or testing of the present application, representative methods and materials are herein described.

The patents listed herein describe the general skill in the art and are hereby incorporated by reference in their entireties for all purposes and to the same extent as if each was specifically and individually indicated to be incorporated by reference. In the case of any conflict between a cited reference and this specification, the specification shall control. In describing embodiments of the present application, specific terminology is employed for the sake of clarity. However, the invention is not intended to be limited to the specific terminology so selected. Nothing in this specification should be considered as limiting the scope of the present invention. All examples presented are representative and non-limiting. The above-described embodiments may be modified or varied, without departing from the invention, as appreciated by those skilled in the art in light of the above teachings.

What is claimed is:

1. A pharmaceutical composition comprising an active ingredient, diheptanoate, and hexano-diheptanoate, wherein the active ingredient is in the form of triheptanoin oil; wherein the triheptanoin oil has a purity greater than 98%; wherein the composition comprises 0.01% to 1.5% diheptanoate by weight; wherein the composition comprises 0.01% to 1.0% hexano-diheptanoate by weight; wherein glycerol is undetectable in the composition; wherein monoheptanoate is undetectable in the composition; wherein the composition comprises no more than 0.2% water by weight; and wherein the composition is a liquid composition for oral administration to a human.
2. The pharmaceutical composition of claim 1, wherein the composition comprises 0.05% to 0.7% diheptanoate by weight.
3. The pharmaceutical composition of claim 1, wherein the triheptanoin oil has a purity greater than 98.5%.
4. The pharmaceutical composition of claim 1, wherein the total amount of diheptanoate and hexano-diheptanoate in the composition is no more than 1.3% by weight.
5. The pharmaceutical composition of claim 1, wherein the composition has an assay value of the triheptanoin oil between 95% to 103% as measured by Ph. Eur. 2.2.28.

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